

The Dynamics of Technology Transfer: Multinational Investment in China and Rising Global Competition*

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Abstract

US multinationals formed joint ventures in China for market access and lower labor costs. However, these ventures transfer technology to Chinese firms, intensifying future competition. While individual multinationals weigh the risks to their own future profits, they disregard the negative impact on other US firms and the broader economy, resulting in an over-investment that may reduce US welfare. In our empirical analysis, US industries with more joint ventures in China experienced worse outcomes. Furthermore, within industries, firms with greater product overlap with the multinationals operating joint ventures in China fared worse. We develop a two-country model with oligopolistic competition, innovation, and joint ventures. For the US as a whole, the short-run gains from joint ventures are outweighed by long-run losses due to rising Chinese competition. Joint ventures raise participants' profits at the expense of non-participants' profits, overall US innovation, and real wages. A ban on joint ventures since 1999 would have boosted US welfare by 1.1 percent. However, a joint venture ban does not always raise welfare. If the ban is implemented in 2025, when the US-China productivity gap has narrowed, it would in fact reduce US welfare by 0.9 percent.

Keywords: Technology transfer, technology diffusion, innovation, competition, joint venture, foreign direct investment

JEL Codes: F23, O25, O33

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1. INTRODUCTION

Intensifying economic rivalry between the United States (US) and China has heightened scrutiny of China's economic policies and business practices. For instance, China has long mandated that multinational enterprises (MNEs) transfer technology as a condition for market access. This often involved the formation of joint ventures, as such arrangements were considered to be more conducive to technology transfer than other forms of foreign direct investment (FDI). This practice has been criticized by some as a form of intellectual property "theft" and has also been cited as a factor exacerbating the trade imbalance between the US and China. In light of these concerns, the US has recently imposed restrictions on outward FDI in critical technologies.¹

However, US firms formed joint ventures with Chinese partners voluntarily to gain access to the Chinese market and cheaper labor, even with the risk of technology leakage. This raises a crucial question: What is the economic justification for government intervention when MNEs voluntarily engage in such partnerships? US firms recognize that by establishing joint ventures, their technology will be leaked to their Chinese partners, enhancing the productivity of their Chinese partners and thereby intensifying global competition in the future. However, they do not care about (i) the profit losses that other US firms will suffer from intensified global competition and (ii) general equilibrium effects. This may lead to an over-investment in joint ventures and excessive technology transfer to China from the perspective of what is socially optimal for the US. Despite this seemingly straightforward economic logic, this specific mechanism has not been formally explored in the literature, which presents an opportunity to provide novel insights.²

Our main contribution is theoretical and quantitative. We develop a two-country endogenous growth model where oligopolistic firms strategically determine their innovation and

¹"As companies negotiate the terms of the joint venture, the foreign side may be asked—or required—to transfer its technology in order to finalize the partnership . . . [F]oreign companies have limited leverage in the negotiation if they wish to access the market. Although this type of technology transfer may not be explicitly mandated in a Chinese law or regulation, it is often an unwritten rule for market access." ([Office of the U.S. Trade Representative, 2018](#)). For example, the CHIPS and Science Act of 2022 prohibits US government funding recipients from making certain investments in China.

²On FDI and technology transfer to China, a former Tesla executive said: "In this game, one American company gets to win. They don't care if all their US competitors lose. It's actually better for them. But on the other side, all the Chinese companies win. They all get to step up and create a massive market where none previously existed." [McGee \(2025, p.283\)](#)

joint venture decisions, and we analyze the full dynamics of the model. Our quantitative analysis demonstrates that, from a US social welfare perspective, the market equilibrium leads to too many joint ventures.

We construct a dataset by integrating firm-level balance sheet data from China’s National Bureau of Statistics, patent data, and ownership structure information from the Orbis database. Data for US firms come from Compustat.

Using this dataset, we document two key motivating facts. First, using an event study design, we match Chinese partner firms that formed joint ventures with MNEs (the treated group) to control firms that never did, via propensity score matching. Following the formation of these joint ventures, the Chinese partner firms experience significant growth in sales and exports, not counting the sales and exports of the joint ventures they co-own. Furthermore, their patenting activities grow similar to that of their MNE partners, which supports the notion of technology diffusion between the partners. The results hold when we restrict the control group to firms with rumored foreign investment deals that did not materialize. This group likely shares the unobserved factors that lead to joint ventures, and hence selection bias is less likely to drive our results.

Second, and more importantly, we provide evidence that joint ventures are associated with negative outcomes for non-participating US firms, both across and within industries. US firms in industries with more joint venture investment in China experienced larger declines in sales, employment, capital, and exports. Within industries, an event study shows that non-participants with greater product overlap with the participating US MNEs experienced worse outcomes. We find no such negative outcomes in response to wholly foreign-owned enterprise activities in China, which involve no local partners and therefore no direct technology diffusion between partners.

Motivated by these two facts, we first build a simple model to characterize the mechanism. The US has a leader firm and a fringe firm, and China has a single firm that is initially less productive than both US firms. We model a joint venture as a mutual agreement between the US leader and the Chinese firm. With a joint venture, the US leader produces and sells in China without trade costs, but its technology can leak to the Chinese firm. The joint venture increases the profits of both the US leader and the Chinese firm, through larger market access for the US leader and higher productivity from leakage for the Chinese firm. However, what

they gain comes at the expense of the US fringe firm's profits.

Such business stealing depends crucially on the assumption that fringe firms produce positive quantities in equilibrium. If we were to follow the common assumption in the literature that fringe firms only pin down the limit price but do not produce, there would be no fringe profit to steal and US leaders would internalize the effect of joint ventures. We show theoretically that the joint venture can reduce US welfare when the business stealing effect is sufficiently strong, although technology leakage to China always lowers prices and raises consumer surplus in the US.

The joint venture has two opposing effects on the US leader's innovation incentive. First, it enlarges the leader's market by letting it serve China without trade costs, which raises its return on innovation (market size effect). However, the improved technology can leak to the Chinese rival, which then competes away the gains in both markets (leakage effect). When leakage is highly probable, the joint venture lowers US innovation, and eventually US welfare.

The simple model has several important limitations. First, the simple model has no general equilibrium forces and no meaningful dynamics. Quasi-linear preferences pin down wages, so joint ventures cannot affect labor costs. Dynamics matter because China's catch-up is gradual, and firms' decisions on joint ventures and innovation are forward looking. Second, our empirical analysis exploits cross-sectional variation across firms and industries, which the simple model cannot address. Third, the welfare effects of joint ventures can be positive or negative depending on the relative strengths of the business stealing, market size, and leakage effects, necessitating a quantitative analysis.

We therefore turn to a dynamic two-country growth model that incorporates such channels. In each industry, the market structure consists of a leader and a fringe firm in each country. All firms produce differentiated varieties, which they sell in both domestic and foreign markets. While firms possess market power within their specific industry, there is a continuum of industries, so no single firm can influence aggregate prices or quantities. Leaders can enhance their productivity through innovation. In addition, US leaders have the option to establish joint ventures in China, partnering with their Chinese counterpart in the same industry. These ventures enable US firms to bypass trade costs and use cheaper Chinese labor.

Even without joint ventures, the model allows for stochastic diffusion of technology both within and between countries, acknowledging that joint ventures are not the sole channel of technology diffusion. The formation of a joint venture in China introduces an additional probability of technology diffusion from the US leader to its Chinese counterpart during their partnership, a feature consistent with our first motivating fact.

The entry of a new joint venture firm immediately intensifies competition within the industry, and the stochastic diffusion of technology to the Chinese leader adds to this pressure over time. As in the simple model, the US leader accounts for these competition effects on its own future profits when deciding whether to form a joint venture. Through bargaining, it also captures part of the joint venture's profit flow and of the benefits that accrue to the Chinese leader. Fringe firms, by contrast, suffer a purely negative effect from intensified Chinese competition, consistent with our second motivating fact. Because US leaders rightly disregard this business stealing effect, the equilibrium number of joint ventures may exceed the US social optimum.

We solve for the model's transitional dynamics from an initial state, in which Chinese firms' productivity lags behind the US level. We calibrate the model to match key empirical moments along this transition. We infer the parameters governing technology diffusion using the corresponding regression coefficients from our empirical analysis. Importantly, the model reproduces our second motivating fact, the negative outcomes for US non-participants, as a non-targeted moment. Furthermore, in our model, the value and, consequently, the likelihood of a US leader forming a joint venture are greater when the US-China technology gap is larger, a pattern we confirm in the data as another non-targeted moment. As joint ventures shrink this gap through diffusion, they erode the US's comparative advantage and worsen its terms of trade, thereby reducing its gains from trade.

As in the simple model, joint ventures have opposing effects on US leaders' innovation efforts. Innovation increases the total joint venture surplus that US leaders capture through bargaining, and joint ventures enlarge the effective market and raise marginal returns to R&D. These are positive effects. On the other hand, by accelerating technology diffusion to Chinese rivals, joint ventures reduce future profits and returns to innovation. In our quantitative analysis, the latter (leakage effect) dominates over the medium run, decreasing US innovation. For Chinese leaders, technology diffusion serves as a substitute for their own

innovation efforts, so they also innovate less when a joint venture is established.

To compute the welfare consequences of joint ventures, we analyze the effect of a policy that bans US firms from forming joint ventures with Chinese firms, effective from 1999. We find that this ban increases US consumption-equivalent welfare by 1.1 percent. For the US, leaders' profits fall by 17 percent in present value terms, while fringe firms' profits rise by 6.5 percent, but the total profit of the corporate sector declines. The real wage increases by 2.1 percent due to higher labor demand in the US and lower US price indices from increased US innovation, driving the overall welfare gain. The ban has a short-run negative effect, as US firms can no longer benefit immediately from lower wages and reduced trade costs in China. However, this effect is outweighed by medium-run benefits, as less technology leakage and higher innovation efforts let the US maintain its technological advantage over China for longer.

When the US prohibits joint ventures, Chinese leader firms compensate for the reduced technology diffusion by increasing their own innovation efforts. However, China's productivity growth is delayed and China's welfare is reduced by 8.9 percent in units of permanent consumption. In China, the profits of both leaders and fringe firms, as well as the real wage, are lower without joint ventures from the US.

Two main margins drive this negative welfare effect of joint ventures on US welfare: the business stealing effect and lower innovation rates due to technology leakage. To isolate each margin, we consider two additional exercises. First, we examine an alternative scenario in which US leaders are required to compensate US fringe firms for their losses from joint ventures through multilateral Nash-in-Nash bargaining. That is, US leaders are compelled to internalize the profit losses of their domestic competitors. With such multilateral bargaining, substantially fewer joint ventures are formed. Moreover, a complete ban of joint ventures decreases US welfare by 0.03 percent, contrary to the benchmark result. That is, the business stealing effect is the main reason why joint ventures reduce welfare in the benchmark.

In the second exercise, we neutralize the innovation response to the leakage effect by exogenously shifting average US innovation rates when joint venture are banned, so that average innovation is the same with and without the ban. In this scenario, the welfare effect of joint ventures is about 80 percent of that in the benchmark, implying that the leakage effect accounts for only 20 percent of the welfare loss from joint ventures.

Our result does not imply that banning joint ventures would *always* enhance welfare for the US. If new joint ventures were prohibited starting in 2025, a time when the US-China technology gap is substantially smaller than in 1999, both the business stealing effect and the leakage effect would be less pronounced. In that scenario, the forgone short-run gains from joint ventures are large relative to the medium-run boost to innovation, so banning joint ventures would in fact reduce US welfare by 0.91 percent.

Related literature. First, our paper contributes to the growth literature on innovation and diffusion in open economies (e.g., [Grossman and Helpman, 1993](#); [Atkeson and Burstein, 2010](#); [Impullitti, 2010](#); [Sampson, 2016, 2023](#); [Arkolakis et al., 2020](#); [Buera and Oberfield, 2020](#); [Perla et al., 2021](#); [Sui, 2025](#); [Rachapalli, 2021](#); [Cai et al., 2022](#); [Peters, 2022](#); [Bai et al., 2025b,c](#); [de Souza et al., 2025](#); [Hémous et al., 2025](#); [Milicevic et al., 2025](#)). Our model builds on [Akcigit et al. \(2023\)](#) and [Choi and Shim \(2024\)](#), in which firms compete with foreign firms through innovation but also benefit from technology diffusion. We extend this framework by incorporating the idea that multinational production facilitates technology diffusion from advanced to developing countries (e.g., [Burstein and Monge-Naranjo, 2009](#); [Holmes et al., 2015](#); [Alviarez et al., 2023](#)). [Akcigit et al. \(2024\)](#) discuss technology leakage in the context of Chinese venture capital investment in the US and national security concerns. [Lam \(2024\)](#) focuses on illegal imitation and studies optimal intellectual property rights protection in China. [Santacreu \(2025\)](#) studies the dynamic effects of intellectual property rights provision. Our contribution lies in studying how technology leakage through joint ventures generates business stealing and shapes innovation responses, and in analyzing the policy implications both theoretically and quantitatively.

[Holmes et al. \(2015\)](#) study the welfare implications of China's quid pro quo policy in the multi-country dynamic general equilibrium framework of [McGrattan and Prescott \(2009\)](#), which features transfers of technology capital. Their technology capital plays a role similar to innovation in our model. In our model, as in theirs, technology leakage to China lowers multinationals' FDI incentives and reduces US welfare. However, we differ in two main respects. First, our model focuses on the business stealing effect and US innovation, which are absent from their framework. In their model, a representative firm maximizes its profit under perfect competition, so the firm's profit coincides with total sectoral (or aggregate)

profit, which is not the case in ours. We find that once this externality is corrected for, the US can gain from joint ventures even under the quid pro quo. Second, we calibrate our model to firm-level microdata and our own reduced form estimates. Our empirical and quantitative analyses also suggest that this business stealing effect contributed to the recent US manufacturing decline ([Autor et al., 2013](#); [Acemoglu et al., 2016](#)).

Regarding China's growth, [Brandt et al. \(2012\)](#) document sizable TFP growth among Chinese firms using firm-level data. [Alder et al. \(2016\)](#) study place-based policy, and [Brandt et al. \(2017\)](#), trade liberalization. [Song et al. \(2011\)](#), [Hsieh et al. \(2015\)](#), [Tombe and Zhu \(2019\)](#), and [Chen et al. \(2021\)](#) study misallocation, and [König et al. \(2022\)](#) quantify its dynamic effects on TFP growth in an endogenous growth model. We instead study joint ventures as a key learning channel for China's catch-up, consistent with recent micro-level evidence of positive spillovers from joint ventures ([Jiang et al., 2024](#); [Bai et al., 2025a](#)). Among recent studies of US-China technology rivalry (e.g., [Ju et al., 2024](#); [Ma et al., 2025a,b](#); [Alp and Ates, 2026](#)), [Ma and Zhang \(2025\)](#) is closest to ours. It documents positive FDI spillovers in Chinese patent data and, using a Ricardian trade–multinationals–innovation model, finds that the quid pro quo policy raised China's welfare. We study the US-side business stealing effect generated by the joint ventures that diffused technology to China and contributed to its growth.

This paper also relates to the large literature on multinational production (e.g., [Helpman et al., 2004](#); [Irarrazabal et al., 2013](#); [Ramondo and Rodríguez-Clare, 2013](#); [Tintelnot, 2017](#); [Arkolakis et al., 2018](#); [Alviarez, 2019](#); [Head and Mayer, 2019](#); [Bilir and Morales, 2020](#); [Garetto et al., 2024](#); [Cai and Xiang, 2025](#)). Much of this work studies the margin between exporting and establishing a foreign affiliate. We highlight a novel margin, the dynamic trade-off between market access and technology leakage, and show that its timing shapes the welfare effects of opening to joint ventures. The joint surplus is largest where China has the most to learn, so joint ventures concentrate in sectors of US comparative advantage and bias China's growth toward them, lowering US gains from trade ([Samuelson, 2004](#)). In [Alviarez \(2019\)](#), a similar erosion arises from the multinational's location choice alone, whereas in ours it also requires the consent of the Chinese firm, whose gain from diffusion rises with the technology gap.

2. BACKGROUND AND DATA

2.1 Quid Pro Quo Policy and the US-China Trade War

After decades of isolation, Deng Xiaoping initiated economic reforms and opened China to foreign investment in 1979 with the “Law on Sino-Foreign Equity Joint Ventures” (henceforth referred to as the JV Law). Joint ventures (JVs) are defined as firms with mixed ownership between foreign and Chinese shareholders, with foreign equity shares of at least 25 percent but less than 100 percent. Firms with foreign equity below 25 percent are classified as domestic firms, while those with 100 percent foreign equity are registered as wholly foreign-owned enterprises (WFOEs). A key difference between JVs and WFOEs is whether local Chinese partners engage in ownership and control. WFOEs are fully owned and run by foreign MNEs. In contrast, JVs require shared ownership between foreign MNEs and local Chinese partners. Foreign firms were often required to transfer technology to their local partners, and profits from JVs were shared based on equity shares. Equity shares were strictly regulated, with minimum requirements and maximum caps on foreign MNE ownership.

The quid pro quo policy required foreign MNEs to transfer technologies, capital equipment, know-how, and product lines through JVs in exchange for access to China’s large consumer market and abundant labor. From 1979 to 1986, JVs were the only legal form of FDI in China. WFOEs were allowed in some sectors from 1986. After China joined the WTO in 2001, the government reformed its FDI policies, liberalized tariffs, and strengthened intellectual property protection. Explicit technology transfer mandates were dropped and JV requirements eased, but equity caps and JV requirements remained in certain high-tech sectors. Despite these reforms, the quid pro quo policy has been at the center of US-China tensions. US policymakers have criticized it as an unfair trade practice and argue that it has persisted in more implicit forms.³

³In its 2017–18 Section 301 investigations, the Office of the United States Trade Representative reported that China implicitly pressured foreign MNEs to form JVs and transfer advanced technologies through both formal regulations and informal administrative barriers. [Holmes et al. \(2015\)](#) also showed that the mandates became implicit but remained in effect. Some described it as “voluntary is the new mandatory” ([McGee, 2025](#)).

2.2 Data

Our main dataset combines firm outcomes, ownership structure, and patent data for Chinese and US manufacturing firms with sector-level data. It covers 1998–2013. All monetary values are in 2007 US dollars. Appendix A provides details.

We obtain Chinese firm outcomes from the Annual Survey of Industrial Enterprises, constructed by the National Bureau of Statistics. The survey covers all state-owned firms and all private firms with sales above 5 million RMB through 2010 and above 20 million RMB from 2011 onward. We apply the 20 million RMB threshold in every year. It reports annual sales, exports, employment, and capital (fixed assets), each firm’s 4-digit 1994 Chinese Industry Classification (CIC) code, and its location. The firms in the survey account for 90 percent of Chinese manufacturing output. The survey also records registration types, which identify JVs, WFOEs, state-owned firms, and private firms. In our definition of JVs and WFOEs, we exclude those involving MNEs from Hong Kong, Macao, and Taiwan, given the special economic and regulatory relations between mainland China and these regions.

We obtain US firm outcomes from Compustat, which covers publicly listed firms. The data report sales, employment, and capital (measured as PPEGT). We also obtain each firm’s total foreign sales (including both exports and sales from foreign affiliates) from the historical geographic segment data and use this variable as a proxy for exports.⁴ Firms are classified by 4-digit 1987 SIC codes, which we aggregate into 383 industries for compatibility with the CIC and HS codes.⁵

To identify ownership links between Chinese partners and foreign MNEs, we use historical ownership data from the Orbis Global database. We clean the data following [Kalemli-Ozcan et al. \(2024\)](#). We match these ownership links with the Chinese firm data using the unified social credit identifier and firm names. The dataset also records the equity share held by each party.

We obtain patent data for Chinese firms from the Google Public Patent Database, covering patents granted by the China National Intellectual Property Administration (CNIPA), and

⁴Under SFAS No. 131, US publicly listed firms must disclose foreign sales when they exceed 10 percent of total sales. This requirement is the source of the historical segment data.

⁵We follow [Autor et al. \(2013\)](#) and aggregate 4-digit SIC codes into 387 industries, which we then consolidate into 383. We map CIC codes to SIC codes using the concordance from [Ma et al. \(2014\)](#).

for US firms from the United States Patent and Trademark Office (USPTO). Among CNIPA's three patent types (innovation, utility model, and design), we follow the literature and include only innovation patents. For US firms, we include only utility patents, which are analogous to CNIPA's innovation patents. We construct firm-level counts of yearly new patents and patent stock across 875 three-digit International Patent Classification (IPC) codes. When constructing patent stock, we use a depreciation rate of 0.30, following [Li and Hall \(2020\)](#).⁶

We obtain sector-level data from the US NBER-CES Manufacturing Industry Database and China's National Bureau of Statistics, and bilateral trade data from UN Comtrade.

3. MOTIVATING FACTS

In this section, we present two motivating facts on joint ventures.

Fact 1. Direct Effects of JV Formation on Chinese Partners

To examine the direct effects, we compare a treated group (Chinese firms that formed JVs with foreign MNEs) to a control group (those that did not form any JVs) before and after they formed their first JV relationship. We construct the control group using propensity score matching. Each year, firms that formed JVs serve as treated observations, and those that never formed JVs serve as control observations. After pooling these observations across all years, we estimate the propensity score for the probability of forming a JV, using a logit model with firm-size related observables as covariates.⁷ For each treated firm, we match up to 4 control firms from the same year and 2-digit industry with the closest propensity score, allowing replacements. The matching procedure yields 176 matches, involving 176 unique treated and 692 unique control firms. The treated and control groups are well-balanced across observables, including those not used for matching (Tables [B4–B5](#)).⁸

⁶We compute $F_{it} = \text{NewPatent}_{it} + 0.7 \cdot F_{i,t-1}$, where F_{it} is firm i 's vector of patent stocks and NewPatent_{it} is the vector of new patents across technological fields. Total patent stock is the sum across all IPC codes.

⁷These observables include log sales, log capital, log employment, exporter and positive-patent-stock dummies, the inverse hyperbolic sines of exports and patent stock, and a state-owned enterprise (SOE) dummy. Larger firms and exporters are more likely to form a JV, and SOEs less likely (Figure [B3](#)).

⁸Figure [B2](#) also shows that control-group sales follow the same trend before and after the event, while only treated-group sales rise after the event.

Using the constructed matches, we estimate the following event study specification:

$$y_{imt} = \sum_{\tau=-5}^7 \beta_{\tau} \left(D_{mt}^{\tau} \times \mathbb{1}[\text{JV Partner}_i] \right) + \delta_{im} + \delta_{mt} + \varepsilon_{imt}, \quad (3.1)$$

where i denotes firm, m match, and t year. D_{mt}^{τ} are event-time dummies. $\mathbb{1}[\text{JV Partner}_i]$ is an indicator for the treated group. We normalize $\beta_{-1} = 0$. δ_{im} and δ_{mt} are match-firm and match-year fixed effects.⁹ ε_{imt} is the error term. Standard errors are two-way clustered at the match and firm levels.

We consider three dependent variables, log sales, the inverse hyperbolic sine of exports, and technological proximity to foreign MNE partners. The first two capture firm size and performance in global markets. The last measures the extent to which Chinese partners became technologically similar to their foreign MNE partners after forming JVs. Following Bloom et al. (2013), we calculate technological proximity using patent data:¹⁰

$$\text{Technological proximity}_{imt} = \frac{F_{imt}^{\top} F_{\text{MNE}(m),t(m)}}{(F_{imt}^{\top} F_{imt})^{0.5} (F_{\text{MNE}(m),t(m)}^{\top} F_{\text{MNE}(m),t(m)})^{0.5}}. \quad (3.2)$$

$F_{imt} = (p_{i1t}, \dots, p_{iKt})$ is a vector whose k -th element is Chinese firm i 's patent stock (from the CNIPA) in the k -th technological field. $F_{\text{MNE}(m),t(m)}$ is the analogous vector for the treated firm's foreign MNE partner of match m (from the USPTO), fixed at the event year $t(m)$.

Figure 1 reports the results. After forming JVs, Chinese partners' sales increased by 20 percent on average, and their export performance improved.¹¹ They also became technologically closer to their foreign MNE partners. These findings suggest that their improved performance is related to technology transfer and diffusion from foreign partners. These results are also consistent with Bai et al. (2025a), who find that Chinese automakers' model strength resembled that of their foreign partners in joint ventures.

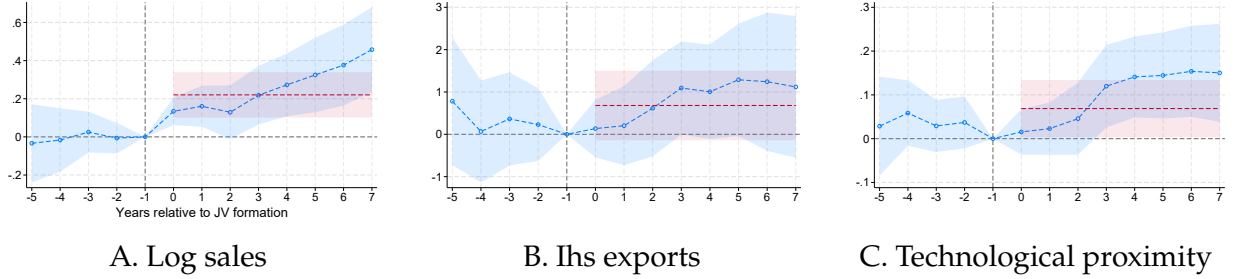
A remaining concern is that matching on observables may not fully remove selection

⁹This is a fully-stacked event study design that does not suffer from the issues raised in the recent staggered diff-in-diff literature (Roth et al., 2023).

¹⁰For the proximity measure to be well-defined, we require that both foreign MNEs and Chinese partners have engaged in patenting. We therefore restrict the sample to Chinese partners that ever patented in the Chinese patent system and foreign MNEs that ever patented at the USPTO.

¹¹The average post-event effects (red lines) are obtained by running $y_{imt} = \beta(D_{mt}^{0 \leq \tau \leq 7} \times \mathbb{1}[\text{JV Partner}_i]) + \sum_{\tau=-5}^{-2} \beta_{\tau}(D_{mt}^{\tau} \times \mathbb{1}[\text{JV Partner}_i]) + \delta_{im} + \delta_{mt} + \varepsilon_{imt}$, where $D_{mt}^{0 \leq \tau \leq 7}$ is an indicator for $0 \leq \tau \leq 7$.

Figure 1: Direct Effects of Joint Venture Formation on Chinese Partners



Notes. This figure illustrates the estimated event study coefficients in equation (3.1). 95% confidence intervals are based on standard errors two-way clustered at the match and firm levels. β_{-1} is normalized to zero. The red lines represent the average pooled diff-in-diff coefficients, with their 95% confidence intervals.

on unobservables. We therefore restrict the control group to firms with rumored foreign investment deals that were never completed, and re-estimate the event study (see Appendix B.1 for details). Because a rumor indicates that an MNE considered the firm a partner, it conveys information about the unobservables driving JV formation, so conditioning on it should absorb part of that selection. The results remain robust to this selected subsample (Figure B1).

JVs also had positive impacts on capital, employment, and patenting activity. Our findings are also robust to alternative numbers of matched controls and depreciation rates for the patent stock. Tables B6–B7 report these results.

Fact 2. Negative Outcomes for US Non-Participants

Next, we examine how the rise in JV activity in China affects US non-participating firms.

Across-Industry Evidence. We consider the long-difference specification over 1999–2012 for US firms, excluding any US MNEs that ever established a JV or a WFOE in China before 2012. This sample restriction isolates the negative effect on US firms that did not themselves invest in China.

$$\Delta y_{fj} = \beta_1 \Delta JV_j + \beta_2 \Delta WFOE_j + \vartheta \text{NTRgap}_j + \mathbf{X}'_{fj} \gamma + \varepsilon_{fj}, \quad (3.3)$$

where f denotes firms and j SIC 4-digit industries. The dependent variable is the DHS growth rate of firm-level outcomes, $\Delta y_{fj} = 100 \times \frac{y_{fj,12} - y_{fj,99}}{0.5(y_{fj,12} + y_{fj,99})}$ (Davis et al., 1998). We include NTRgap_j to control for sectoral reductions in US-China trade policy uncertainty

post-WTO (Pierce and Schott, 2016).¹² $\mathbf{X}_{fj}$ is a vector of additional controls. ε_{fj} is the error term. Regressions are weighted by firms' initial sales. Standard errors are clustered at the SIC 3-digit level.

ΔJV_j measures industry-level JV exposure in China, defined as the change in total domestic sales of JV affiliates from MNEs of all source countries in industry j over 1999–2012, normalized by total industry j sales in 1998¹³:

$$\Delta JV_j = \frac{\Delta JV \text{ dom. sales}_j}{\text{Total sales}_{j,98}} = \frac{\sum_{g \in \mathcal{J}_{j,12}} \text{Sales}_{gj,12}^{\text{dom}} - \sum_{g \in \mathcal{J}_{j,99}} \text{Sales}_{gj,99}^{\text{dom}}}{\text{Total sales}_{j,98}}, \quad (3.4)$$

where $\mathcal{J}_{j,t}$ is the set of JV affiliates in industry j in year t . In the numerator, we use domestic sales rather than total sales because these JV affiliates could have engaged in processing trade, serving export markets without engaging in complex manufacturing activity. We include JV affiliates from MNEs of all source countries in the numerator because negative competition effects may arise not only from US MNEs but also from other countries' MNEs.

To compare how JVs and WFOEs differ in their effects on US non-participants, we include the sectoral WFOE exposure, defined analogously to the JV exposure:

$$\Delta WFOE_j = \frac{\Delta WFOE \text{ dom. sales}_j}{\text{Total sales}_{j,98}} = \frac{\sum_{g \in \mathcal{W}_{j,12}} \text{Sales}_{gj,12}^{\text{dom}} - \sum_{g \in \mathcal{W}_{j,99}} \text{Sales}_{gj,99}^{\text{dom}}}{\text{Total sales}_{j,98}}. \quad (3.5)$$

Table 1 reports the results. Column 1 reports the estimate for sales growth, which is negative and significant at the 1 percent level. A 1 percentage point increase in industry j 's JV exposure is associated with 0.14 percentage point lower sales growth for US firms in that industry over the same period. In column 2, we add industry-level controls for technological trends, pretrends, and Chinese subsidy intensity.¹⁴ If these factors were driving our results, the estimate would change substantially once we add these controls. Instead, it remains

¹²NTRgap _{j} is the potential increase in US tariffs on Chinese goods had Congress failed to renew China's Normal Trade Relations status annually prior to the granting of Permanent Normal Trade Relations.

¹³This is a standard industry-level FDI exposure measure in the literature (e.g., Aitken and Harrison, 1999).

¹⁴The technological-trend controls, following Acemoglu et al. (2016), are 1996 US import penetration, 1993 production-worker shares, 1990 computer and high-tech investment shares, and 1-digit industry dummies. The pretrend controls are US sectoral employment and value-added growth (1993–1998) from the NBER manufacturing database and China's sectoral value-added growth (1995–1997) from China's National Bureau of Statistics. Consistent with this, US and China's pretrends are uncorrelated with JV exposure (Table B9). The subsidy control is changes in share of total subsidies to gross output and the Herfindahl index of subsidy distribution (1999–2012), following Aghion et al. (2015). Years of these controls vary with data availability.

Table 1: Across-Industry Evidence. JV and Negative Outcomes for US Non-Participants (OLS)

Dep. var.	Δ Sale		Δ Emp.		Δ Capital		Δ Export	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔJV_j	-14.27*** (2.62)	-17.42*** (2.09)	-15.51*** (3.33)	-18.86*** (2.38)	-11.76*** (3.24)	-15.72*** (2.86)	-14.96*** (5.23)	-26.72*** (3.06)
$\Delta WFOE_j$	9.07 (6.04)	2.41 (4.20)	17.83** (8.19)	2.95 (3.63)	7.30 (8.94)	-3.65 (5.45)	16.50 (14.36)	4.01 (5.50)
NTRgap _j	-2.06** (0.79)	-2.22*** (0.57)	-2.12** (0.91)	-2.27*** (0.61)	-2.15*** (0.80)	-2.24*** (0.54)	-2.87* (1.49)	-2.97*** (0.87)
Controls		✓		✓		✓		✓
Mean dep. var.	7.84	7.84	-11.27	-11.27	9.75	9.75	27.79	27.79
# clusters	105	105	105	105	105	105	99	99
N	949	949	949	949	949	949	767	767

Notes. Standard errors clustered at the SIC 3-digit level are reported in parentheses. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. This table reports OLS estimates of equation (3.3). ΔJV_j and $\Delta WFOE_j$ are defined in equations (3.4) and (3.5). In columns 1–2, 3–4, 5–6, and 7–8, the outcomes are DHS growth of US firms' sales, employment, capital, and exports (1999–2012). The NTR gap is the tariff increase on Chinese goods from a failed NTR renewal. The even columns add US sectoral employment and value-added growth (1993–1998), US import penetration (1996), US production-worker shares (1993), US computer and high-tech investment shares (1990), China's sectoral value-added growth (1995–1997), China's changes in the subsidy-to-gross-output ratio and the Herfindahl index of subsidies, and 1-digit industry dummies. All regressions are weighted by initial sales.

stable. JV exposure is also negatively correlated with growth in employment, capital, and exports (cols. 3–8). It is likewise negatively correlated with the outcomes of firms in other countries (excluding China and the US), further supporting the view that JV investment in China intensified global competition (Table B8).

In contrast to the consistently negative and significant JV estimates, the WFOE coefficients are imprecisely estimated, without stable patterns across outcomes. This difference is consistent with WFOEs having no local Chinese partner to absorb technology. This suggests that the negative effects originate mainly from technology transfer to local partners rather than from operating a production facility in China per se.

To address endogeneity, we consider an IV strategy as a robustness check. The IV is industry-level changes in FDI by Korean and Japanese MNEs into India. From the India side, this allows us to extract the push factors behind Korean and Japanese outward FDI, plausibly exogenous to China's sectoral productivity growth and to policies targeting particular sectors. They are relevant because the JV and WFOE exposures include affiliates of MNEs from all countries, including Korea and Japan. Appendix B.2 discusses the IV strategy and its

identifying assumptions in detail. The IV estimates are similar in magnitude to the OLS estimates. The results also remain robust to including firms that formed JVs, clustering at the 4-digit industry level, restricting the sample to the pre-Great Recession period over 1999–2007, and using employment-based weights (Table B10).

Within-Industry Evidence. We provide further evidence using variation in product overlap across firms within industries. Using the sample of US firms that have never engaged in either a JV or a WFOE, we estimate the following event study specification:

$$y_{fjt} = \sum_{\tau=-5}^7 \beta_{\tau} (\text{TNIC}_{fk} \times D_{j\tau t}^{\text{JV}}) + \sum_{\tau=-5}^7 \gamma_{\tau} (\text{TNIC}_{fk'} \times D_{j\tau t}^{\text{WFOE}}) + \delta_f + \delta_{jt} + \varepsilon_{fjt}, \quad (3.6)$$

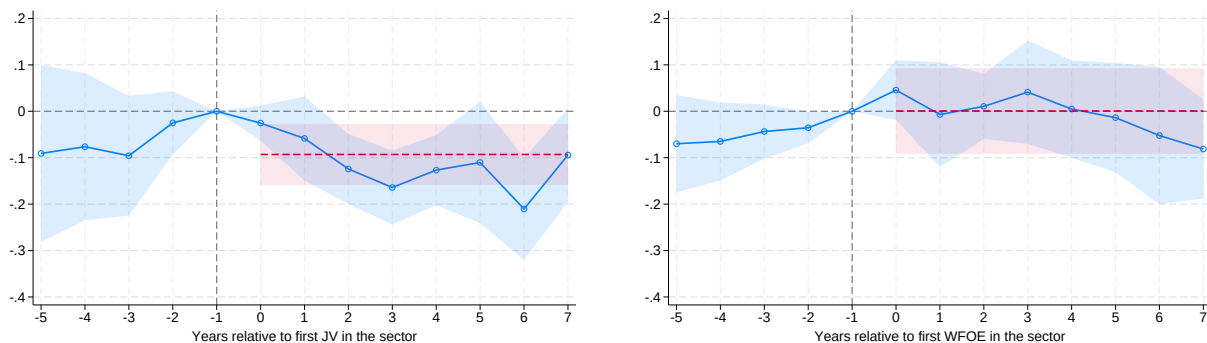
where j SIC 4-digit denotes industries. Log sales is the baseline firm outcome. $D_{j\tau t}^{\text{JV}} = \mathbb{1}[t - t_j^{\text{JV}} = \tau]$ are event-time dummies that equal 1 when year t is τ years from sector j 's first-JV event year t_j^{JV} . $D_{j\tau t}^{\text{WFOE}}$ is defined analogously for WFOE. We include 4-digit SIC industry-by-year fixed effects δ_{jt} to absorb time-varying industry-level shocks, and firm fixed effects δ_f . Standard errors are clustered at the SIC 3-digit level.

Here k and k' are the first US publicly listed firms to form, respectively, a new additional JV and WFOE in f 's sector over 1999–2012, and their formation years define the event years. Because the sample contains only never-participating firms and δ_{jt} absorbs each sector's event timing, β_{τ} and γ_{τ} are identified from within-industry-year variation in exposure rather than from comparisons of earlier- to later-treated firms, avoiding the staggered difference-in-differences problem. Over 1999–2012, 87 industries had a first-JV or -WFOE event (38 a first JV, 79 a first WFOE, and 30 both). We restrict observations to the event-time window -5 to 7 for both event types, and the estimation sample consists of 2,330 firms.

TNIC_{fk} is the standardized pre-event average of the text-based network industry classification (TNIC) similarity between firms f and k (Hoberg and Phillips, 2016). TNIC is a time-varying pairwise measure of product overlap, obtained as the cosine similarity of word-frequency vectors from the business descriptions in firms' annual 10-K filings.¹⁵ Interacting

¹⁵Hoberg and Phillips (2016) show that the Compustat historical business-segment data rarely reclassify firms and effectively hold a firm's product portfolio fixed over time, whereas TNIC provides a time-varying, firm-specific, continuous pairwise measure. Bloom et al. (2013) also consider similar product overlap to study business stealing effects of R&D.

Figure 2: Within-Industry Event Study. JV and Negative Outcomes for US Non-Participants



A. Log sales, JV

B. Log sales, WFOE

Notes. This figure reports the event study estimates of equation (3.6). 95% confidence intervals, based on standard errors clustered at the SIC 3-digit level, are reported. β_{-1} and γ_{-1} are normalized to zero. Panels A and B report the coefficients associated with the first-JV and first-WFOE events over 1999–2012. The red lines represent the average pooled diff-in-diff coefficients, with their 95% confidence intervals.

pre-event $TNIC_{fk}$ with the first-JV (or -WFOE) indicators generates firm-specific exposure. Firms with higher $TNIC_{fk}$ are more likely to be adversely affected by the first JV, because greater product overlap exposes them to more intense competition once it forms.

Figure 2 reports the results. After the first-JV event, sales of firms with greater product overlap decline more over time. The estimates imply that a 1 standard deviation increase in $TNIC_{fk}$ is associated, on average, with a 10 percent larger decline in sales. Consistent with the across-industry evidence, we find declines after the first-JV event but not after the first-WFOE event, again supporting the view that a JV intensifies future global competition, as a local Chinese partner can absorb technology from MNEs.

One concern is that our results reflect mechanical effects of participants' expansion rather than business stealing tied to JVs. We view this alternative hypothesis as unlikely. We plot raw sales of JV participants, non-participants whose product overlap is above the 75th percentile, and non-participants below the 75th percentile. If this alternative explanation were true, we would expect to observe an increase in JV participants' sales. However, we do not find such results. Instead, after the JV event, only the sales of non-participants above the 75th percentile begin declining (Figure B4). Also, when we include the pre-event levels and growth rates of the first-JV and first-WFOE MNEs' sales and TFPR, each interacted with the corresponding $TNIC$ exposure and post-event dummies, so that our estimates are net of participants' pre-event size and growth, our results remain robust to the inclusion of these controls (col. 2 of

Table B11).¹⁶

As a robustness check, following Bloom et al. (2013), we add technological proximity (based on equation (3.2)) to the JV or WFOE participants as a second exposure measure, to separate business stealing from technology spillovers. Only the TNIC estimates stay negative, significant, and stable, which supports the business stealing interpretation (Figure B5). The baseline results also hold when we include the participants' controls, replace the continuous exposure with a 75th-percentile dummy, use employment and capital as outcomes, and restrict the sample to sectors with both JV and WFOE events (Table B11). The last check shows that sectoral composition does not drive our results.

Taking Stock. Together, the two facts support the view that JV investment in China strengthened Chinese firms through technology diffusion and, as their growth intensified global competition, harmed other US firms that did not engage in JV investment. We acknowledge clear limitations in our empirical analyses. JV investment is endogenous, and we cannot completely rule out that unobserved shocks drive the negative US outcomes we document, although the difference in magnitude between the JV and WFOE estimates suggests this is unlikely. We therefore do not interpret our findings as causal. Instead, we build a model consistent with these two facts to perform counterfactuals, and we use these reduced-form correlations as targeted or non-targeted moments to calibrate and validate the model.

4. SIMPLE MODEL

Motivated by these two facts, we develop a simple static model of joint ventures and innovation, which highlights the business stealing channel due to technology transfer (or leakage). All proofs and detailed expressions are collected in Appendix C.

Setup. There are two countries, the US and China, indexed by $c \in \{H, F\}$. Three firms operate with no entry. They are a US leader, a US fringe, and a Chinese firm, indexed by $i \in \{h, \tilde{h}, f\}$. Each produces a unique variety under monopolistic competition, and every variety is tradable subject to an iceberg cost $\tau > 1$. Labor is the only factor of production. Country

¹⁶TFPR is obtained from İmrohoroglu and Tüzel (2014). Due to potential measurement errors in TFPR growth, we use the average of two consecutive growth rates.

c is populated by a mass L_c of households, each supplying one unit of labor inelastically, immobile across countries, and owning the domestic firms. We normalize $L_H = 1$.

There is a freely traded outside good whose price is normalized to 1. The outside good is produced one for one from labor in both countries. We assume that production of this good is positive in both countries, so free trade pins both wages to 1.

Household. Households have quasi-linear preferences, $U_c = C_c^0 + \ln C_c$, where C_c^0 is consumption of the outside good and C_c is a composite of varieties,

$$C_c = \left[\sum_{i \in \{h, \tilde{h}, f\}} \psi_i^{\frac{1}{\sigma}} c_{ci}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad \psi_h = \psi_f = 1, \quad \psi_{\tilde{h}} = \psi > 0,$$

where $\sigma > 1$ is the elasticity of substitution and the ψ_i are demand shifters, normalized to 1 for the US leader and the Chinese firm. The budget constraint is $C_c^0 + P_c C_c = 1 + \frac{\Pi_c}{L_c}$, where P_c is the price index of the composite and Π_c collects the profits of domestic firms.

Firms, Innovation, and Joint Venture. Each firm's production function is linear in labor, $y_i = z_i l_i$, so delivered unit costs are $1/z_i$ at home and τ/z_i abroad. Firms charge the constant markup $\frac{\sigma}{\sigma-1}$ over unit cost. With quasi-linear utility, total differentiated expenditure in market c equals L_c , and firm i 's operating profit there is $\pi_{ci} = \frac{1}{\sigma} s_{ci} L_c$, where s_{ci} is its market share. Firm i 's total operating profit is the sum of its profits across markets: $\Pi_i = \sum_c \pi_{ci}$.

A JV allows the US leader to produce inside China and avoid the trade cost, but its technology may leak to the Chinese rival. The US leader and the US fringe start with productivity 1, $z_h = z_{\tilde{h}} = 1$.¹⁷ The leader can innovate to improve its technology. It chooses the success probability x at cost $\frac{\alpha}{2} x^2$ in units of labor. Upon success, its productivity improves to $z_h = \gamma > 1$. The Chinese firm starts with productivity $1/b$, where $b > 1$ is the initial technology gap. If the technology leaks under the JV, which happens with probability $\phi \in (0, 1)$, the Chinese

¹⁷We call the non-participating firm the fringe, but it is not necessarily small. In the model, the fringe differs from the leader only in that it does not form a JV. The US firms whose sales decline in Fact 2 are publicly listed.

firm's productivity rises to the US leader's:

$$z_f = \begin{cases} 1/b & \text{no JV,} \\ 1/b & \text{JV, no diffusion (prob. } 1 - \phi), \\ z_h & \text{JV, diffusion (prob. } \phi). \end{cases} \quad (4.1)$$

The Chinese firm reaches productivity level of γ when the US leader's innovation succeeds and its technology leaks.

The US leader and the Chinese firm form the JV through Nash bargaining over a transfer T that maximizes the Nash product $(\Delta\mathbb{E}[\Pi_h] - \frac{\alpha}{2}\Delta(x^2) + T)^\xi (\Delta\mathbb{E}[\Pi_f] - T)^{1-\xi}$ subject to the participation constraints, $\Delta\mathbb{E}[\Pi_h] - \frac{\alpha}{2}\Delta(x^2) + T \geq 0$ and $\Delta\mathbb{E}[\Pi_f] - T \geq 0$. Δ denotes the change induced by the JV, so $\Delta\mathbb{E}[\Pi_i]$ is firm i 's change in expected operating profit, and $\frac{\alpha}{2}\Delta(x^2) = \frac{\alpha}{2}[(x^J)^2 - (x^N)^2]$ is the change in the leader's innovation spending. For tractability, we give the US leader zero bargaining power, $\xi = 0$. A JV is established if the total bargaining surplus $S = \Delta\mathbb{E}[\Pi_h] + \Delta\mathbb{E}[\Pi_f] - \frac{\alpha}{2}\Delta(x^2) > 0$.

Timing. The model has three stages. First, the leader and the Chinese firm bargain over the JV. Second, the leader chooses its innovation rate, given bargaining outcomes.¹⁸ Third, innovation and technology leakage are realized, firms produce, and markets clear.

Equilibrium. In equilibrium, households maximize utility, firms maximize expected profits, and markets clear in each state. If a JV is established, the US leader's participation constraint binds, $\Delta\mathbb{E}[\Pi_h] - \frac{\alpha}{2}\Delta(x^2) = -T$, and the Chinese firm receives the entire surplus S .¹⁹ Total operating profit in each market and state is fixed at $\sum_i \pi_{ci} = \frac{L_c}{\sigma}$, so the JV, innovation, and diffusion only reshuffle profits across firms. The leader's and the Chinese firm's combined gain in operating profit from the JV equals the fringe's loss, which is the business stealing effect. Proposition 1 shows that a JV is formed when the fringe's profits are large enough.

¹⁸Innovation is not contractible. After bargaining, the leader chooses the innovation rate that maximizes its own profit rather than the joint surplus.

¹⁹The transfer can take either sign. When b is close to one, the leader loses little from diffusion and gains from producing in China, so it pays the Chinese firm, $T < 0$. When the expected loss from diffusion is larger than the gain from local production, the Chinese firm pays the leader, $T > 0$.

Proposition 1 (JV formation). (i) Let $\omega \in \{0, 1\}$ and $d \in \{0, 1\}$ denote the innovation and diffusion outcomes. In every state, the leader's and the Chinese firm's combined gain from the JV equals the fringe's loss $\Delta\Pi_h(\omega, d) + \Delta\Pi_f(\omega, d) = -\Delta\Pi_{\tilde{h}}(\omega, d) > 0$, and the total surplus is $S = -\Delta\mathbb{E}[\Pi_{\tilde{h}}] - \frac{\alpha}{2}\Delta(x^2)$.

(ii) There exists a cutoff $\bar{\psi}$ such that for $\psi > \bar{\psi}$, the JV is established in equilibrium.

Welfare. Let $P_H^J(\omega, d)$ and $P_H^N(\omega)$ denote the US price index with and without the JV in each state. The change in US welfare from the JV can be decomposed as

$$\Delta W_H = \underbrace{\mathbb{E}_{x^N} \left[\phi \ln \frac{P_H^N(\omega)}{P_H^J(\omega, 1)} \right]}_{\text{consumer surplus gain} > 0} + \underbrace{\mathbb{E}_{x^N} [-B(\omega)]}_{\text{business stealing} < 0} + \underbrace{(x^J - x^N) [U^J(1) - U^J(0)]}_{\text{innovation margin} \leq 0}, \quad (4.2)$$

where $\mathbb{E}_{x^N}[X(\omega)] \equiv x^N X(1) + (1 - x^N)X(0)$ is the expectation over innovation success at the no-JV innovation rate, $B(\omega) = \Pi_{\tilde{h}}^N(\omega) - \phi \Pi_{\tilde{h}}^J(\omega, 1) - (1 - \phi) \Pi_{\tilde{h}}^J(\omega, 0) > 0$ is the fringe's expected profit loss in innovation state ω , and

$$U^J(\omega) = -\phi \ln P_H^J(\omega, 1) - (1 - \phi) \ln P_H^J(\omega, 0) + \phi \Pi_{\tilde{h}}^J(\omega, 1) + (1 - \phi) \Pi_{\tilde{h}}^J(\omega, 0)$$

is US household utility under the JV in innovation state ω .

The first term is positive because diffusion lowers the price of US imports. The second term is the business stealing loss. The sign of the third term depends on whether the JV raises or lowers the leader's innovation.

To focus on business stealing, we turn off the innovation margin by setting $\alpha \rightarrow \infty$, so there is no innovation. When China's market size L_F is sufficiently large, the business stealing effect outweighs the consumer surplus gain, and the JV lowers US welfare. Foreign market size matters because the fringe's profits in China, and therefore its loss from business stealing, grow with L_F . A higher fringe demand shifter ψ also amplifies the effect because the fringe's share of sector profits, which neither bargaining party internalizes, becomes larger. A larger technology gap b means a bigger catch-up jump, which reduces the fringe's profit more and can amplify the effect further. These results are summarized in Proposition 2.

Proposition 2 (Business Stealing). Suppose $\alpha \rightarrow \infty$. Let $\bar{L}_F \equiv \frac{\phi \tau^{2(1-\sigma)}(1-b^{1-\sigma})}{(\sigma-1)[\phi(2-\tau^{1-\sigma}-b^{1-\sigma})+(1-\phi)(1-\tau^{1-\sigma})]}$.

- (i) (Welfare) If $L_F \leq \bar{L}_F$, the business stealing effect never dominates and the JV always raises US welfare. If $L_F > \bar{L}_F$, there exists $\check{\psi}$ such that the JV lowers US welfare for $\psi > \check{\psi}$;
- (ii) (Comparative statics) Suppose $L_F > \frac{\tau^{2(1-\sigma)}}{\sigma - 1}$.²⁰ Then, there exists $\tilde{\psi}$ such that the JV raises US welfare, $\Delta W_H > 0$, if $\psi \in (0, \tilde{\psi})$, and lowers it, $\Delta W_H \leq 0$, if $\psi \in [\tilde{\psi}, \infty)$. Also, there exists $\hat{\psi}$ such that a larger gap b lowers the US welfare effect of the JV for $\psi > \hat{\psi}$.

Next, we focus on the innovation margin by setting $\psi \rightarrow 0$ and $b \rightarrow 1$. The first limit makes the fringe negligible, so there is no business stealing, and the second closes the initial technology gap. The JV has two opposing effects on the leader's innovation incentive: a market size effect (Bustos, 2011) and a leakage effect (Helpman, 1993).²¹ The JV lets the leader serve China without the trade cost. This gain is larger when the trade cost τ is higher and the foreign population L_F is larger. However, the improved technology leaks to the Chinese rival with probability ϕ , which reduces the leader's profits in both markets. The leader does not internalize the consumer gain from innovation, so innovation is below the US social optimum with and without the JV. The gap is larger under the JV because JV-induced diffusion makes the US better off through lower import prices. When the leakage effect is strong enough, the JV lowers US innovation and thus US welfare, and the magnitude of this loss increases with the diffusion probability ϕ .

Proposition 3 (Innovation). Suppose $\gamma < \tau$, $\alpha \geq \frac{1+L_F}{\sigma}$, and take the limit $\psi \rightarrow 0$, $b \rightarrow 1$.²²

- (i) (Welfare and innovation) At the equilibrium rates, the marginal welfare value of innovation is strictly positive with and without the JV, and it is strictly larger under the JV;
- (ii) (Market size vs. leakage) There exists $\bar{\phi}(\tau, L_F) \in (0, 1)$, strictly increasing in both the trade cost τ and the foreign population L_F , such that for $\phi > \bar{\phi}(\tau, L_F)$, the JV is established and strictly decreases US innovation. When ϕ is sufficiently high above $\bar{\phi}(\tau, L_F)$, the JV lowers US welfare.

Taking Stock. The simple model highlights two negative welfare consequences of JVs: the business stealing effect and lower US innovation due to leakage. However, it abstracts from general equilibrium forces, since the wage is pinned down by quasilinear utility, and from

²⁰This bound exceeds $\bar{L}_F$, so under it the JV lowers US welfare for every sufficiently large ψ .

²¹In the context of intellectual property rights, Helpman (1993) shows that imitation (or diffusion) potentially reduces the expected return to R&D investment because innovators anticipate that competitors will erode their market position by imitating their innovation.

²² $\gamma < \tau$ guarantees that the JV raises the leader's marginal return to innovation absent diffusion and $\alpha \geq \frac{1+L_F}{\sigma}$ keeps optimal innovation rates interior.

dynamics. Dynamics are central to understanding China's catch-up, because firms' JV and innovation decisions are forward looking. We therefore turn to the quantitative model in the next section, which incorporates both.

5. QUANTITATIVE FRAMEWORK

To conduct counterfactual analysis, we develop a two-country growth model with oligopolistic competition, innovation, and joint ventures. JVs are the only form of FDI in the model, and we do not separately model acquisitions of existing firms or the creation of WFOEs.

5.1 Setup

The world consists of two large countries, indexed by $c \in \{H, F\}$ and corresponding to the US and China. Time is continuous, indexed by subscript $t \in [0, \infty)$. There are two sectors, tradable and non-tradable. The tradable sector comprises a unit-mass continuum of products (or industries) $j \in [0, 1]$. Each firm produces a unique variety within a product. Varieties are tradable across countries subject to iceberg trade costs $\tau \geq 1$ and import tariffs $t_{ct} \geq 0$. The non-tradable sector consists of a representative firm in each country. Each country has a representative household that owns all domestic firms, supplies labor inelastically, and is immobile across countries. There is no trade in assets, ruling out international borrowing and lending.

For each product j in the tradable sector, there are three types of firms: leaders (Home and Foreign), fringe firms (Home and Foreign), and JVs (Foreign). JVs can be established through mutual agreements between the leaders from the two countries. We assume that only Home leaders form JVs in Foreign (and not vice versa), so the set of operating firms in Foreign varies by product and over time, while firm composition in Home remains fixed. For Home, $\mathcal{I}_H = \{h, \tilde{h}\}$ across all products. For Foreign, $\mathcal{I}_{Fjt} = \{f, \tilde{f}, v\}$ if a JV is formed and $\mathcal{I}_{Fjt} = \{f, \tilde{f}\}$ otherwise. h and f denote leaders, $\tilde{h}$ and $\tilde{f}$ fringe firms, and v a JV. Only JVs have entry and exit margins. $\mathcal{I}_{jt} = \mathcal{I}_H \cup \mathcal{I}_{Fjt}$ is the set of firms supplying product j .

5.2 Household

A representative household in Home maximizes utility subject to a budget constraint:

$$U_{Ht} = \int_t^\infty \exp(-\rho(s-t)) \ln C_{Hs} ds, \quad (5.1)$$

$$\text{s.t. } r_{Ht} A_{Ht} + w_{Ht} L_H + T_{Ht} = P_{Ht} C_{Ht} + \dot{A}_{Ht},$$

where C_{Ht} is final consumption good with price index P_{Ht} , $\rho > 0$ is discount rate, r_{Ht} is interest rate, L_H is labor endowment, and w_{Ht} is wage. T_{Ht} is lump-sum transfer from the government, A_{Ht} is assets owned by households, and $\dot{A}_{Ht}$ is time derivative of A_{Ht} . Its Euler equation pins down the interest rate in a no-asset-trade equilibrium: $\frac{\dot{C}_{Ht}}{C_{Ht}} = r_{Ht} - \left(\rho - \frac{\dot{P}_{Ht}}{P_{Ht}}\right)$.

A household has Cobb-Douglas preference over the tradable and non-tradable sector output (C_{Ht}^T and C_{Ht}^N) with the expenditure share of the tradable sector β : $C_{Ht} = (C_{Ht}^T)^\beta (C_{Ht}^N)^{1-\beta}$. Its price index is $P_{Ht} = \left(\frac{P_{Ht}^T}{\beta}\right)^\beta \left(\frac{P_{Ht}^N}{1-\beta}\right)^{1-\beta}$, where P_{Ht}^T and P_{Ht}^N denote the price indices of tradable and non-tradable sector output, respectively.

5.3 Sectors

Tradable sector output aggregates varieties produced by each firm across products:

$$Y_{Ht}^T = \exp \left(\int_0^1 \ln \left(\left(\sum_{i \in \mathcal{I}_{jt}} \psi_i \right)^{-\frac{1}{\sigma}} \left(\sum_{i \in \mathcal{I}_H} \psi_i^{\frac{1}{\sigma}} y_{ijt}^{\frac{\sigma-1}{\sigma}} + \sum_{i \in \mathcal{I}_{Fjt}} \psi_i^{\frac{1}{\sigma}} (y_{ijt}^*)^{\frac{\sigma-1}{\sigma}} \right) \right)^{\frac{\sigma}{\sigma-1}} dj \right).$$

y_{ijt} and y_{ijt}^* are the quantities of varieties produced by domestic and foreign firms, and the superscript “*” indicates exported varieties. ψ_i is firm i 's demand shifter. Leaders in both countries and the JV share a common demand shifter, $\psi_h = \psi_f = \psi_v = 1 - \psi$, while fringe firms share a different common value, $\psi_{\tilde{h}} = \psi_{\tilde{f}} = \psi$, with $\psi \in (0, 1)$. This parameterization normalizes the two shifter values to sum to 1. Varieties are imperfectly substitutable within products, with an elasticity of substitution $\sigma \in (1, \infty)$.

To prevent the new variety created by a JV from mechanically raising utility, we neutralize the love of variety by normalizing the product-level aggregator with the sum of demand shifters $\sum_{i \in \mathcal{I}_{jt}} \psi_i$ (Benassy, 1996). In the quantitative section we also report results

that preserve love of variety. The price index is

$$P_{Ht}^T = \exp \left(\int_0^1 \ln \left(\frac{1}{\sum_{i \in \mathcal{I}_{jt}} \psi_i} \left(\sum_{i \in \mathcal{I}_H} \psi_i p_{ijt}^{1-\sigma} + \sum_{i \in \mathcal{I}_{Fjt}} \psi_i (p_{ijt}^*)^{1-\sigma} \right) \right)^{\frac{1}{1-\sigma}} dj \right),$$

where p_{ijt} and p_{ijt}^* are prices charged by domestic and foreign firms.

In the non-tradable sector, a perfectly competitive representative firm produces the sectoral good with a linear technology, $Y_{Ht}^N = Z_{Ht}^N L_{Ht}^N$, where Z_{Ht}^N is exogenous productivity that grows at rate g^N . With perfect competition, $P_{Ht}^N = \frac{w_{Ht}}{Z_{Ht}^N}$.

5.4 Firms

Production and Market Structure. A firm produces a unique variety using a production function linear in labor: $\mathcal{Y}_{ijt} = z_{ijt} l_{ijt}$, where z_{ijt} denotes productivity and l_{ijt} labor inputs. Because its output can be sold in both markets, it is subject to a resource constraint: $\mathcal{Y}_{ijt} = y_{ijt} + \tau y_{ijt}^*$.

Within each product, the two leaders (Home and Foreign) and the JV (if established) engage in Bertrand competition, charging variable markups over their marginal cost. With the CES aggregator, their markups become functions of their market shares ([Atkeson and Burstein, 2008](#)). Home leaders' prices in Home and Foreign markets are

$$p_{hjt} = \frac{1 - \frac{\sigma-1}{\sigma} s_{hjt}}{\frac{\sigma-1}{\sigma} (1 - s_{hjt})} \frac{w_{Ht}}{z_{hjt}} \quad \text{and} \quad p_{hjt}^* = \frac{1 - \frac{\sigma-1}{\sigma} s_{hjt}^*}{\frac{\sigma-1}{\sigma} (1 - s_{hjt}^*)} \frac{\tau(1 + t_{Ft})w_{Ht}}{z_{hjt}}, \quad (5.2)$$

where s_{hjt} and s_{hjt}^* are its sales shares within product j in the Home and Foreign markets. Its operating profits in the two markets are

$$\pi_{hjt} = \frac{s_{hjt}}{\sigma - (\sigma - 1)s_{hjt}} P_{Ht}^T C_{Ht}^T \quad \text{and} \quad \pi_{hjt}^* = \frac{s_{hjt}^*}{\sigma - (\sigma - 1)s_{hjt}^*} P_{Ft}^T C_{Ft}^T, \quad (5.3)$$

and its total operating profit is $\Pi_{hjt} = \pi_{hjt} + \pi_{hjt}^*$.

Fringe firms, unlike the others, charge the constant monopolistic-competition markup $\frac{\sigma}{\sigma-1}$, so their prices are $p_{\tilde{h}jt} = \frac{\sigma}{\sigma-1} \frac{w_{Ht}}{z_{\tilde{h}jt}}$ and $p_{\tilde{h}jt}^* = \frac{\sigma}{\sigma-1} \frac{\tau(1+t_{Ft})w_{Ht}}{z_{\tilde{h}jt}}$, with operating profits $\pi_{\tilde{h}jt} = \frac{1}{\sigma} s_{\tilde{h}jt} P_{Ht}^T C_{Ht}^T$ and $\pi_{\tilde{h}jt}^* = \frac{1}{\sigma} s_{\tilde{h}jt}^* P_{Ft}^T C_{Ft}^T$.

Innovation. Leaders in both countries can improve productivity through successive innovations. They choose the Poisson arrival rate of innovation, x_{ijt} , subject to the convex cost function

$$l_{ijt}^r = \alpha_{cr} \frac{(x_{ijt})^\gamma}{\gamma}, \quad \gamma > 1,$$

where l_{ijt}^r is the R&D labor employed by firm i , α_{cr} scales the innovation cost in country c , and γ governs its curvature. Upon a successful innovation, which arrives at rate x_{ijt} , productivity jumps to $z_{ij,t+\Delta t} = \lambda z_{ijt}$, where $\lambda > 1$ is the step size.

Fringe firms do not innovate. Their productivity improves only through technology diffusion from the domestic leader within the same product.

Joint Venture. A Home leader may partner with the Foreign leader in the same product to establish a JV in Foreign that produces a new variety using Foreign labor. The JV serves the Foreign market without incurring trade costs or Foreign import tariffs but pays trade costs and Home tariffs when exporting back to Home. A Home leader's incentive to form a JV rises with Foreign market size, the Home-Foreign wage gap, and trade costs, capturing the proximity-concentration trade-off (Helpman et al., 2004).

The JV does not innovate. Its productivity is tied to the Home leader's: $z_{vjt} = \frac{z_{hjt}}{\tau^z}$, where $\tau^z > 1$ is the productivity loss from producing abroad, due to foreign regulations and higher communication costs (Holmes et al., 2015). The fact that the JV carries the headquarters' productivity reflects the nonrivalry of ideas.²³

We assume that JVs maximize their own profits rather than the joint profit with their parent firms.²⁴ The Home leader receives a fraction κ of the JV's total profit, $\kappa \Pi_{vjt}$, and the Foreign leader retains the remainder, $(1 - \kappa) \Pi_{vjt}$. This fraction is fixed across JVs and taken as given by firms. Other more complex contractual arrangements could change welfare and JV formation, but we keep the split constant because it lets us focus on the business stealing effect. It is also consistent with the JV Law, which required MNEs and Chinese partners to

²³This nonrivalry is also consistent with Burstein and Monge-Naranjo (2009) and Holmes et al. (2015), who model firm-embedded productivity (e.g., managerial practices and intangible capital) as shareable across countries. Alviarez et al. (2023) find that such shareable firm-specific productivity explains one-third of the cross-country variance in output per worker. Bilir and Morales (2020) document productivity spillovers from MNE-headquarters R&D to foreign affiliates.

²⁴This can be micro-founded through an agency problem in which the JV manager maximizes only the JV's profit, not the joint profit with its parents.

share JV profits in proportion to their equity shares.²⁵

A Home leader chooses the Poisson arrival rate d_{hjt} of a JV opportunity, which requires l_{hjt}^d workers:²⁶

$$l_{hjt}^d = \alpha_{Hd} \frac{(d_{hjt})^\gamma}{\gamma}, \quad \gamma > 1.$$

When an opportunity arrives, the Home and Foreign leaders engage in Nash bargaining over a one-time fee C_{jt} paid by the Home leader (or received, if negative), which we detail in the next subsection. The fee ensures mutual gains for the two leaders, with the surplus split according to their bargaining power.²⁷ Once established, a JV operates until it exits exogenously at Poisson rate χ .

We assume a JV can be formed only when the US leader is more productive than the Chinese leader, $z_{hjt} > z_{fjt}$, because the JV Law specifies that the technology from the foreign partner must be advanced. Relaxing this assumption leaves the results unchanged (Table D4).

Technology Diffusion. There are three types of technology diffusion. The first is direct diffusion through JVs, between the two partners. While the JV operates, the lagging partner, either the Home or the Foreign leader, learns from its more advanced partner and catches up to that partner's productivity with Poisson rate ϕ , consistent with Fact 1.²⁸

The second and third, domestic and international diffusion, are standard in the literature. Fringe firms catch up to the domestic leader's productivity at Poisson rate δ^D (domestic diffusion).²⁹ JVs also indirectly benefit Foreign fringe firms through this domestic diffusion. Finally, the lagging leader within product j catches up to the frontier leader's productivity

²⁵Article 4 of the JV Law required that "the parties to the venture shall share the profits, risks and losses in proportion to their respective contributions to the registered capital." We treat κ as the cross-JV average equity share. In practice, the 25 percent minimum foreign stake under the JV Law and the 50 percent caps imposed in some industries, such as automobiles, kept shares in a narrow range.

²⁶ l_{hjt}^d represents expenses for training local managers or legal processing costs of setting up a new firm. We impose the same curvature as in innovation. In principle we could allow two different curvature parameters, but calibrating the JV cost curvature would require information on the cost of setting up a JV, which is rarely available in the data.

²⁷This one-time fee generalizes the fixed or sunk costs typically assumed in the FDI literature. The size of the cost and the party that bears it are determined endogenously through the Nash bargaining between Home and Foreign leaders, based on the technology gap. Because Home leaders form JVs only when the added profit exceeds the fee, and because formation is probabilistic, our model captures the extensive margin of JVs observed in the data.

²⁸Immediate catch-up means firms further behind gain more from diffusion, the advantage of backwardness (Gerschenkron, 1962), documented at the micro level by Choi and Shim (2024).

²⁹Lucas and Moll (2014); Perla and Tonetti (2014); König et al. (2022) assume similar domestic diffusion.

at Poisson rate δ^F (international diffusion). The rate δ^F captures diffusion across countries through channels other than JVs.³⁰

5.5 Equilibrium

State Variable. The vector of technology gaps, $\mathbf{m}_{jt} = \{m_{jt}^F, m_{jt}^{DH}, m_{jt}^{DF}\}$, is the payoff-relevant state variable, which determines each firm's profit within product j conditional on JV status and other aggregates. The gaps are defined by

$$\frac{z_{hjt}}{z_{fjt}} = \lambda^{m_{jt}^F}, \quad \frac{z_{hjt}}{z_{\tilde{h}jt}} = \lambda^{m_{jt}^{DH}}, \quad \frac{z_{fjt}}{z_{\tilde{f}jt}} = \lambda^{m_{jt}^{DF}}. \quad (5.4)$$

m_{jt}^F is the cross-country gap between the two leaders, positive when the Home leader is ahead. m_{jt}^{DH} and m_{jt}^{DF} are the within-country gaps between each leader and its fringe. All three gaps are bounded by $\bar{m}$: $(m_{jt}^F, m_{jt}^{DH}, m_{jt}^{DF}) \in \{-\bar{m}, \dots, 0, \dots, \bar{m}\}^3$. These bounds keep the state space finite and computation feasible. The JV's productivity is tied to the Home leader's, so we need not track it separately. Because products are symmetric, we drop the product subscript j from $\mathbf{m}_{jt}$ and from firm-level variables.

Value Function. Let $V_{ht}(\mathbf{m}; \mathcal{J})$ denote the value of the Home leader h in a given product, given the state $\mathbf{m}$ and JV status $\mathcal{J} \in \{0, 1\}$. For expositional purposes, we present the value function only for the case where the Home leader is ahead of the Foreign leader, $m^F > 0$. Appendix D presents the remaining cases.

Without a JV, the Home leader's value function is

$$\begin{aligned} r_{Ht} V_{ht}(\mathbf{m}; 0) - \dot{V}_{ht}(\mathbf{m}; 0) = & \max_{x_{ht}, d_{ht}} \left\{ \Pi_{ht}(\mathbf{m}) - \alpha_{Hr} \frac{(x_{ht})^\gamma}{\gamma} w_{Ht} - \alpha_{Hd} \frac{(d_{ht})^\gamma}{\gamma} w_{Ht} \right. \\ & + x_{ht} \left(V_{ht}(\mathbf{m} + (1, 1, 0); 0) - V_{ht}(\mathbf{m}; 0) \right) + x_{ft} \left(V_{ht}(\mathbf{m} + (-1, 0, 1); 0) - V_{ht}(\mathbf{m}; 0) \right) \\ & \left. + d_{ht} \left(V_{ht}(\mathbf{m}; 1) - V_{ht}(\mathbf{m}; 0) - C_t(\mathbf{m}) \right) + \sum_{\mathbf{m}'} \mathbb{T}(\mathbf{m}'; \mathbf{m}) \left(V_{ht}(\mathbf{m}'; 0) - V_{ht}(\mathbf{m}; 0) \right) \right\}, \end{aligned} \quad (5.5)$$

³⁰See, for example, Buera and Oberfield (2020) and de Souza et al. (2025) on international trade, Rachapalli (2021) on supply chains, and Santacreu (2025) and Choi and Shim (forthcoming) on formal licensing.

where $\mathbb{T}(\mathbf{m}'; \mathbf{m})$ gives the diffusion transition rates from $\mathbf{m}$ to $\mathbf{m}'$,

$$\mathbb{T}(\mathbf{m}'; \mathbf{m}) = \begin{cases} \delta^F & \text{if } \mathbf{m}' = \{0, m^{DH}, m^F + m^{DF}\} \\ \delta^D & \text{if } \mathbf{m}' = \{m^F, 0, m^{DF}\} \\ \delta^D & \text{if } \mathbf{m}' = \{m^F, m^{DH}, 0\} \\ 0 & \text{Otherwise.} \end{cases}$$

The first line is the flow profit, operating profit net of innovation and JV-formation costs. The second line reflects innovation by the two leaders. On the third line, the Home leader forms a JV at rate d_{ht} and pays the fee $C_t(\mathbf{m})$ and the last term captures exogenous diffusion.

With a JV in place, the value function is

$$\begin{aligned} r_{Ht} V_{ht}(\mathbf{m}; 1) - \dot{V}_{ht}(\mathbf{m}; 1) = \max_{x_{ht}} & \left\{ \Pi_{ht}(\mathbf{m}) - \alpha_{Hr} \frac{(x_{ht})^\gamma}{\gamma} w_{Ht} + \kappa \Pi_{vt}(\mathbf{m}) \right. \\ & + x_{ht} \left(V_{ht}(\mathbf{m} + (1, 1, 0); 1) - V_{ht}(\mathbf{m}; 1) \right) + x_{ft} \left(V_{ht}(\mathbf{m} + (-1, 0, 1); 1) - V_{ht}(\mathbf{m}; 1) \right) \\ & + \phi \left(V_{ht}(0, m^{DH}, m^F + m^{DF}; 1) - V_{ht}(\mathbf{m}; 1) \right) + \chi \left(V_{ht}(\mathbf{m}; 0) - V_{ht}(\mathbf{m}; 1) \right) \\ & \left. + \sum_{\mathbf{m}'} \mathbb{T}(\mathbf{m}'; \mathbf{m}) \left(V_{ht}(\mathbf{m}'; 1) - V_{ht}(\mathbf{m}; 1) \right) \right\}. \end{aligned} \quad (5.6)$$

With a JV in place, the Home leader's flow profit also includes its share of the JV's profit, $\kappa \Pi_{vt}(\mathbf{m})$, and it forms no new JV ($d_{ht} = 0$). On the third line, the first term is JV diffusion: at rate ϕ , the Foreign leader closes its gap with the Home leader. The second term captures the JV's exogenous dissolution at rate χ .

Optimal Innovation and Joint Venture Rates. The optimal innovation rate is

$$x_{ht}(\mathbf{m}; \mathcal{J}) = \left(\frac{V_{ht}(\mathbf{m} + (1, 1, 0); \mathcal{J}) - V_{ht}(\mathbf{m}; \mathcal{J})}{\alpha_{Hr} w_{Ht}} \right)^{\frac{1}{\gamma-1}}, \quad \mathcal{J} \in \{0, 1\}, \quad (5.7)$$

and the optimal JV rate is

$$d_{ht}(\mathbf{m}) = \begin{cases} \left(\max \left\{ \frac{V_{ht}(\mathbf{m}; 1) - V_{ht}(\mathbf{m}; 0) - C_t(\mathbf{m})}{\alpha_{Hd} w_{Ht}}, 0 \right\} \right)^{\frac{1}{\gamma-1}} & \text{if } m^F > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (5.8)$$

The rate d_{ht} is bounded below by 0 because Home leaders are better off not investing when they expect a JV to bring no additional surplus. When $m^F \leq 0$ (i.e., $z_h \leq z_f$), JVs are not allowed and Home leaders set $d_{ht} = 0$.

Bargaining and Joint Venture Fees. Once a JV opportunity arises at rate d_{ht} , the two leaders bargain over a fee $C_t(\mathbf{m})$:

$$C_t(\mathbf{m}) = \operatorname{argmax}_C \left\{ (\Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C)^\xi (\Delta^{\text{JV}} V_{ft}(\mathbf{m}) + C)^{1-\xi} \right\} = (1 - \xi) \Delta^{\text{JV}} V_{ht}(\mathbf{m}) - \xi \Delta^{\text{JV}} V_{ft}(\mathbf{m})$$

$$\text{s.t. } \Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C \geq 0, \quad \Delta^{\text{JV}} V_{ft}(\mathbf{m}) + C \geq 0 \quad (5.9)$$

where ξ is the Home leader's bargaining power. $\Delta^{\text{JV}} V_{it}(\mathbf{m}) = V_{it}(\mathbf{m}; 1) - V_{it}(\mathbf{m}; 0)$ is the value change for firm $i \in \{h, f\}$ after forming a JV. Under this fee the Home leader captures ξ and the Foreign leader $1 - \xi$ of the joint surplus $\Delta^{\text{JV}} V_{ht}(\mathbf{m}) + \Delta^{\text{JV}} V_{ft}(\mathbf{m})$, so both participation constraints hold whenever that surplus is non-negative.

When the Foreign leader lags behind, it benefits from diffusion and typically pays for the JV, so $C_t(\mathbf{m}) \leq 0$. The fee compensates the Home leader for the profits it expects to lose as the Foreign leader catches up. But the Home leader does not take into account the Home fringe's profit loss from the JV, so it may form too many JVs relative to the US social optimum, as in the simple model.³¹

Combining equations (5.8) and (5.9) gives the optimal JV rate,

$$d_{ht}(\mathbf{m}) = \begin{cases} \left(\max \left\{ \frac{\xi (\Delta^{\text{JV}} V_{ht}(\mathbf{m}) + \Delta^{\text{JV}} V_{ft}(\mathbf{m}))}{\alpha_{Hd} w_{Ht}}, 0 \right\} \right)^{\frac{1}{\gamma-1}} & \text{if } m^F > 0 \\ 0 & \text{otherwise} \end{cases}.$$

The JV rate rises with the joint surplus $\Delta^{\text{JV}} V_{ht}(\mathbf{m}) + \Delta^{\text{JV}} V_{ft}(\mathbf{m})$, of which the Home leader keeps a share ξ . A wider technology gap m^F raises the JV's profit and the Foreign leader's gain from catching up, despite the Home leader's loss. Thus, the surplus, and therefore the JV rate, increases with the gap.

³¹From China's perspective there may instead be too few, because the Foreign leader ignores that the technology it acquires eventually diffuses to Foreign fringe firms.

Market Clearing. Asset markets clear: $A_{Ht} = \int_0^1 \sum_{i \in \mathcal{I}_H} V_{ijt} dj$, where the right-hand side is the total value of Home firms. Goods markets clear:

$$\sum_{i \in \mathcal{I}_H} p_{ijt} y_{ijt} + \sum_{i \in \mathcal{I}_{Fjt}} p_{ijt}^* y_{ijt}^* = P_{Ht}^T C_{Ht}^T \quad \forall j \in [0, 1], \quad Y_{Ht}^T = C_{Ht}^T, \quad Y_{Ht}^N = C_{Ht}^N.$$

Labor markets clear:

$$L_H = \int_0^1 \left(\sum_{i \in \mathcal{I}_H} l_{ijt} + \alpha_{Hr} \frac{(x_{hjt})^\gamma}{\gamma} + \alpha_{Hd} \frac{(d_{hjt})^\gamma}{\gamma} \right) dj + L_{Ht}^N.$$

The balance of payments holds:

$$\int_0^1 \left(\sum_{i \in \mathcal{I}_{Fjt}} \frac{p_{ijt}^* y_{ijt}^*}{1 + t_{Ht}} + d_{hjt} C_{jt} \right) dj = \int_0^1 \left(\sum_{i \in \mathcal{I}_H} \frac{p_{ijt} y_{ijt}}{1 + t_{Ft}} + \kappa \Pi_{vjt} \right) dj.$$

Similar asset, goods, and labor market clearing conditions hold in Foreign.

Equilibrium. The distribution of products across states, $\mu_t(\mathbf{m}; \mathcal{J})$, evolves according to firms' optimal innovation and JV decisions (see equation (D.8) for its law of motion). We define a Markov perfect equilibrium and a balanced growth path below. The two countries share a common growth rate on the balanced growth path. However, given initial conditions, their growth rates differ along the transition, and JVs accelerate China's convergence.

Definition 1. A Markov perfect equilibrium consists of sequences of prices $\{r_{ct}, w_{ct}, p_{ijt}, p_{ijt}^*, P_{ct}^T, P_{ct}^N, P_{ct}\}$ and allocations $\{l_{ijt}, x_{ijt}, d_{ijt}, y_{ijt}, y_{ijt}^*, L_{ct}^N, C_{ct}^N, C_{ct}^T, C_{ct}\}$ such that (i) the representative household maximizes utility; (ii) firms maximize the present discounted value of profits; (iii) goods, labor, and asset markets clear in every country c and period t ; and (iv) $\mu_t(\mathbf{m}; \mathcal{J})$ evolves according to firms' optimal innovation (x_{ijt}) and joint venture (d_{ijt}) investments.

Definition 2. A balanced growth path is the equilibrium in Definition 1 in which $\{C_{ct}, A_{ct}\}$ grow at a constant rate g , and r_{ct} and $\mu_t(\mathbf{m}; \mathcal{J})$ stay constant over time.

5.6 Welfare

We measure welfare in consumption-equivalent terms. Let ϑ be the proportional increase in baseline consumption that gives the household the same lifetime utility as under the counterfactual:

$$\int_t^\infty \exp(-\rho(s-t)) \ln((1+\vartheta)C_{Hs}^b) ds = U_{Ht}^c, \quad (5.10)$$

where C_{Hs}^b is baseline consumption at time s and U_{Ht}^c is lifetime utility under the counterfactual.

5.7 Taking Stock

What is new in our model is the business stealing effect of JVs on Home fringe firms. Technology diffusion from JVs makes Foreign firms more productive, taking market shares from Home firms in the global market. Home leaders account for their own lost profits when they form a JV, but not for the profits the Home fringe loses. The simple model shows that two parameters govern how far leaders over-invest in JVs relative to the Home optimum: the diffusion rate ϕ and the fringe demand shifter ψ . Higher diffusion means more future competition for the fringe. When ψ is low, leaders hold most of the market, so their own losses make up most of the industry's loss and their JV decisions align closer with the Home optimum. If the Home fringe barely produced, the over-investment would largely disappear.

JVs also have growth effects through Home leaders' innovation. A JV allows the leader to serve the Foreign market without paying trade costs, which expands its market and raises the return to innovation, especially when trade costs are high and the Foreign population is large. Leakage through direct diffusion works the other way. Which force dominates is a quantitative question.

6. TAKING THE MODEL TO THE DATA

We map each product in the tradable sector to a 4-digit SIC manufacturing industry. We solve the model's transition dynamics from 1997, when no JVs exist, until the economy converges to a balanced growth path. Calibrating along the transition, rather than at a steady state, matters

Table 2: Calibration Summary

Parameter	Value	Description	Source / Main target
Panel A. Directly from the Data			
L_F/L_H	2.83	Labor supply of China / US	Human-capital adj. pop. (Lee and Lee, 2016)
β	0.40	Tradable consumption share	1997 US BEA IO table
χ	0.08	JV exit rate	Avg. exit rate in CN (Chen et al., 2023)
$\mathcal{V}$	0.70	Variance of initial gap	Variance of productivity ratio in 1999
κ, ξ	0.54	US JV profit share	Avg. equity share of MNE
g^N	0.017	Non-tradable prod. growth rate	US GDP per capita growth rate (2011–2019)
t_t^H, t_t^F		US/CN import tariff rates	Avg. import tariff rates
τ^z	1.7	JV iceberg technology cost	Avg. TFPQ ratio (MNE / JV affiliate)
Panel B. Externally Calibrated			
ρ	0.03	Time preference	Literature
σ	4	Elast. of subst. across varieties	Broda and Weinstein (2006)
γ	2	Innovation/JV cost curvature	Acemoglu et al. (2018)
τ	2.85	Iceberg trade cost	Bai et al. (2024)
Panel C. Internally Calibrated by SMM			
α_{Hr}	0.67	US R&D cost scale	Avg. R&D / sales of US firms
α_{Fr}	0.94	CN R&D cost scale	Long-run avg. gap= 0
α_{Hd}	2.04	US JV scale	Avg. JV sales shares
λ	1.12	Step size	US GDP per capita growth rate
b	19.7	Avg. technology gap	1999 mfg. value-added / emp. CN/US ratio
δ	0.029	Domestic & Intl. diffusion	2020 mfg. value added / emp. CN/US ratio
ψ	0.75	Fringe demand shifter	Mfg. US Compustat firm sales / gross output
ϕ	0.15	Direct diffusion	Direct effect on Chinese parents, Fig. 1

because China grew rapidly over our sample period. Initial technology gaps between US and Chinese leaders are drawn from a normal distribution $N(b, \mathcal{V})$ with mean b and variance $\mathcal{V}$. A positive b means that, on average, US leaders start with higher productivity than Chinese leaders.

Because we lack direct evidence on the rate of domestic diffusion δ^D , we set $\delta^D = \delta^F = \delta$ and check robustness to alternative values.

We calibrate 23 parameters in three groups. We take 11 directly from the data, calibrate 4 externally from the literature, and estimate 8 jointly by simulated method of moments (SMM). For each guess of the jointly estimated parameters, we solve the model's transition, compute the implied moments, and measure their distance from the data. We select the parameters that minimize this distance.

We take the 11 parameters $\{L_H, L_F, \beta, \chi, \kappa, \xi, g^N, \mathcal{V}, t_t^H, t_t^F, \tau^z\}$ directly from the data.

Home labor is normalized to $L_H = 1$, and Foreign labor is set to $L_F = 2.83$ based on the human-capital-adjusted population of China relative to the US (Lee and Lee, 2016). The consumption share of tradables β is set to 0.4 from the 1997 US BEA Input-Output table. The exit rate of JVs χ is set to 0.08 from the average exit rate of Chinese firms in Chen et al. (2023). The tariffs t_t^H and t_t^F are computed as import-weighted annual averages across manufacturing.

The US JV profit share κ is set to 0.54, the average equity share of MNEs in JV firms in Orbis, consistent with the JV Law’s mandate of equity-proportional profit sharing. Lacking direct information on licensing fees between US and Chinese firms, we set the US leaders’ bargaining power ξ to the same value, 0.54.³² The non-tradable growth rate g^N is set to match the average growth rate of US real GDP per capita over 2011–2019. The variance $\mathcal{V}$ is set to 0.7, the variance of US-China labor productivity ratios across manufacturing industries in 1999.

The JV iceberg cost τ^z captures the JV’s productivity loss. We compute the TFPQ ($= \frac{\text{Sale}^{\sigma/(\sigma-1)}}{\text{Emp}}$) of US MNEs and of their affiliates and take the ratio. The ratio is 1.7, so we set $\tau^z = 1.7$, which implies that JV productivity is about 40 percent lower than that of the US leaders.

The 4 parameters $\{\rho, \sigma, \gamma, \tau\}$ are externally calibrated from the literature. We set the discount rate to $\rho = 0.03$. We set the elasticity of substitution to $\sigma = 4$ following Broda and Weinstein (2006). We set $\gamma = 2$, the curvature of the innovation cost function, following Acemoglu et al. (2018). The iceberg trade cost between the US and China is set to $\tau = 2.85$ following Bai et al. (2024).

The 8 parameters $\Theta = \{\alpha_{Hr}, \alpha_{Fr}, \alpha_{Hd}, \lambda, b, \phi, \psi, \delta\}$ are jointly estimated by minimizing the distance between model moments $M_m(\Theta)$ and their data counterparts M_m^D :

$$\min_{\Theta} \sum_{m=1}^8 \left(\frac{M_m^D - M_m(\Theta)}{\frac{1}{2}(M_m^D + M_m(\Theta))} \right)^2.$$

Although all parameters are jointly identified, we discuss below the moment most informa-

³²To pin down ξ , we would need direct data on royalty payments between US and Chinese firms. Our value of 0.54 falls within the range of comparable models with bargaining between innovators and adopters. It lies between 0.25 in Santacreu (2025), chosen as a common heuristic in licensing negotiations, and 0.85 in Choi and Shim (2024), who calibrate the bargaining parameter to match Korean firms’ payments for technology transfers.

tive about each one.

The US R&D cost scale parameter α_{Hr} is calibrated to match the average R&D-to-sales ratio among manufacturing firms in Compustat. China's α_{Fr} is calibrated so that both countries reach the same average productivity level on the balanced growth path.³³ The JV cost scale parameter α_{Hd} is calibrated to match the share of JV sales in total firm sales in China. This moment is informative because higher JV costs reduce JV investment and therefore the JV share in the Chinese economy. The step-size parameter λ governs the long-run growth rate, so it is calibrated to match the average US GDP per capita growth rate over 2011–2019.

The initial gap b is calibrated to match the 1999 ratio of value-added per employee between China and the US. The international diffusion parameter δ^F is calibrated to match the 2020 ratio of value-added per employee between the two countries, which we take as the long-run target. This moment is informative, as a higher δ^F implies faster Chinese convergence.

We calibrate the leader's demand shifter $1 - \psi$ to the share of US Compustat firm sales in US gross manufacturing output (Brault and Khan, 2024). Here we treat publicly listed manufacturing firms in Compustat as the empirical counterpart of the model's leaders. These firms have the scale to form joint ventures abroad, and they account for most US outbound FDI to China in our Orbis data: 72 percent of JV-affiliate sales and 82 percent of WFOE-affiliate sales by US firms in 1999.³⁴

We calibrate the direct diffusion parameter ϕ from Fact 1 by running a pooled diff-in-diff regression analogous to equation (3.1) on model-generated data and matching the average of the post-event coefficients. Specifically, we run $y_{imt} = \beta(1[\text{Post}_{mt}] \times 1[\text{JV Partner}_i]) + \delta_{im} + \delta_{mt} + \varepsilon_{imt}$, where each match consists of a leader and a fringe firm in the same industry. The estimated β equals 0.2 in our data, which we target in our calibration.³⁵

Table 2 summarizes the estimation results, and Panel A of Table 3 shows that the model moments closely match their data counterparts. The estimate $b = 19.7$ implies that US leaders

³³This calibration yields $\alpha_{Fr} > \alpha_{Hr}$ because China has a larger labor endowment. If $\alpha_{Hr} = \alpha_{Fr}$, average Chinese productivity on the balanced growth path would exceed US productivity.

³⁴Because the fringe is calibrated to non-Compustat firms, the model does not capture business stealing effects among Compustat firms themselves, which our Fact 2 within-industry evidence documents. Our welfare losses are therefore a lower bound, as a larger fringe demand shifter amplifies the negative welfare effect, as shown in the simple model. The comparative statics based on the full quantitative model in the next section also confirms this.

³⁵To replicate the regression in the model, we simulate 100,000 products whose technology gap between the US and China is randomly drawn from the calibrated initial distribution. Since innovation and diffusion are stochastic, products become heterogeneous in their technology gaps over time.

Table 3: Targeted and Non-Targeted Moments in the Model and the Data

Moment	Model	Data
Panel A. Targeted Moments		
Direct effect on CN partners (Panel A, Figure 1)	0.202	0.200
Avg. US GDP per capita growth, 2011–2019	0.017	0.017
Avg. R&D-to-sales ratio, Compustat mfg. firms	0.078	0.073
JV sales shares in CN	0.113	0.110
Mfg. value added per emp. ratio (CN / US), 1999	0.083	0.084
Mfg. value added per emp. ratio (CN / US), 2020	0.385	0.382
Leader firms' sales share	0.280	0.280
Long-run avg. productivity ratio	0.996	1.000
Panel B. Non-Targeted Moments		
US negative outcome (col. 1, Table 1)	−15.43	−14.27
Sectoral positive corr. between JV exposure & initial gap	0.122	0.169
JV fee / JV sales ratio (%) (Branstetter et al., 2006)	1.69	1.50

are initially 9 times as productive as Chinese leaders on average. The estimates $\phi = 0.151$ and $\delta^F = \delta^D = 0.029$ imply that forming a JV raises the diffusion rate between the two leaders from 0.029 to 0.18, more than sixfold.

The model also matches three non-targeted moments (Panel B of Table 3). The most important is the across-industry decline in sales among US non-participants (Fact 2). Estimating equation (3.3) on model-generated data gives a coefficient of −15.43, close to the empirical estimate of −14.27 in column 1 of Table 1.

Second, it predicts a positive relationship between the initial US-China technology gap and sectoral JV exposure, as in the data. This pattern arises in the model because a wider gap raises the joint surplus from a JV, since the JV earns more when the Chinese leader lags further behind. Technological diffusion through JVs therefore directs China's productivity growth toward the industries in which the US initially holds a large comparative advantage, eroding that advantage over time and reducing the gains from trade ([Samuelson, 2004](#)).³⁶

Finally, the average ratio of JV fee flow to JV sales flow is 1.69 percent, which is in line

³⁶ [Alviarez \(2019\)](#) obtains a similar erosion of Ricardian comparative advantage through a different mechanism. Multinational production concentrates in industries where the host's local producers are relatively unproductive, so the technology it transfers compresses the host's cross-sector productivity dispersion. In her model, selection follows from the multinational's location choice alone. In ours, a JV requires agreement between two leaders, and a wider gap raises their joint surplus.

Table 4: Welfare Effects of Banning Joint Venture Investments in 1999

Δ US welfare (%)	Δ China welfare (%)
1.08	−8.90

Notes. This table reports the welfare effect of banning new JV investments from 1999 onward, relative to the baseline, in consumption-equivalent percent (equation (5.10)).

with the range reported by [Branstetter et al. \(2006\)](#) from US BEA microdata.³⁷

7. QUANTITATIVE EXERCISES

7.1 Did US Multinationals Transfer Too Much Technology to China?

Using the calibrated model, we first examine whether the US transferred too much technology to China through JVs. In the counterfactual, the US government bans new JV investments from 1999 onward. JVs formed before 1999 remain in place until they exit exogenously. The ban is unanticipated, and we compare welfare under the ban with the baseline.

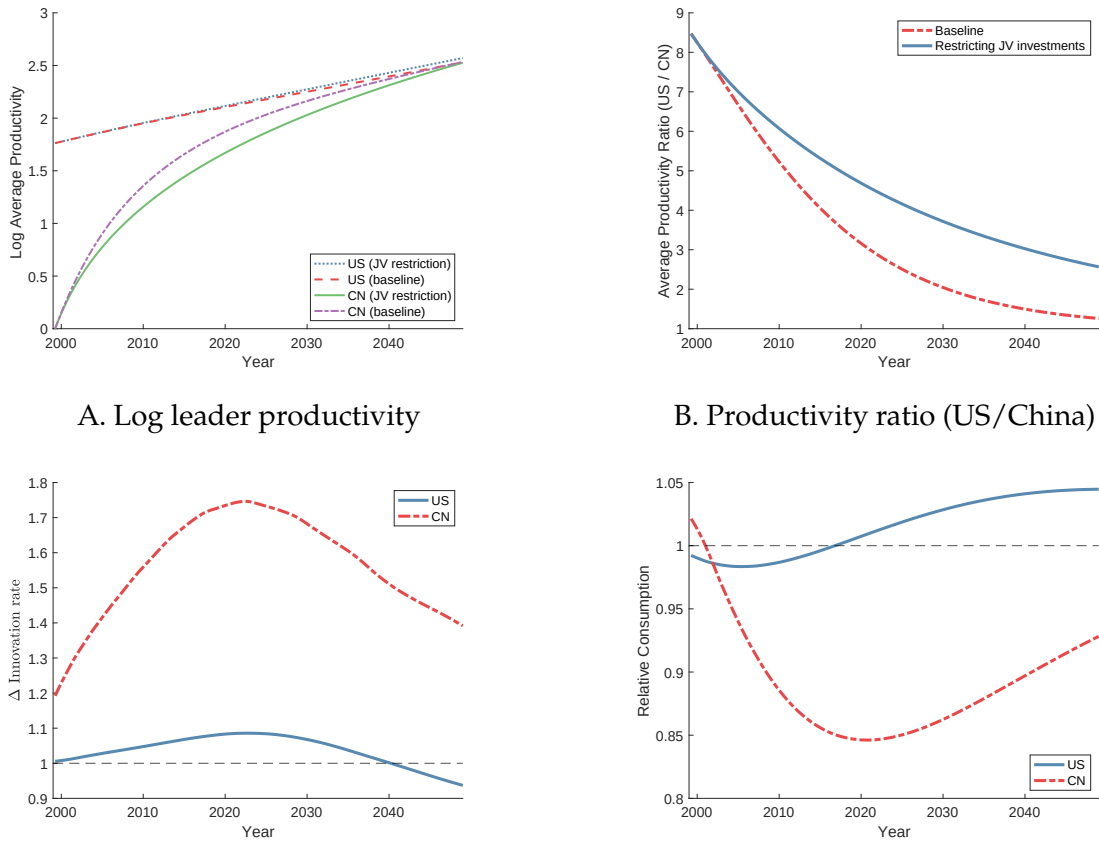
The JV ban in 1999 raises US welfare by 1.1 percent and reduces China’s welfare by 8.9 percent (Table 4). Under the ban, Chinese leaders’ productivity grows more slowly without diffusion from JVs, so the US-China technology gap widens and China’s convergence slows (Panels A and B of Figure 3).³⁸ Panel C shows that innovation rises in both countries relative to the baseline. US innovation rises for reasons we discuss below. Chinese firms raise R&D to substitute the lost diffusion, but the extra innovation efforts make up only part of it.

Panel D of Figure 3 plots consumption under the ban relative to the baseline. US consumption falls at first, as US leaders lose JV fees and profits. Higher innovation then makes US firms more competitive globally, and US consumption rises above the baseline after about 18 years. China’s consumption falls as its productivity growth slows, reaching its minimum

³⁷The BEA’s US Direct Investment Abroad surveys require US multinationals to report the affiliate-level value of royalties paid by foreign affiliates to parent firms for the sale or use of intangible property (e.g., technology licensing fees and use of trademarks). [Branstetter et al. \(2006\)](#) report that annual royalties as a share of affiliate sales average around 1–1.5 percent. Because the JV fee is a one-time payment, we convert it to an instantaneous-flow equivalent. The corresponding ratios of fees to US and Chinese leaders’ sales are 2.78 percent and 3.67 percent. The ratio is lower for the JV because it has the largest sales, operating with the US leader’s productivity, lower Chinese wages, and no iceberg trade costs on Chinese sales.

³⁸Chinese productivity grows at 7.8 percent per year on average in our baseline over 1999–2025, close to the 8 percent rate [Brandt et al. \(2012\)](#) estimate from firm-level TFP for 1998–2007.

Figure 3: Effects of Banning Joint Venture Investments in 1999: Dynamics of Productivity, Innovation, and Consumption



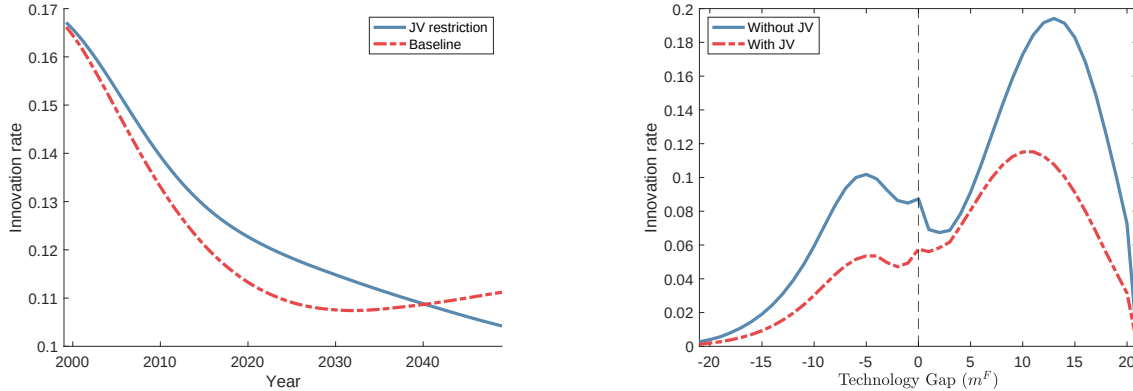
C. Innovation rate ratio (counterfactual / baseline) D. Consumption (counterfactual / baseline)
Notes. This figure compares dynamics under the counterfactual (JV ban in 1999) to the baseline. Panel A plots average log leader productivity for the US and China. Panel B plots the US–China productivity ratio. Panel C plots the ratio of average innovation rates in the counterfactual relative to the baseline for each country. Panel D plots consumption in the counterfactual relative to the baseline.

about 22 years after the ban. It then gradually recovers toward the baseline level as the technology gap narrows and diffusion through JVs matters less.

Three forces shape US innovation under the JV ban. The *option value* and *market size* effects raise innovation when JVs are available, while the *leakage* effect lowers it. The option value effect is absent in the static simple model. It reflects the incentive to innovate before forming a JV. Higher productivity raises the expected JV surplus, and US leaders capture part of that surplus through the Nash bargaining fee. The market size effect arises because JVs enlarge the leader’s effective market and thus raise the marginal return to R&D.

The leakage effect works the other way. JVs accelerate technology diffusion to Chinese

Figure 4: US Innovation Rate over Time and Technology Gap



A. US average innovation rate

B. US innovation rate over m^F (baseline)

Notes. Panel A plots the average US innovation rate under the baseline and the counterfactual (JV ban in 1999). Panel B plots the US innovation rate in the baseline as a function of the US–China leader technology gap m^F , evaluated in 2025. $m^F > 0$ denotes the case where US leaders have higher productivity than Chinese leaders. In Panel B, the technology gaps between domestic firms are set to $m^{DF} = m^{DH} = \bar{m}$.

rivals, eroding future profits and dampening returns to innovation. Any extra innovation surplus that benefits the Chinese partner cannot be recouped once the JV fee is paid. In our calibration, the leakage effect dominates, and products with JVs exhibit lower innovation intensities than those without, conditional on technology gaps (Panel B of Figure 4).³⁹ As JVs are established in more products, the overall mix shifts toward products with lower innovation rates due to the leakage effect. This composition shift outweighs the option value and market size effects in the medium run, raising average innovation rates under the JV ban.⁴⁰ In the long run, however, the leakage effect weakens as the productivity gap narrows.

Next, we decompose the welfare effect of banning JVs by income source (Table 5). Real income is the sum of discounted leader profits (own profits + JV profits + JV fees), fringe profits, and labor income (real wage), each normalized by the country’s baseline real income.

³⁹The two peaks in Panel B arise from escape-competition effects (Akçigit et al., 2023). The peak at $m^F \approx -5$ reflects a “defensive” motive: US and Chinese leaders are neck-and-neck in the US market (after accounting for wage differentials and trade costs), so even a small productivity gain sharply raises domestic profits and prompts US leaders to innovate more to defend their domestic market share. The peak at $m^F \approx 13$ reflects an “expansary” motive: US leaders are neck-and-neck in the Chinese market, so they raise R&D to expand abroad. The asymmetry between the peaks (−5 vs. 13) comes from China’s lower wages. One difference from Akçigit et al. (2023) is that JV-driven diffusion pushes the technology gap toward zero, where innovation rates fall below the two peaks.

⁴⁰This result is consistent with the observation by Andy Grove, a former CEO of Intel: “Our pursuit of our individual businesses . . . often involves transferring manufacturing and a great deal of engineering out of the country . . . We don’t just lose jobs—we lose our hold on new technologies and ultimately damage our capacity to innovate” (McGee, 2025, p. 113).

Table 5: Baseline vs. Joint Venture Ban in 1999

	Baseline	Ban on JV	Changes (%)
US leader profit (own + JV + JV fee)	0.047	0.039	-17.00
Own profit	0.034	0.038	11.38
JV profit	0.007	0.000	n/a
JV fee revenue	0.005	0.000	n/a
US fringe profit	0.066	0.070	6.53
US labor income	0.888	0.906	2.07
US total real income	1.000	1.015	1.47

Notes. This table reports net present values of real profits and labor income, deflated by the US price index and normalized by baseline total real income, under the counterfactual that JVs are banned from 1999 onward versus the baseline. Leader profits include own profits, JV profits, and JV fees.

Table 6: Elimination of the Quid Pro Quo Component of Joint Ventures: $\phi = 0.15$ vs. $\phi = 0$

Δ US welfare (%)	Δ China welfare (%)
3.26	-3.65

Notes. This table reports consumption-equivalent welfare changes from eliminating the technology-transfer component of joint ventures, measured as welfare under $\phi = 0$ relative to the baseline calibrated value $\phi = 0.15$.

For the US, leader profits fall in the counterfactual because JV profits and JV fees from Chinese partners are lost. Fringe profits rise both because JVs do not operate and because reduced technology diffusion weakens competition from Chinese firms. The real wage rises due to higher domestic labor demand and lower prices driven by higher US innovation.⁴¹

The large general equilibrium amplification on the US side, operating through labor demand and wages, arises from US leaders' failure to internalize domestic fringe firms' profits. It is the aggregate consequence of business stealing compounding across all products. We confirm this in the next subsection and show that when leaders internalize fringe profits, fewer JVs are established and the real wage decreases following a JV ban.

On the Chinese side, both leader and fringe profits decline after a JV ban because of reduced diffusion, and labor income also falls as labor demand weakens (see Table D1 for the China decomposition). Slower diffusion also pushes price levels higher in China.

We interpret the direct diffusion parameter ϕ as the model counterpart of the quid pro quo policy. It is the extra rate of US-to-China diffusion triggered by JV formation, beyond

⁴¹ Absent innovation effects, JVs do not necessarily lower the price index because JV decisions are driven not only by lower production costs but also by JV fees.

domestic and international diffusion, so setting $\phi = 0$ isolates the mandated technology-transfer component of the policy. Relative to the baseline with $\phi = 0.15$, eliminating it raises US welfare by 3.26 percent and lowers Chinese welfare by 3.65 percent (Table 6). These results are consistent with [Holmes et al. \(2015\)](#), who find that the quid pro quo policy benefits China at the expense of the US.⁴²

To further assess robustness, we consider alternative modeling assumptions (Table D4). First, we preserve love-of-variety effects rather than shutting them down. Because JVs introduce additional varieties, banning JVs now results in a love-of-variety loss, which reduces the welfare gains from the JV ban. Nevertheless, the gains remain positive at 0.75 percent. Second, we impose constant markups, as under monopolistic competition. Third, we allow JVs even when US leaders are less productive than their Chinese counterparts. In both cases, the results change little. Finally, sensitivity checks with respect to key parameters (ϕ , δ^F , δ^D , b , and κ) yield similar results.

7.2 Inspecting the Mechanism

We have shown that, because of the business stealing effect and the innovation losses from the leakage effect, JVs are established more often than is socially optimal for the US, resulting in a welfare loss. In this section, we consider alternative scenarios and parameterizations to quantify the relative importance of each margin. In each scenario, we report the effect of removing JVs relative to that scenario's own baseline equilibrium in which JVs are allowed, in the spirit of a difference-in-differences approach.

Coordinated Joint Venture Decisions: Multilateral Bargaining. To illustrate how the business stealing effect leads to such over-investment, we consider coordinated JV decisions, under which a JV can be established only if US leaders compensate fringe firms for their future losses (i.e., leaders internalize fringe profits). We add a bargaining problem between US leaders and fringe firms, in addition to the one between US and Chinese leaders (see Appendix D.1 for details). As a result, a JV is established only when the joint surplus of the US leader and the fringe firms is positive. We solve these two bargaining problems jointly

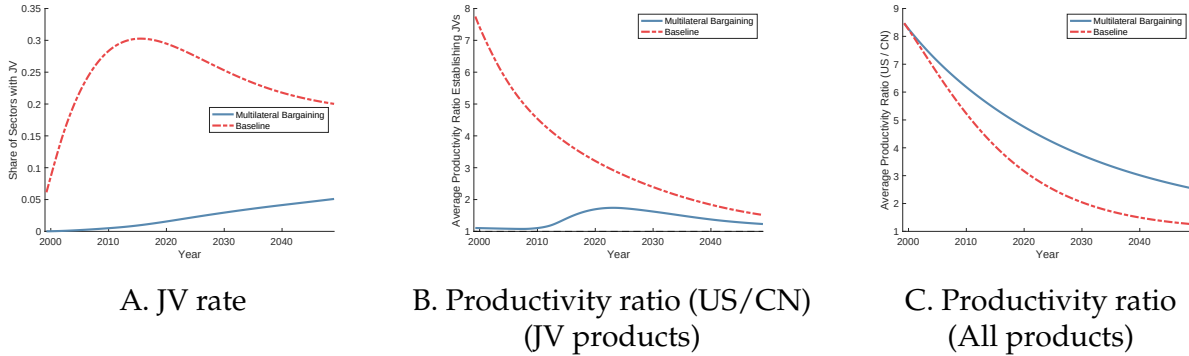
⁴²China's loss in our model (3.65 percent) is close to their 4.69 percent, but our US gain (3.26 percent) is far larger than their 0.45 percent, because our model adds strategic competition and innovation decisions that are absent from theirs.

Table 7: The Effects of Banning Joint Venture Investments in 1999 under Different Scenarios

	Δ Fringe profit (%)	Δ Innovation rate (%)	Δ Real wage (%)	Δ Welfare (%)
Panel A. JV ban in 1999				
(i) Baseline	6.53	3.62	2.07	1.08
(ii) Multilateral bargaining	-0.38	-0.87	-0.05	-0.03
(iii) Fixing innovation	6.54	0.00	2.19	0.85
(iv-a) No leakage ($\phi = 0$)	3.69	-8.26	-2.64	-1.83
(iv-b) + Larger profit share ($\phi = 0, \kappa = 1$)	1.86	-10.53	-5.18	-3.69
Panel B. JV ban in 2025				
(v) Baseline	1.95	-13.16	-1.75	-0.91

Notes. Each row reports the effect of a JV ban relative to the corresponding no-ban baseline, under different scenarios or parameterizations. Profits and the real wage are net present values deflated by the US price index, and welfare is in consumption-equivalent units. The innovation rate is the average intensity over the first 50 years. All columns are percent changes. Rows (i)–(iv) ban new JV investment from 1999 onward and evaluate outcomes from 1999, while row (v) bans from 2025 and evaluates from 2025. Row (i) is our main counterfactual. In row (ii), US leaders must compensate US fringe firms for their future profit losses in order to establish a JV, through multilateral Nash-in-Nash bargaining. In row (iii), we adjust the US innovation rate under the counterfactual so that average innovation matches the baseline in each period, which shuts off the innovation margin. Row (iva) sets $\phi = 0$, eliminating direct diffusion through JVs and hence the leakage effect, and row (ivb) additionally sets $\kappa = 1$, assigning US leaders the entire JV profit.

Figure 5: Coordinated Joint Venture Decisions: Baseline vs. Multilateral Bargaining



Notes. This figure reports the dynamics of the share of products with JVs (Panel A) and the productivity ratio between US and China in sectors with JVs (Panel B) and all products (Panel C), in the baseline and in the counterfactual with coordinated JV decisions.

using the Nash-in-Nash concept (Horn and Wolinsky, 1988), assuming that US leaders have full bargaining power in the leader-fringe negotiation.⁴³

Coordinated JV decisions raise US welfare by 1.2 percent relative to the uncoordinated baseline. By requiring leaders to compensate their own fringe firms, coordination makes JVs more costly and better aligns leaders' interests with the US social optimum. As Panel A of Figure 5 shows, fewer JVs are formed under coordination. In particular, JVs are not

⁴³The modified bargaining outcomes are $C = (1 - \xi)\{\Delta^{JV}V_{ht}(\mathbf{m}) + \Delta^{JV}V_{ht}(\mathbf{m})\} - \xi\Delta^{JV}V_{ft}(\mathbf{m})$ and $C^E = -\Delta^{JV}V_{ht}(\mathbf{m})$, where C^E is the fee paid by a US leader to a fringe firm in the same industry.

established in industries with larger productivity gaps because the total surplus for US firms (inclusive of fringe firms) is negative (Panel B). With fewer JVs, technology leakage to China slows, and China's convergence rate declines (Panel C).

Coordination therefore alters the welfare impact of the JV ban. Row (ii) of Table 7 shows how different margins change when JVs are banned under this scenario (see Table D3 for the Chinese side). Fringe firms' profits fall rather than rise. Business stealing is already muted under coordination, so the small operating-profit gain from the ban is outweighed by the compensation fees the fringe no longer receives.⁴⁴ In addition, the innovation rate decreases slightly. Under coordination, fewer JVs are established in sectors with large productivity gaps, making the leakage effect negligible. As a result, the option value channel dominates the leakage channel, and the JV ban lowers the innovation rate. This decrease in innovation raises the US price index, which outweighs the increase in US labor demand and slightly reduces the real wage. Overall, welfare decreases slightly, showing that correcting business stealing is enough to offset the negative welfare effects of JVs.

Fixing Innovation. US welfare losses from JVs come from business stealing and lower innovation rates. To quantitatively assess the importance of each channel, we fix the average US innovation rate in the counterfactual by adjusting x_{ht} by ι_t , so that average US innovation rates are the same in the baseline and the counterfactual in each period.⁴⁵ Row (iii) in Table 7 reports the results. When JVs are banned while the average innovation rates stay the same, fringe firms' profits increase by as much as in the baseline case. The real wage still increases due to higher labor demand. Overall, banning JVs increases welfare by 0.85 percent, which is about 80 percent of the welfare gain from banning JVs under the baseline. This result shows that the business stealing margin and the resulting general equilibrium effects through labor demand are quantitatively more important than the innovation losses from the leakage effect, which account for the remaining 20 percent.

Shutting Down Technology Leakage. To further examine the mechanisms through which JVs affect US innovation, we consider alternative parameterizations. Row (iv-a) of Table 7

⁴⁴Excluding the fees, fringe firms' operating profit rises by only 0.18 percent.

⁴⁵Specifically, we set the innovation rate in the counterfactual to $\hat{x}_{hjt} = x_{hjt}^c(\mathbf{m}_{jt}) \times \iota_t$, and choose ι_t such that $\int_0^1 \hat{x}_{hjt} dj = \int_0^1 x_{hjt}^b dj$, where x_{hjt}^c and x_{hjt}^b denote the optimal innovation rate in the counterfactual and baseline, respectively.

shows that the innovation result reverses when we shut down direct diffusion from JVs ($\phi = 0$). In this case, JVs raise innovation through the market size and option value channels, so banning JVs lowers the innovation rate and thereby raises the price index. This increase in the price index outweighs the increase in labor demand, reducing the real wage and resulting in lower welfare under the JV ban. In addition, when we raise the profit share of US leader by setting $\kappa = 1$, so that US leaders capture all the profits generated by JVs, JVs increase innovation rate even further. Under this scenario, banning JVs lowers innovation incentives further and leads to a larger welfare loss, reported in row (iv-b). Figure D1 graphically illustrates the innovation responses over time and technology gaps m^F in both cases.

Comparative Statics. Proposition 2 suggests that the welfare effect of banning JVs increases with foreign market size L_F , and the fringe demand shifter ψ , because larger values of each amplify the business stealing effect. Figure D3 confirms this pattern in the full quantitative model.

7.3 What if the US Bans Joint Venture Investments in 2025?

Next, in light of recent policy debates, we consider banning JVs in 2025 rather than 1999. In contrast to the 1999 case, a ban in 2025 reduces US welfare by 0.9 percent, measured in 2025 (row (v) of Table 7). The reason is that the US-China technology gap has narrowed substantially by 2025 (Figure D2). With a smaller gap, less technology leaks through JVs, which weakens both the business stealing effect and the leakage effect on innovation. The option-value channel then dominates, and JVs raise the innovation rate. Banning them therefore lowers innovation, raises the price index, and reduces the real wage and welfare. The welfare gain from a ban thus falls as the technology gap narrows and eventually turns negative (Panel C of Figure D3). The timing of a ban matters for its welfare consequences, consistent with Proposition 2(ii).

8. CONCLUSION

The intensifying economic and geopolitical rivalry between the US and China has spurred ongoing debates over whether US firms transfer too much technology to China and whether

policies should be implemented to curb such transfers. Our model provides a justification for policy interventions, demonstrating that leading US firms may over-invest in joint ventures in China. This is because they do not account for the business stealing effects on other US firms and aggregate innovation and real wages. Despite this seemingly straightforward economic logic, this specific mechanism has not been formally explored in the literature.

Our findings show that a ban on joint ventures beginning in 1999 would have raised US welfare. However, we also identify alternative policies that could generate even greater welfare gains, such as mandating that leaders compensate fringe firms for future profit losses. While these policies may be challenging to implement and enforce, they highlight the specific market inefficiency at play.

Furthermore, our research does not suggest that banning joint ventures will always improve US welfare. By 2025, the US-China technology gap is small enough that banning joint ventures would actually reduce US welfare. This finding underscores the importance of a dynamic approach to policy, as the effectiveness of any intervention is highly dependent on the current economic and technological context.

Our analysis serves as a useful first step and opens several avenues for future research. On the empirical side, our novel findings could be refined to better understand the role of firm heterogeneity. For instance, future work could examine how the characteristics of different US leader firms, such as their size or R&D intensity, influence their joint venture decisions and the extent of technology leakage. Similarly, it would be valuable to explore how the absorptive capacities of Chinese firms affect their ability to benefit from diffusion. In addition, future research could explore alternative measures and methodologies to better identify the causal effects of technology transfer through joint ventures.

On the modeling side, we do not separately model other forms of FDI, such as acquisitions, WFOEs, or licensing contracts, all of which also facilitate international technology diffusion. While the underlying mechanism may be qualitatively similar, it would be a valuable extension to quantitatively assess the roles of these alternative channels. Another compelling avenue for future research is to explore the interaction between optimal tariffs and joint venture policies. Finally, a three-country model that includes China and two advanced countries (for example, the US and Japan) would allow for the analysis of strategic interactions between the leader firms of the advanced nations. This framework would likely reveal a prisoner's

dilemma-like dynamic, where each firm's unilateral incentive to form joint ventures in China results in a collectively suboptimal outcome of excessive technology leakage. Such a model could also be used to explore the strategic interactions between the two advanced-country governments regarding joint venture policies. We see this as a promising direction for our future research.

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Supplemental Appendix

A. APPENDIX: DATA

Annual Survey of Industrial Enterprises. We drop observations with missing or negative values for sales, capital (fixed assets), or employment, and retain only manufacturing firms with CIC 4-digit codes between 1300 and 4400. The Annual Survey of Industrial Enterprises covers all state-owned, private firms, and JV and WFOE affiliates with annual sales above 5 million RMB before 2010 and 20 million RMB thereafter. To ensure consistency, we apply the 20 million RMB threshold throughout the sample period. Industry codes follow CIC 1994 from 1998 to 2001 and CIC 2002 from 2002 to 2013. We harmonize industry classifications using concordance tables from the Industrial Classification for National Economic Activities and the CIC 1994–2002 concordance provided by [Brandt et al. \(2012\)](#).

Compustat. We drop observations with missing or negative values for sales, capital (PPEGT), or employment. We restrict our sample to manufacturing firms with SIC 4-digit codes between 2000 and 3999. We also obtain each firm’s total foreign sales (including both exports and sales from foreign affiliates) from the historical geographic segment data. For Global Compustat, used only in the robustness check in Table B8, we apply the same cleaning procedure as for US Compustat. We follow [Autor et al. \(2013\)](#) and aggregate these codes up to 383 4-digit codes for compatibility with the CIC and HS codes.

Sectoral data. We map Compustat data to Comtrade and the NBER-CES database using industry codes. Comtrade data, obtained from BACI ([Gaulier and Zignago, 2012](#)), are converted from HS 6-digit codes to 1987 SIC 4-digit codes following [Pierce and Schott \(2012\)](#). We obtain the SIC 4-digit level NTRgap_{*j*} from [Che et al. \(2022\)](#).

Mapping between the Chinese balance sheet and the Orbis Global datasets. The matching proceeds in two steps. First, we use the Legal Entity Identifier, a standard unique identifier, to link across datasets. For firms without a Legal Entity Identifier, we apply fuzzy matching

based on firm names in Chinese characters using Orbis’s built-in bulk matching algorithm. To ensure reliability, we retain only “A-level” matches, which represent the highest match quality according to Orbis’s classification. Lower-confidence matches (B or C levels) are excluded to reduce matching errors.

Mapping between the Chinese balance sheet and CNIPA patent data sets. We clean firm names in both datasets by removing non-distinctive terms (e.g., “Co., Ltd.” or “Limited Liability Company”). In the patent data, where multiple applicants are listed in a single field separated by colons, we extract and standardize each applicant name. Then, we match the two datasets based on cleaned names.

Mapping between US Compustat and the Orbis Global datasets. To merge the Compustat and Orbis databases, we again use Orbis’s built-in bulk matching algorithm, which relies on fuzzy matching. We apply the same criterion, A-level confidence, as in the mapping between the Chinese balance sheet and the Orbis dataset.

Mapping between the US Compustat and USPTO. We use [Kogan et al. \(2017\)](#) data in which the assignees of the patents in USPTO are matched with Compustat firm identifier.

Mapping between the Orbis Global datasets and USPTO. When the Orbis firm identifier is matched with Compustat firm identifier, we use [Kogan et al. \(2017\)](#) data to map into USPTO assignee IDs. The remaining firms are merged using a fuzzy matching algorithm. We first clean the firm names as in the previous step, and then apply fuzzy matching. To ensure correct matching, we only keep the pairs with similarity score higher than 0.9.

Concordance between SIC and CIC. We construct the concordance between the 4-digit CIC and 1987 SIC codes in two steps. First, we map CIC 2002 to NAICS 1997 using the concordance table from [Ma et al. \(2014\)](#), and then apply the 1997 NAICS to 1987 SIC concordance from the US Census. Each unique 4-digit CIC code maps to multiple 4-digit 1987 SIC codes. For these multi-mappings, we assign weights based on 1995 gross output from the NBER-CES Manufacturing Industry Database, so that larger industries receive greater weight.

Second, using this mapping, we construct the JV exposure at the 4-digit SIC level. The denominator of the exposure is $\text{Total sales}_{j,98} = \sum_{h \in \text{CIC}(j)} \omega_h^j \sum_{g \in \mathcal{F}_{h,98}} \text{Sale}_{gh,98}$, where $\mathcal{F}_{h,98}$ is the set of firms with CIC code h in 1998, $\text{CIC}(j)$ is the set of 4-digit CIC codes mapped to SIC j , and ω_h^j is the weight assigned to CIC h within SIC j . The numerator is computed analogously over JV affiliates:

$$\Delta \text{JV domestic sales}_j = \sum_{h \in \text{CIC}(j)} \omega_h^j \left(\sum_{g \in \mathcal{J}_{h,12}} \text{Sale}_{gh,12}^{\text{dom}} - \sum_{g \in \mathcal{J}_{h,99}} \text{Sale}_{gh,99}^{\text{dom}} \right),$$

where $\mathcal{J}_{h,t}$ is the set of JV affiliates with CIC code h in year t .

B. APPENDIX: EMPIRICS

B.1 Fact 1: Control Group based on FDI Rumors

We measure rumored foreign investment using Orbis Crossborder Investment, the licensed Bureau van Dijk deal database formerly distributed as Zephyr, which previous studies have also used (e.g. [Alperovych et al., 2020](#); [Andres et al., 2025](#)). It records cross-border and domestic transactions worldwide, including joint ventures, acquisitions, minority stakes, buy-outs, and mergers, with the identity and nationality of the target, the investor, and the ultimate group investor, the deal value where disclosed, and BvD identifiers that link each party to company records.

The database records a deal at every stage rather than only when it completes. Each deal carries separate dates for when it was first rumored, announced, expected to complete, completed, withdrawn, and postponed, along with a status field. A deal that never concluded therefore remains in the data with a rumor date and no completion date, which is what lets us identify firms that foreign investors evaluated but never partnered with. Our extract contains 19,016 deals with a Chinese party between 1995 and 2025, of which 10,649 were rumored but never completed.

We link Orbis firms to the Chinese firm-level data using the concordance between Orbis

Table B1: Balance of Matched Sample with Rumored Never-JV Controls. Direct Effects of Joint Venture Formation on Chinese Partner Firms

	JV				Non-JV				(Col. 1 - Col. 5)	
	Mean (1)	Median (2)	SD (3)	N (4)	Mean (5)	Median (6)	SD (7)	N (8)	F-stat (9)	p-val (10)
Log sale	17.42	17.26	1.67	580	17.46	17.38	1.50	557	0.07	0.79
Log emp.	6.36	6.25	1.38	580	6.34	6.10	1.26	557	0.01	0.92
Log sales per emp.	11.06	10.96	1.15	580	11.12	11.06	1.07	557	0.25	0.62
Log capital	16.27	16.16	1.86	580	16.30	16.24	1.77	557	0.02	0.90
Log capital per emp.	9.91	9.86	1.23	580	9.96	9.86	1.40	557	0.08	0.78
Ihs export	9.50	14.33	8.30	580	10.50	15.12	8.24	557	0.80	0.37
Dum export	0.58	1	0.49	580	0.63	1	0.48	557	0.55	0.46
Ihs cumulative patent	0.65	0	1.39	580	0.68	0	1.36	557	0.04	0.84
Dum. patent stock	0.26	0	0.44	580	0.27	0	0.45	557	0.10	0.76
Ihs yearly new patent	0.44	0	1.15	580	0.42	0	1.08	557	0.01	0.92
Dum. yearly new patent	0.16	0	0.37	580	0.17	0	0.37	557	0.04	0.84
Dum. SOE	0.27	0	0.44	580	0.18	0	0.39	557	2.09	0.15

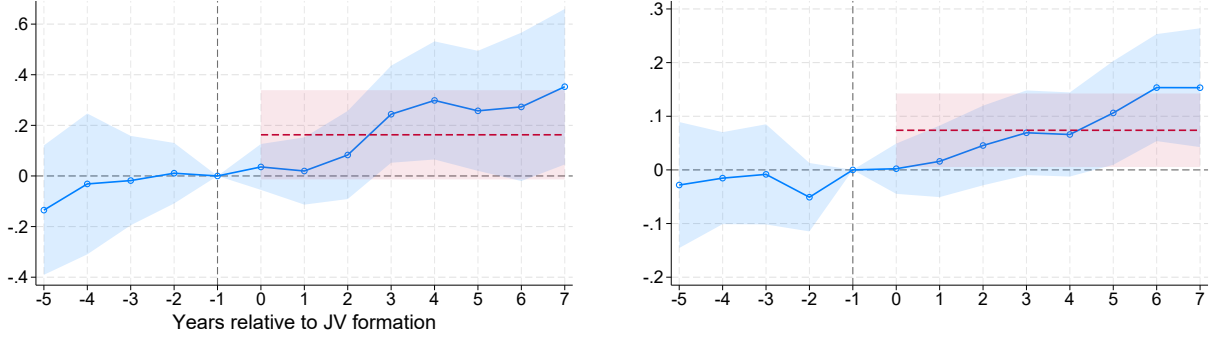
Notes. This table presents descriptive statistics for treated and control firms from five to one years before the event, where controls are drawn from firms that appeared as rumored joint-venture targets before the event but never formed a joint venture. Column 9 reports F -statistics for the mean differences between treated and control groups, while Column 10 provides the corresponding p -values, computed using standard errors clustered at the firm and match levels. All monetary values are expressed in 2007 US dollars.

BvD identifiers and Chinese national identifiers. Specifically, we use either the National Organization Code or the Unified Social Credit Code reported in Orbis and map its nine-character organization-code component to the corresponding firm in the industrial survey. This yields 2,984 Chinese firms with a mapped rumor history.

For a treated event in year t , an eligible control must (i) have entered the rumor database by $t - 1$, (ii) never have established a JV over the sample period, and (iii) have a complete set of matching variables at $t - 1$. These conditions leave 843 firms eligible for at least one treated event. For each treated firm we then match 1 control firm on the estimated propensity score, rather than the 4 we use in the baseline, because the rumored pool is much smaller. This selects 115 distinct controls. Both groups are well balanced in terms of observables (Table B1).

Using this constructed sample, we estimate the same event study as in equation (3.1). Figure B1 reports the results. Every pooled coefficient lies within one standard error of its baseline counterpart.

Figure B1: Robustness. Control Group based on FDI Rumors. Direct Effects of Joint Venture Formation on Chinese Partners



A. Log sales

B. Technological proximity

Notes. The figure illustrates the event study estimation results for equation (3.1). 95% confidence intervals are based on standard errors two-way clustered at the match and firm levels. The control group is restricted to a set of firms who had rumors of establishing joint ventures with foreign multinationals (Appendix B.1). β_{-1} is normalized to zero. The red lines represent the average pooled diff-in-diff coefficients, with their 95% confidence intervals.

B.2 Fact 2: Instrumental Variable Strategy

The OLS estimates can be biased due to endogeneity as unobservable factors may affect both FDI flows and firm growth. We propose an IV strategy to alleviate this concern. Unlike the OLS regressions in equation (3.3), we focus on total FDI rather than separating JVs and WFOEs, because our IV strategy, detailed below, exploits exogenous variation in total FDI rather than JV and WFOE separately. Therefore, we focus on the combined effects and run the following specification, analogous to equation (3.3):

$$\Delta y_{fj} = \beta_1 \Delta \text{FDI}_j + \vartheta \text{NTRgap}_j + \mathbf{X}'_{fj} \gamma + \varepsilon_{fj}, \quad (\text{B.1})$$

where the FDI exposure is defined analogously to the JV and WFOE exposures:

$$\Delta \text{FDI}_j = \frac{\Delta \text{FDI dom. sales}_j}{\text{Total sales}_{j,98}} = \frac{\sum_{g \in \mathcal{J}_{j,12} \cup \mathcal{W}_{j,12}} \text{Sales}_{gj,12}^{\text{dom}} - \sum_{g \in \mathcal{J}_{j,99} \cup \mathcal{W}_{j,99}} \text{Sales}_{gj,99}^{\text{dom}}}{\text{Total sales}_{j,98}}, \quad (\text{B.2})$$

We construct the IV as the ratio of the total sales of FDI in India affiliated with MNEs

from Japan or South Korea, relative to China's total sector sales in 1998:

$$IV_j^{JP,KR} = \frac{\Delta \text{India FDI (S. Korea \& Japan) sales}_j}{\text{Total sales}_{j,98}} = \frac{\sum_{g \in \mathcal{J}_{j,12}^{IN,JP,KR}} \text{Sale}_{gj,12} - \sum_{g \in \mathcal{J}_{j,99}^{IN,JP,KR}} \text{Sale}_{gj,99}}{\text{Total sales}_{j,98}}, \quad (\text{B.3})$$

$\mathcal{J}_{jt}^{IN,JP,KR}$ is the set of FDI affiliates in India associated with MNEs from South Korea or Japan. We obtain data on Indian firm balance sheets and ownership from the Prowess database, supplemented with the ownership information from Orbis. The dataset covers over 70 percent of the Indian manufacturing sector and is representative of large and medium-sized firms. While it may exclude some small firms, this is unlikely to be a major concern, as we focus only on the sales of FDI affiliates in India, typically larger than domestic Indian firms.

The IV strategy aims to isolate variation in China's FDI exposure that is plausibly exogenous to factors specific to the US and China. For example, consider exogenous productivity shocks in South Korea or Japan that increase overall FDI by those two countries. By using their FDI affiliates in India, the IV extracts these exogenous shocks. The explicit identifying assumption is that any unobservables that affect US FDI in China are uncorrelated with the IV. We choose India for its attractiveness to FDI due to its large market size, low wages, and strong economic growth potential, a condition similar to China's, and South Korea and Japan because they were the two largest sources of FDI in China.

There can be two main potential threats to identification, namely, export platform and technological changes. First, regarding export platform, demand shocks in China may induce South Korean and Japanese MNEs to invest in India to serve the Chinese market, and vice versa. Similarly, US demand shocks may cause South Korean and Japanese MNEs to invest in China or India to serve the US market. In both cases, the exclusion restriction is violated because both demand shocks influence FDI flows into India from the two countries. The second concern is technological changes that are skill-biased or reduce communication costs between headquarters and affiliates. These shocks may make certain industries more attractive for FDI, potentially correlating FDI by MNEs in the US, South Korea, or Japan.

We investigate these concerns by inspecting pretrends and industry-level balance, fol-

lowing [Borusyak et al. \(2022\)](#) (Table B2). First, pre-1999 5-year growth (1993–1998) is not meaningfully correlated with the IV. Overall imports (excluding China, India, Japan, and Korea) and imports from China do not show any pretrends. Although the IV has weak positive correlations with gross output and employment at the 10 percent significance level, these relationships are in the opposite direction. We also find no significant correlation with the pre-1999 5-year growth of US firm-level outcomes.

We assess industry-level balance by checking the correlation between our IV and initial industry characteristics that could be related to unobserved shocks. The export platform is unlikely to be a significant concern, as there are no significant correlations between bilateral import penetration (import-to-domestic absorption ratio) and the IV for China, India, and the US. Moreover, sectors with a higher IV are not necessarily those in which China initially had higher productivity or those more exposed to FDI, supported by the lack of correlation between the IV and Chinese import penetration in the US, FDI affiliates' initial sales shares in China, or their numbers relative to the total firm numbers. While our research design does not require industries to be identical in levels, the absence of correlations with these variables supports the plausibility of the exclusion restriction.

Three variables related to technological change are significantly correlated with the IV, overall US import penetration (excluding China, India, Japan, and Korea), production workers' employment shares, and computer investment share. This raises concerns for omitted-variable bias from unobservable technological changes especially in labor-intensive sectors that are often characterized by higher foreign import penetration, larger production worker shares, and lower computer investment. However, if such unobservables were driving our results, they would likely appear as negative pretrends in gross output or employment, which we do not observe. Also, including them as additional controls leaves our estimates unchanged, suggesting that omitted variable bias due to these unobservables is unlikely to be a major concern.

Table B3 reports the results. The IV estimates are qualitatively similar to the OLS estimates. The IV is strong. The results remain robust to the additional controls.

Table B2: Pretrend and Shock Balance Test of the IV

Balance variable	IV _j ^{JP,KR}			N
	Coef.	SE	p-val.	
<i>Panel A. Industry-level Pretrend</i>				
Δ Log gross output 1993–1998	0.11	(0.06)	[0.06]	383
Δ Log emp. 1993–1998	0.10	(0.06)	[0.10]	383
Δ Log PPI 1993–1998	0.05	(0.05)	[0.35]	383
Δ US import (ex. CN, IN, JP, KR) / absorption 1996-1998	0.04	(0.02)	[0.12]	383
Δ US-CN import / absorption 1996-1998	−0.05	(0.06)	[0.41]	383
<i>Panel B. Industry-level balance</i>				
US-CN import / absorption 1996	−0.06	(0.04)	[0.14]	383
US-IN import / absorption 1996	−0.02	(0.01)	[0.24]	383
CN-IN import / absorption 1996	−0.01	(0.04)	[0.77]	383
IN-CN import / absorption 1996	−0.02	(0.02)	[0.13]	383
JV sales share 1998	−0.04	(0.04)	[0.28]	383
Number of JV firms to total number of firms ratio 1998	0.02	(0.01)	[0.12]	383
US share (ussh) 1996	−0.06	(0.05)	[0.26]	383
US import (ex. CN, IN, JP, KR) / absorption 1996	0.14	(0.04)	[0.00]	383
Ratio of capital to wage-bills 1993	0.01	(0.07)	[0.91]	383
Ratio of wage bills to value-added 1993	0.03	(0.05)	[0.56]	383
R&D intensity 1993	0.08	(0.11)	[0.47]	383
Production workers' share of employment 1993	0.27	(0.06)	[0.00]	383
High-tech investment shares 1990	−0.13	(0.09)	[0.15]	383
Computer investment shares 1990	−0.18	(0.05)	[0.00]	383
<i>Panel C. Firm-level Pretrend</i>				
Δ Sale 1993–1998	−2.51	(17.45)	[0.89]	723
Δ Emp. 1993–1998	−0.68	(23.08)	[0.98]	723
Δ Capital 1993–1998	18.31	(12.88)	[0.16]	723
Δ Export 1993–1998	−19.20	(35.88)	[0.59]	565

Notes. Standard errors clustered at the SIC 3-digit level are reported in parentheses. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. This table reports OLS estimates from regressing each balance variable on the IV based on South Korea and Japan's FDI in India. In Panels A and B, each observation is a 4-digit SIC industry, all variables are standardized, and regressions are weighted by initial gross output. R&D intensity is the 1993 sectoral mean of R&D-to-sales ratios, calculated from Compustat. High-tech and computer investment shares are obtained from [Acemoglu et al. \(2016\)](#), varying at the SIC 3-digit level. In Panel C, each observation is a US firm, the dependent variables are DHS growth of sales, employment, capital, and exports over 1993–1998, and regressions include the NTR gap control and are weighted by initial sales.

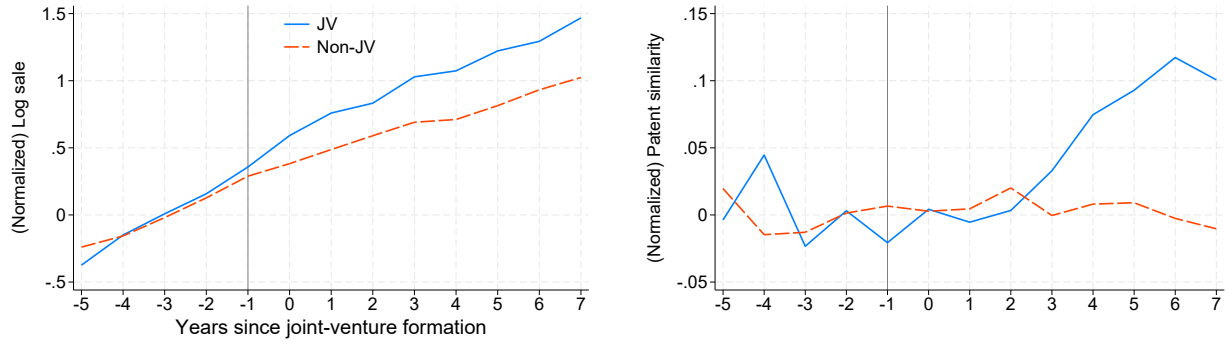
Table B3: Negative Outcomes for US Non-Participants (IV)

Dep. var.	Δ Sale		Δ Emp		Δ Capital		Δ Export	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Second stage</i>								
ΔFDI_j	-15.33*** (3.59)	-18.21*** (3.89)	-16.79*** (4.26)	-18.42*** (4.08)	-10.59*** (3.28)	-12.80*** (3.63)	-13.29** (5.96)	-27.00*** (5.83)
NTRgap _j	-1.91* (0.97)	-2.15*** (0.64)	-1.88 (1.20)	-2.14*** (0.71)	-1.89** (0.93)	-2.05*** (0.61)	-2.45 (1.74)	-2.79*** (1.02)
AR	18.75	19.46	18.57	18.16	10.69	11.52	5.18	18.72
AR <i>p</i> -val.	[0.00]	[0.00]	[0.00]	[0.00]	[0.00]	[0.00]	[0.02]	[0.00]
Mean dep. var.	7.84	7.84	-11.27	-11.27	9.75	9.75	27.79	27.79
<i>Panel B. First stage</i>								
IV	12.97*** (1.23)	12.27*** (1.09)	12.97*** (1.23)	12.27*** (1.09)	12.97*** (1.23)	12.27*** (1.09)	12.80*** (1.29)	12.02*** (1.10)
NTRgap _j	-0.01 (0.01)	-0.02* (0.01)	-0.01 (0.01)	-0.02* (0.01)	-0.01 (0.01)	-0.02* (0.01)	-0.01 (0.01)	-0.02* (0.01)
First-stage <i>F</i>	112.03	126.78	112.03	126.78	112.03	126.78	98.60	120.40
Add. ctrl.		✓		✓		✓		✓
# clusters	105	105	105	105	105	105	99	99
N	949	949	949	949	949	949	767	767

Notes. Standard errors clustered at the SIC 3-digit level are reported in parentheses. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. This table reports IV estimates of equation (B.1) for US firms. ΔFDI_j is defined in equation (B.2) and is instrumented by the IV based on South Korea and Japan's FDI in India. In columns 1–2, 3–4, 5–6, and 7–8, the dependent variables are the DHS growth rates of sales, employment, capital, and exports over 1999–2012. Panel A reports the second-stage estimates, and Panel B the first stage, where the dependent variable is the endogenous regressor ΔFDI_j . The NTR gap is the tariff increase on Chinese goods from a failed NTR renewal. The even columns include the same set of additional controls as in Table 1. AR and AR *p*-val. are the Anderson-Rubin statistic and its *p*-value. First-stage *F* is the Kleibergen-Paap F-statistic. All regressions are weighted by initial sales.

B.3 Additional Figures and Tables

Figure B2: Raw Plots. Direct Effects of Joint Venture Formation on Chinese Partner Firms

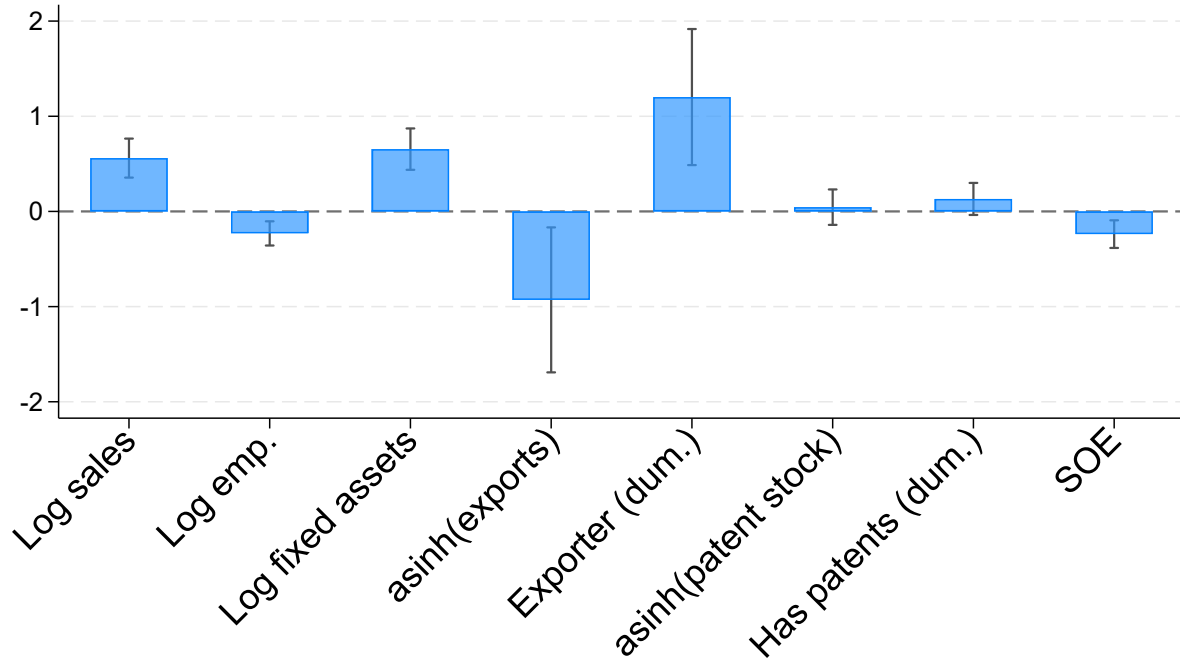


A. Log sales

B. Technological proximity

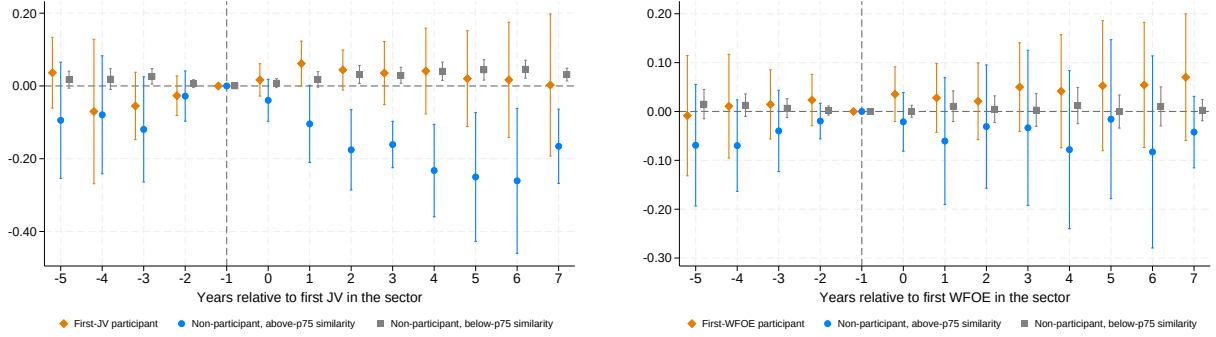
Notes. The figure plots mean trends of log sales and technological proximity (equation (3.2)) for treated and control groups, residualized on match fixed effects. All values are normalized by each group's pre-event average.

Figure B3: Propensity Score Matching. Logit Estimation Result



Notes. The figure plots the estimated coefficients from the logit regression used to construct the propensity score for matching. 95% confidence intervals, with standard errors clustered at the two-digit industry level, are reported. The dependent variable equals one for firms that form a joint venture and zero for candidate control firms, and all covariates are measured in the year before the joint venture. Every covariate is standardized to mean zero and unit standard deviation, so each bar measures the change in the log-odds of forming a JV associated with a one-standard-deviation increase in that covariate and is comparable across covariates. The covariates are log sales, log employment, log fixed assets, the inverse hyperbolic sine (asinh) of exports, an indicator for being an exporter, the asinh of the cumulative patent stock (depreciated at 30% per year), an indicator for holding any patents, and an indicator for state-owned enterprises (SOE).

Figure B4: Raw Plots of Log Sales for JV Participants and Non-Participants. Participants' Expansion Does Not Explain Non-Participants' Decline

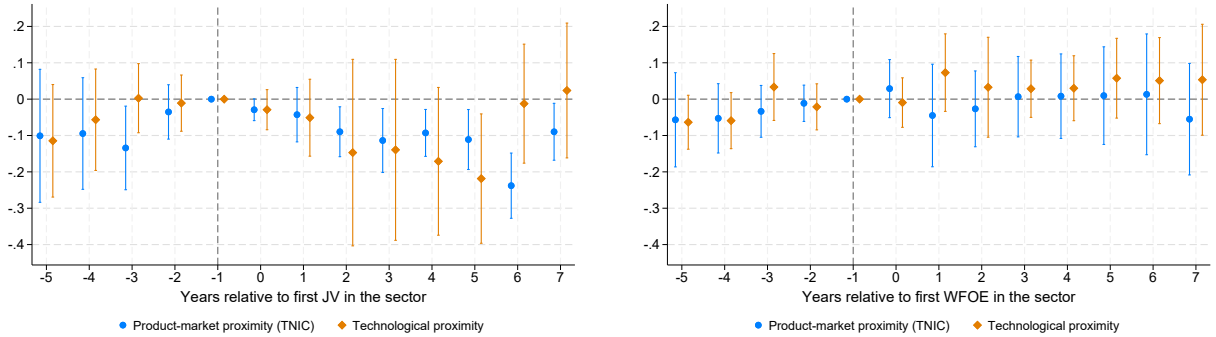


A. JV

B. WFOE

Notes. Panel A plots log sales of JV participants, non-participants whose product overlap to the participant in the same sector is above the 75th percentile, and non-participants whose product overlap is below the 75th percentile. Log sales are residualized on firm and SIC 4-digit industry $\times$ year fixed effects, and each series is normalized to zero in the year before the event. 95% confidence intervals, based on standard errors clustered at the SIC 3-digit level, are reported. Panel B plots the analogous figure for WFOE.

Figure B5: Robustness: Technological Similarity. JV and Negative Outcomes for US Firms



A. JV

B. WFOE

Notes. The figure plots the estimated event study coefficients from equation (3.6). 95% confidence intervals, with standard errors clustered at the SIC 3-digit industry level, are reported. β_{-1} and γ_{-1} are normalized to zero. Each panel reports coefficients for the first-JV or first-WFOE event over 1999–2012. The dependent variable is log sales. Unlike the baseline, we use pre-event average technological similarity (equation 3.2) as an exposure, based on USPTO patent data, instead of product overlap. In all panels except A and B, the participants' controls are included.

Table B4: Balance of Matched Sample. Direct Effects of Joint Venture Formation on Chinese Partner Firms

	JV				Non-JV				(Col. 1 - Col. 5)	
	Mean (1)	Median (2)	SD (3)	N (4)	Mean (5)	Median (6)	SD (7)	N (8)	<i>t</i> -stat (9)	<i>p</i> -val (10)
Log sale	17.42	17.28	1.67	629	17.46	17.29	1.63	2,506	0.36	0.55
Log emp.	6.39	6.27	1.42	629	6.42	6.37	1.50	2,506	0.15	0.70
Log sales per emp.	11.03	10.92	1.14	629	11.04	10.93	1.25	2,506	0.01	0.90
Log capital	16.28	16.17	1.85	629	16.24	16.16	1.90	2,506	0.26	0.61
Log capital per emp.	9.90	9.85	1.22	629	9.82	9.79	1.34	2,506	0.97	0.33
Ihs export	9.62	14.33	8.21	629	9.86	14.16	8.22	2,506	0.17	0.68
Dum export	0.57	1	0.50	629	0.58	1	0.49	2,506	0.12	0.73
Ihs cumulative patent	0.64	0	1.36	629	0.66	0	1.34	2,506	0.02	0.90
Dum patent stock	0.26	0	0.44	629	0.28	0	0.45	2,506	0.28	0.60
Ihs yearly new patent	0.42	0	1.12	629	0.42	0	1.11	2,506	0.01	0.94
Dum yearly new patent	0.16	0	0.37	629	0.17	0	0.37	2,506	0.09	0.77
Dum SOE	0.27	0	0.44	629	0.31	0	0.46	2506	1.4	0.24

Notes. This table presents descriptive statistics for treated and control firms from five to one years before the event. Column 9 reports *t*-statistics for the mean differences between treated and control groups, while Column 10 provides the corresponding *p*-values (in brackets), computed using standard errors clustered at the firm and match levels. All monetary values are expressed in 2007 US dollars.

Table B5: Balance Test. Direct Effects of Joint Venture Formation on Chinese Partner Firms

Dep. var.	Dummies of JV status											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Log sale	-0.003 (0.005)											
Log emp		-0.003 (0.007)										
Log sales per emp			-0.001 (0.009)									
Log capital				0.002 (0.004)								
Log capital per emp					0.008 (0.008)							
lhs export						-0.001 (0.001)						
Dum export							-0.008 (0.022)					
lhs cumulative patent stock								-0.001 (0.009)				
Dum cumulative patent stock									-0.015 (0.029)			
lhs yearly patent										-0.001 (0.010)		
Dum yearly patent											-0.009 (0.029)	
Dum SOE												-0.033 (0.028)
Mean dep. var.	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
# clusters (match)	176	176	176	176	176	176	176	176	176	176	176	176
# clusters (pair)	868	868	868	868	868	868	868	868	868	868	868	868
N	3135	3135	3135	3135	3135	3135	3135	3135	3135	3135	3135	3135

Notes. Standard errors in parentheses are clustered at the match and firm levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table presents the covariate balance test for the event study sample, covering five to one years before the event. The dependent variable is a dummy indicating treatment status. The regressors include log sales, log employment, log sales per employment, log fixed assets, log fixed assets per employment, and the inverse hyperbolic sine transformation and dummies of exports, cumulative patent stock, yearly new patents, and dummies of State Owned Enterprises (SOE).

Table B6: Direct Effects of Joint Venture Formation on Chinese Partner Firms

Dep. var.	Baseline outcomes			Alternative outcomes				
	Log sales	Ihs export	Technological proximity	Log capital	Log emp.	Dum. export	Ihs patent stock	Ihs annual patent
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
5 years before	−0.03 (0.10)	0.78 (0.77)	0.03 (0.06)	0.02 (0.11)	−0.01 (0.12)	0.07 (0.05)	−0.12 (0.16)	−0.13 (0.17)
4 years before	−0.02 (0.08)	0.07 (0.61)	0.06 (0.04)	−0.04 (0.09)	0.04 (0.09)	0.00 (0.04)	0.01 (0.12)	0.12 (0.12)
3 years before	0.03 (0.05)	0.36 (0.56)	0.03 (0.03)	−0.05 (0.09)	0.02 (0.08)	0.02 (0.04)	0.01 (0.10)	0.06 (0.11)
2 years before	−0.01 (0.04)	0.23 (0.44)	0.04 (0.03)	0.06 (0.06)	−0.00 (0.05)	0.01 (0.03)	−0.02 (0.06)	0.01 (0.08)
1 year before								
Year of the event	0.13*** (0.04)	0.14 (0.35)	0.02 (0.03)	0.11** (0.05)	0.04 (0.04)	−0.00 (0.03)	0.04 (0.04)	0.12 (0.08)
1 year after	0.16*** (0.05)	0.20 (0.48)	0.02 (0.03)	0.16*** (0.05)	0.14** (0.06)	0.01 (0.03)	0.14* (0.08)	0.17* (0.10)
2 years after	0.13* (0.07)	0.62 (0.58)	0.05 (0.04)	0.25*** (0.07)	0.12 (0.09)	0.04 (0.04)	0.25** (0.11)	0.25** (0.12)
3 years after	0.22*** (0.08)	1.09* (0.56)	0.12** (0.05)	0.31*** (0.08)	0.07 (0.11)	0.05 (0.04)	0.30** (0.12)	0.35*** (0.12)
4 years after	0.27*** (0.08)	1.01* (0.57)	0.14*** (0.05)	0.35*** (0.08)	0.14* (0.09)	0.06 (0.04)	0.44*** (0.14)	0.47*** (0.14)
5 years after	0.32*** (0.10)	1.29* (0.68)	0.14*** (0.05)	0.37*** (0.09)	0.05 (0.13)	0.08* (0.05)	0.43*** (0.15)	0.37*** (0.14)
6 years after	0.38*** (0.11)	1.24 (0.83)	0.15*** (0.05)	0.43*** (0.10)	0.07 (0.15)	0.06 (0.05)	0.55*** (0.16)	0.62*** (0.17)
7 years after	0.46*** (0.11)	1.12 (0.85)	0.15*** (0.06)	0.54*** (0.12)	0.15 (0.15)	0.09* (0.05)	0.45** (0.19)	0.40** (0.19)
Fixed effects	Firm-match, Match-year							
Mean dep. var.	17.77	10.39	0.22	16.44	6.48	0.58	0.95	0.62
# Cluster (match)	176	176	106	176	176	176	176	176
# Cluster (firm)	859	859	321	859	859	859	859	859
N	7,457	7,457	1,746	7,457	7,457	7,457	7,457	7,457

Notes. Standard errors, shown in parentheses, are clustered at the match and firm levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the estimated event study coefficients of equation (3.1). β_{-1} is normalized to zero. The dependent variables are log sales (col. 1), the inverse hyperbolic sine of exports (col. 2), technological proximity from equation (3.2) (col. 3), log capital (col. 4), log employment (col. 5), an export dummy (col. 6), and the inverse hyperbolic sine of cumulative patent stock (col. 7) and of yearly new patents (col. 8). All specifications include match-firm and match-year fixed effects. In column 3, the sample size decreases because technological proximity is only well-defined for firms with positive patent stock in both the treated and control groups.

Table B7: Robustness. Alternative Number of Matches and Depreciation Rates for Patent Stock. Direct Effects of Joint Venture Formation on Chinese Partners

Robustness	Alt. number of matches						Alt. dep. rate		
	2	3	5	2	3	5	0	0.2	0.5
	Log sales			Technological proximity					
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
5 years before	-0.16 (0.13)	-0.17 (0.11)	-0.09 (0.10)	0.05 (0.06)	-0.04 (0.08)	-0.02 (0.09)	0.03 (0.05)	0.03 (0.06)	0.02 (0.06)
4 years before	-0.10 (0.09)	-0.13 (0.08)	-0.09 (0.08)	0.06 (0.04)	0.02 (0.03)	0.04 (0.04)	0.03 (0.03)	0.05 (0.04)	0.06 (0.04)
3 years before	-0.03 (0.07)	-0.02 (0.06)	-0.00 (0.05)	0.07* (0.04)	0.04 (0.03)	0.04 (0.03)	0.03 (0.03)	0.03 (0.03)	0.02 (0.03)
2 years before	-0.01 (0.05)	-0.02 (0.04)	-0.02 (0.04)	0.06 (0.04)	0.02 (0.03)	0.04 (0.03)	0.03 (0.03)	0.03 (0.03)	0.04 (0.03)
1 year before									
Event year	0.11** (0.04)	0.13*** (0.04)	0.11*** (0.03)	0.04 (0.04)	0.02 (0.04)	0.02 (0.03)	0.03 (0.02)	0.02 (0.03)	0.01 (0.03)
1 year after	0.14** (0.06)	0.16*** (0.05)	0.16*** (0.05)	0.03 (0.04)	0.03 (0.03)	0.00 (0.03)	0.03 (0.03)	0.02 (0.03)	0.02 (0.03)
2 years after	0.09 (0.07)	0.12* (0.07)	0.13* (0.07)	0.05 (0.05)	0.04 (0.04)	0.01 (0.04)	0.04 (0.03)	0.04 (0.04)	0.05 (0.04)
3 years after	0.15* (0.09)	0.22** (0.08)	0.22*** (0.08)	0.12** (0.06)	0.11** (0.05)	0.09** (0.04)	0.08** (0.04)	0.11** (0.05)	0.13*** (0.05)
4 years after	0.16* (0.09)	0.21** (0.09)	0.20** (0.08)	0.12* (0.06)	0.11** (0.05)	0.09** (0.04)	0.10** (0.04)	0.13*** (0.05)	0.14*** (0.05)
5 years after	0.27** (0.11)	0.30*** (0.11)	0.27*** (0.09)	0.13** (0.06)	0.12** (0.06)	0.09** (0.04)	0.11*** (0.04)	0.14*** (0.05)	0.14*** (0.05)
6 years after	0.28** (0.13)	0.25** (0.12)	0.23** (0.10)	0.11 (0.07)	0.11* (0.06)	0.08* (0.04)	0.11*** (0.04)	0.14*** (0.05)	0.16*** (0.05)
7 years after	0.35** (0.14)	0.32** (0.13)	0.31*** (0.11)	0.14* (0.07)	0.13** (0.06)	0.10** (0.05)	0.11** (0.05)	0.15** (0.06)	0.13** (0.05)
FE	Firm-match, Match-year								
Mean dep. var.	17.82	17.74	17.70	0.22	0.21	0.19	0.24	0.23	0.21
# clusters (match)	176	176	176	70	89	111	106	106	106
# clusters (firm)	523	692	1,030	176	253	412	321	321	321
N	4,462	6,046	9,055	971	1,462	2,500	1,746	1,746	1,746

Notes. Standard errors, shown in parentheses, are clustered at the match and firm levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the estimated event study coefficients of equation (3.1). β_{-1} is normalized to zero. In columns 1–3 and 4–9, the dependent variables are log sales and technological proximity (equation (3.2)), respectively. In columns 1–6 and 7–9, we consider alternative numbers of matches and depreciation rates used for computing patent stock, respectively. All specifications include match-firm and match-year fixed effects. In Columns 4–9, the sample size decreases due to firms with zero patent stock, as technological proximity is only well-defined for firms with positive patent stock in both the treated and control groups.

Table B8: Negative Outcomes for Global Firms (OLS)

Dep. var.	ΔSale		ΔEmp		$\Delta\text{Capital}$	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔJV_j	-5.24** (2.35)	-4.18** (2.01)	-0.62 (1.45)	-2.09 (1.87)	-3.54* (2.07)	-4.07** (1.78)
ΔWFOE_j	7.12 (5.53)	7.82 (7.07)	9.72* (5.35)	12.16** (4.88)	5.10 (6.31)	8.93 (6.10)
NTRgap_j	-1.12* (0.59)	-1.10*** (0.35)	-0.50 (0.33)	-0.48* (0.29)	-0.84** (0.41)	-0.82*** (0.29)
Controls		✓		✓		✓
Mean dep. var.	-1.36	-1.36	7.22	7.22	-9.48	-9.48
# clusters	117	117	117	117	117	117
N	1,416	1,416	1,416	1,416	1,416	1,416

Notes. Standard errors clustered at the SIC 3-digit level are reported in parentheses. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. This table reports OLS estimates of equation (3.3). The estimation sample includes global firms, excluding those from China and the US. ΔJV_j and ΔWFOE_j are defined in equations (3.4) and (3.5). In columns 1–2, 3–4, and 5–6, the dependent variables are DHS growth rates of sales, employment, and capital over 1999–2012. The NTR gap is the tariff increase on Chinese goods from a failed NTR renewal. The even columns include additional controls as in Table 1. All regressions are weighted by initial employment.

Table B9: Pretrends Cannot Explain Negative Outcomes for US Firms (OLS)

Panel A. China Pretrends (Sector-Level)						
Dep. var.	Δ Gross output (1995–1997)		Δ Value-added (1995–1997)			
	(1)	(2)	(3)	(4)		
ΔJV_j	–0.01 (0.01)	–0.01 (0.01)	–0.00 (0.01)	–0.00 (0.01)		
$\Delta WFOE_j$		0.05 (0.04)		0.00 (0.05)		
$NTRgap_j$		0.00 (0.00)		0.00 (0.00)		
Mean dep. var.	0.24	0.24	0.31	0.31		
# clusters	157	157	157	157		
N	379	379	379	379		

Panel B. US Pretrends (Firm-Level)						
Dep. var.	Δ Sale (1993–1998)		Δ Emp (1993–1998)		Δ Capital (1993–1998)	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔJV_j	–0.29 (1.23)	–1.36 (1.92)	–0.22 (1.15)	–1.61 (1.61)	0.85 (0.96)	0.07 (1.26)
$\Delta WFOE_j$		–4.16 (8.69)		–4.71 (6.51)		–5.14 (11.13)
$NTRgap_j$		–0.26 (0.34)		–0.34 (0.29)		–0.17 (0.19)
Mean dep. var.	17.11	17.11	5.83	5.83	19.03	19.03
# clusters	97	97	97	97	97	97
N	594	594	594	594	594	594

Notes. Standard errors clustered at the CIC and SIC 3-digit level are reported in parentheses in Panels A and B, respectively. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. ΔJV_j and $\Delta WFOE_j$ are defined in equations (3.4) and (3.5). In Panel A, the dependent variables in columns 1–2 and 3–4 are DHS growth of China’s sectoral gross output and value-added (1995–1997). In Panel B, the dependent variables in columns 1–2, 3–4, and 5–6 are DHS growth of US firms’ sales, employment, and capital (1993–1998). The NTR gap is the tariff increase on Chinese goods from a failed NTR renewal. All regressions are weighted by initial gross output in Panel A and initial sales in Panel B.

Table B10: Robustness. Negative Outcomes for US Firms (OLS)

Robustness	Alt. sample full-sample	Alt. clustering 4-digit level	Alt. sample period 1999–2007	Alt. weight emp.
	(1)	(2)	(3)	(4)
ΔJV_j	–14.00*** (2.34)	–14.27*** (2.64)	–15.55*** (4.71)	–11.25*** (2.78)
$\Delta WFOE_j$	8.51 (7.77)	9.07 (5.91)	12.40** (5.70)	1.16 (6.71)
$NTRgap_j$	–1.80** (0.79)	–2.06** (0.79)	–1.20** (0.53)	–0.89 (0.74)
Mean dep. var.	8.69	7.84	16.46	6.03
# clusters	105	173	110	105
N	1,017	949	1,368	949

Notes. Standard errors are reported in parentheses. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. This table reports OLS estimates of equation (3.3) for US firms. The dependent variable is the DHS growth rate of sales. ΔJV_j and $\Delta WFOE_j$ are defined in equations (3.4) and (3.5) and exclude each firm's own JV and WFOE affiliates. Each column reports one robustness check. Column 1 uses the full sample, including US firms that formed a JV or WFOE in China. Column 2 clusters standard errors at the SIC 4-digit level. Column 3 restricts the sample period to 1999–2007 and computes both JV and WFOE exposures over 1999–2007. Column 4 weights by initial employment. Unless noted otherwise, standard errors are clustered at the SIC 3-digit level and regressions are weighted by initial sales. All columns include the NTR gap control.

Table B11: Robustness. Within-Industry Event Study. JV and Negative Outcomes for US Non-Participants

Robustness	Baseline	With controls	p75 exposure	Alt. outcomes		Both-event sectors
Dep. var.	Log sales (1)	Log sales (2)	Log sales (3)	Log emp. (4)	Log capital (5)	Log sales (6)
5 years before	−0.09 (0.10)	−0.09 (0.10)	−0.10 (0.15)	−0.08 (0.06)	−0.06 (0.06)	−0.05 (0.06)
4 years before	−0.08 (0.08)	−0.07 (0.08)	−0.07 (0.17)	−0.05 (0.05)	−0.05 (0.06)	−0.06 (0.06)
3 years before	−0.10 (0.07)	−0.09 (0.07)	−0.14 (0.14)	−0.04 (0.03)	−0.06 (0.05)	−0.09* (0.05)
2 years before	−0.03 (0.03)	−0.02 (0.04)	−0.02 (0.07)	−0.00 (0.01)	−0.02 (0.02)	−0.01 (0.04)
1 year before Event year	−0.03 (0.02)	−0.00 (0.03)	−0.07 (0.04)	−0.03 (0.04)	−0.03 (0.02)	−0.05 (0.03)
1 year after	−0.06 (0.05)	−0.07 (0.06)	−0.13 (0.08)	−0.02 (0.04)	−0.04 (0.03)	−0.09 (0.09)
2 years after	−0.12*** (0.04)	−0.14*** (0.05)	−0.24** (0.10)	−0.05 (0.04)	−0.12*** (0.03)	−0.21** (0.08)
3 years after	−0.16*** (0.04)	−0.18*** (0.05)	−0.20*** (0.07)	−0.11*** (0.03)	−0.14*** (0.03)	−0.19** (0.08)
4 years after	−0.13*** (0.04)	−0.14*** (0.05)	−0.33*** (0.10)	−0.12*** (0.03)	−0.19*** (0.04)	−0.13 (0.08)
5 years after	−0.11 (0.07)	−0.12 (0.07)	−0.44** (0.19)	−0.13** (0.05)	−0.23*** (0.07)	−0.18* (0.09)
6 years after	−0.21*** (0.06)	−0.22*** (0.06)	−0.49*** (0.14)	−0.14*** (0.04)	−0.26*** (0.05)	−0.34*** (0.11)
7 years after	−0.09* (0.05)	−0.10* (0.06)	−0.22* (0.12)	−0.07* (0.04)	−0.24*** (0.05)	−0.20** (0.09)
Participants' controls		✓				
Mean dep. var.	4.13	4.13	4.13	−0.96	3.39	3.74
# clusters	51	51	51	51	51	23
N	15408	15408	15408	15408	15408	7103

Notes. Standard errors clustered at the SIC 3-digit level are reported in parentheses. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$. This table reports the JV coefficients β_τ of equation (3.6); the WFOE terms are included in every regression but not reported. β_{-1} is normalized to zero. Column 1 is the baseline specification of Figure 2. Column 2 adds the participants' controls, interacted with the corresponding TNIC exposure and post-event indicator. Column 3 replaces the continuous exposure with an indicator for product-market proximity above the 75th percentile. Columns 4 and 5 use log employment and log capital as outcomes. Column 6 restricts the sample to industries that experience both JV and WFOE events, with exposure standardized within that sample.

C. PROOFS OF THE SIMPLE MODEL

Price Index. Throughout the appendix, $\omega \in \{0, 1\}$ indicates whether innovation succeeded and $d \in \{0, 1\}$ whether the technology diffused. Superscripts N and J refer to the no-JV and JV regimes, and JV objects carry the diffusion outcome as a second argument. Price indices and profits are thus $P_c^N(\omega)$ and $\Pi_i^N(\omega)$ without the JV, and $P_c^J(\omega, d)$ and $\Pi_i^J(\omega, d)$ with it. For firm-level objects we write $\Delta\Pi_i(\omega, d) = \Pi_i^J(\omega, d) - \Pi_i^N(\omega)$ for the change from the JV at a fixed state. Wages equal one and all firms charge the same markup, so each firm's market share is its demand-shifter-weighted cost term divided by the market aggregate. Dropping the common markup factor, the aggregates are

$$\begin{aligned} (P_H^N(\omega))^{1-\sigma} &= \gamma^{(\sigma-1)\omega} + \psi + \tau^{1-\sigma}b^{1-\sigma}, & (P_F^N(\omega))^{1-\sigma} &= \tau^{1-\sigma}\gamma^{(\sigma-1)\omega} + \tau^{1-\sigma}\psi + b^{1-\sigma}, \\ (P_H^J(\omega, 0))^{1-\sigma} &= \gamma^{(\sigma-1)\omega} + \psi + \tau^{1-\sigma}b^{1-\sigma}, & (P_F^J(\omega, 0))^{1-\sigma} &= \gamma^{(\sigma-1)\omega} + \tau^{1-\sigma}\psi + b^{1-\sigma}, \\ (P_H^J(\omega, 1))^{1-\sigma} &= (1 + \tau^{1-\sigma})\gamma^{(\sigma-1)\omega} + \psi, & (P_F^J(\omega, 1))^{1-\sigma} &= 2\gamma^{(\sigma-1)\omega} + \tau^{1-\sigma}\psi. \end{aligned}$$

The leader's weight is $\gamma^{(\sigma-1)\omega}$ at home and, in China, $\tau^{1-\sigma}\gamma^{(\sigma-1)\omega}$ without the JV and $\gamma^{(\sigma-1)\omega}$ with it. The fringe carries ψ at home and $\tau^{1-\sigma}\psi$ in China. The Chinese firm's weights are $b^{1-\sigma}$ at home and $\tau^{1-\sigma}b^{1-\sigma}$ in the US, and diffusion replaces them with the leader's $\gamma^{(\sigma-1)\omega}$ and $\tau^{1-\sigma}\gamma^{(\sigma-1)\omega}$. Firm i 's operating profit in market c is $\pi_{ci} = \frac{1}{\sigma}s_{ci}L_c$ with s_{ci} its weight over the market aggregate, and consumer surplus in market c is $-\ln P_c$ up to constants that cancel from all comparisons.

Optimal profits and innovation. A firm's total operating profit is the sum of π_{ci} over the two markets, and each term is the firm's weight over the market aggregate times $\frac{L_c}{\sigma}$. In terms

of the aggregates above,

$$\begin{aligned}\Pi_h^N(\omega) &= \frac{1}{\sigma} \left[\frac{\gamma^{(\sigma-1)\omega}}{(P_H^N(\omega))^{1-\sigma}} + \frac{L_F \tau^{1-\sigma} \gamma^{(\sigma-1)\omega}}{(P_F^N(\omega))^{1-\sigma}} \right], & \Pi_h^J(\omega, d) &= \frac{1}{\sigma} \left[\frac{\gamma^{(\sigma-1)\omega}}{(P_H^J(\omega, d))^{1-\sigma}} + \frac{L_F \gamma^{(\sigma-1)\omega}}{(P_F^J(\omega, d))^{1-\sigma}} \right], \\ \Pi_{\tilde{h}}^N(\omega) &= \frac{1}{\sigma} \left[\frac{\psi}{(P_H^N(\omega))^{1-\sigma}} + \frac{L_F \tau^{1-\sigma} \psi}{(P_F^N(\omega))^{1-\sigma}} \right], & \Pi_{\tilde{h}}^J(\omega, d) &= \frac{1}{\sigma} \left[\frac{\psi}{(P_H^J(\omega, d))^{1-\sigma}} + \frac{L_F \tau^{1-\sigma} \psi}{(P_F^J(\omega, d))^{1-\sigma}} \right], \\ \Pi_f^N(\omega) &= \frac{1}{\sigma} \left[\frac{\tau^{1-\sigma} b^{1-\sigma}}{(P_H^N(\omega))^{1-\sigma}} + \frac{L_F b^{1-\sigma}}{(P_F^N(\omega))^{1-\sigma}} \right], & \Pi_f^J(\omega, 0) &= \frac{1}{\sigma} \left[\frac{\tau^{1-\sigma} b^{1-\sigma}}{(P_H^J(\omega, 0))^{1-\sigma}} + \frac{L_F b^{1-\sigma}}{(P_F^J(\omega, 0))^{1-\sigma}} \right],\end{aligned}$$

and after diffusion the Chinese firm carries the leader's technology,

$$\Pi_f^J(\omega, 1) = \frac{1}{\sigma} \left[\frac{\tau^{1-\sigma} \gamma^{(\sigma-1)\omega}}{(P_H^J(\omega, 1))^{1-\sigma}} + L_F \frac{\gamma^{(\sigma-1)\omega}}{(P_F^J(\omega, 1))^{1-\sigma}} \right].$$

Substituting the aggregates expresses every profit as an explicit function of $(\sigma, \tau, \psi, b, \gamma, L_F)$ and the state. Consider next the innovation choice. The transfer is sunk at that point, so in each regime the leader simply maximizes x times its expected profit gain from success, net of the cost $\frac{\alpha}{2}x^2$. Define the marginal returns

$$D^N \equiv \Pi_h^N(1) - \Pi_h^N(0), \quad D^J(d) \equiv \Pi_h^J(1, d) - \Pi_h^J(0, d). \quad (\text{C.1})$$

The first-order conditions give the optimal innovation rates

$$x^N = \frac{D^N}{\alpha}, \quad x^J = \frac{(1 - \phi)D^J(0) + \phi D^J(1)}{\alpha}, \quad (\text{C.2})$$

whenever the rates on the right are below one. Note that each marginal return in equation (C.1) is a difference of market shares times $\frac{L_c}{\sigma}$, summed over the two markets, and so cannot exceed $\frac{1+L_F}{\sigma}$. The condition $\alpha \geq \frac{1+L_F}{\sigma}$ in Proposition 3 therefore guarantees interior rates. No such condition is needed in the proof of Proposition 1, where the returns vanish as $\psi \rightarrow \infty$ and the rates are interior for every sufficiently large ψ .

Proof of Proposition 1.

Proof. (i) Shares within each market sum to one, so total operating profit equals $\frac{L_C}{\sigma}$ in every state, and $\Delta\Pi_h(\omega, d) + \Delta\Pi_f(\omega, d) = -\Delta\Pi_{\tilde{h}}(\omega, d)$ holds state by state. We now verify that the fringe strictly loses in every JV state. With $d = 0$ only the Chinese aggregate changes, rising from $\tau^{1-\sigma}\gamma^{(\sigma-1)\omega} + \tau^{1-\sigma}\psi + b^{1-\sigma}$ to $\gamma^{(\sigma-1)\omega} + \tau^{1-\sigma}\psi + b^{1-\sigma}$, so the fringe's Chinese share falls and its US share is unchanged. With $d = 1$ both aggregates rise. At home the diffused rival exports at $\tau^{1-\sigma}\gamma^{(\sigma-1)\omega}$ in place of $\tau^{1-\sigma}b^{1-\sigma}$, and the aggregate gains $\tau^{1-\sigma}(\gamma^{(\sigma-1)\omega} - b^{1-\sigma}) > 0$ because $\gamma^{(\sigma-1)\omega} \geq 1 > b^{1-\sigma}$. In China the leader gains $\gamma^{(\sigma-1)\omega} - \tau^{1-\sigma}\gamma^{(\sigma-1)\omega} > 0$ from localizing and the rival gains $\gamma^{(\sigma-1)\omega} - b^{1-\sigma} > 0$ from diffusion. The fringe therefore loses in both markets when $d = 1$, and its total profit strictly falls in every JV state.

For the surplus expression, note that total operating profit equals the constant $\frac{1+L_F}{\sigma}$ in every state, so its expectation equals the same constant at any innovation rate. Differencing across regimes then gives $\Delta\mathbb{E}[\Pi_h] + \Delta\mathbb{E}[\Pi_f] = -\Delta\mathbb{E}[\Pi_{\tilde{h}}]$, even though the JV and no-JV expectations are taken at different innovation and diffusion rates. Substituting this into the definition of S in the text yields $S = -\Delta\mathbb{E}[\Pi_{\tilde{h}}] - \frac{\alpha}{2}\Delta(x^2)$.

(ii) As $\psi \rightarrow \infty$, every fringe share tends to one, and every fringe profit difference across regimes and innovation states is of order $1/\psi$, as are the marginal returns D^N and $D^I(d)$ in equation (C.1). By equation (C.2), the innovation choices are then interior for every sufficiently large ψ , with $x^N, x^I = O(1/\psi)$. The change in R&D spending, $\frac{\alpha}{2}[(x^I)^2 - (x^N)^2]$, and the innovation-rate corrections to the fringe terms are then $O(1/\psi^2)$, and they vanish even after scaling by ψ . Hence ψS converges to the ψ -scaled expected fringe loss at $\omega = 0$. The home loss with diffusion is $\frac{\psi}{\sigma} \left[\frac{1}{1+\psi+\tau^{1-\sigma}b^{1-\sigma}} - \frac{1}{1+\tau^{1-\sigma}+\psi} \right]$, with ψ -scaled limit $\frac{\tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma}$. The corresponding Chinese losses are $\frac{L_F\tau^{1-\sigma}\psi}{\sigma} \left[\frac{1}{\tau^{1-\sigma}+\tau^{1-\sigma}\psi+b^{1-\sigma}} - \frac{1}{2+\tau^{1-\sigma}\psi} \right]$ with diffusion and $\frac{L_F\tau^{1-\sigma}\psi}{\sigma} \left[\frac{1}{\tau^{1-\sigma}+\tau^{1-\sigma}\psi+b^{1-\sigma}} - \frac{1}{1+\tau^{1-\sigma}\psi+b^{1-\sigma}} \right]$ without it, and their ψ -scaled limits are $\frac{L_F(2-\tau^{1-\sigma}-b^{1-\sigma})}{\sigma\tau^{1-\sigma}}$ and $\frac{L_F(1-\tau^{1-\sigma})}{\sigma\tau^{1-\sigma}}$. Weighting the diffusion and no-diffusion states by ϕ and $1 - \phi$, we obtain

$$\psi S \rightarrow \phi \frac{\tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma} + \frac{L_F}{\sigma\tau^{1-\sigma}} \left[\phi(2-\tau^{1-\sigma}-b^{1-\sigma}) + (1-\phi)(1-\tau^{1-\sigma}) \right] > 0,$$

so there exists $\bar{\psi}$ with $S > 0$ for every $\psi > \bar{\psi}$. □

Proof of Proposition 2.

Proof. The marginal returns in equation (C.1) are bounded by $\frac{1+L_F}{\sigma}$, so as $\alpha \rightarrow \infty$ both innovation rates in equation (C.2) vanish. All innovation terms then drop out of equation (4.2), and $\Delta W_H = G - B_H - B_F$ with

$$G = \phi \ln \frac{P_H^N(0)}{P_H^J(0,1)} = \frac{\phi}{\sigma-1} \ln \frac{1 + \tau^{1-\sigma} + \psi}{1 + \psi + \tau^{1-\sigma} b^{1-\sigma}},$$

$$B_H = \frac{\phi \psi}{\sigma} \left[\frac{1}{(P_H^N(0))^{1-\sigma}} - \frac{1}{(P_H^J(0,1))^{1-\sigma}} \right] = \frac{\phi \psi}{\sigma} \left[\frac{1}{1 + \psi + \tau^{1-\sigma} b^{1-\sigma}} - \frac{1}{1 + \tau^{1-\sigma} + \psi} \right],$$

$$B_F = \frac{L_F \tau^{1-\sigma} \psi}{\sigma} \left[\phi \left(\frac{1}{(P_F^N(0))^{1-\sigma}} - \frac{1}{(P_F^J(0,1))^{1-\sigma}} \right) + (1-\phi) \left(\frac{1}{(P_F^N(0))^{1-\sigma}} - \frac{1}{(P_F^J(0,0))^{1-\sigma}} \right) \right]$$

$$= \frac{L_F \tau^{1-\sigma} \psi}{\sigma} \left[\phi \frac{2 - \tau^{1-\sigma} - b^{1-\sigma}}{(\tau^{1-\sigma} + \tau^{1-\sigma} \psi + b^{1-\sigma})(2 + \tau^{1-\sigma} \psi)} + (1-\phi) \frac{1 - \tau^{1-\sigma}}{(\tau^{1-\sigma} + \tau^{1-\sigma} \psi + b^{1-\sigma})(1 + \tau^{1-\sigma} \psi + b^{1-\sigma})} \right],$$

where B_H and B_F denote the fringe's profit loss in the US and Chinese markets.

Throughout the proof, write $X = \phi(2 - \tau^{1-\sigma} - b^{1-\sigma}) + (1-\phi)(1 - \tau^{1-\sigma})$ for the bracket in the definition of $\bar{L}_F$, and note that X rearranges to $(1 - \tau^{1-\sigma}) + \phi(1 - b^{1-\sigma})$.

(i) The bound $\ln x \geq 1 - 1/x$ gives $G \geq \frac{\phi}{\sigma-1} \frac{\tau^{1-\sigma}(1-b^{1-\sigma})}{1+\tau^{1-\sigma}+\psi}$. Combining the two fractions in B_H puts the same factor $\frac{\phi \tau^{1-\sigma}(1-b^{1-\sigma})}{1+\tau^{1-\sigma}+\psi}$ in front of B_H , leaving $\frac{\psi}{\sigma(1+\psi+\tau^{1-\sigma}b^{1-\sigma})}$, so

$$G - B_H \geq \frac{\phi \tau^{1-\sigma}(1-b^{1-\sigma})}{1 + \tau^{1-\sigma} + \psi} \left[\frac{1}{\sigma-1} - \frac{\psi}{\sigma(1 + \psi + \tau^{1-\sigma} b^{1-\sigma})} \right] > \frac{\phi \tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma(\sigma-1)(1 + \tau^{1-\sigma} + \psi)},$$

where the last step uses $\psi < 1 + \psi + \tau^{1-\sigma} b^{1-\sigma}$. Next, since $2 + \tau^{1-\sigma} \psi \geq 1 + \tau^{1-\sigma} \psi + b^{1-\sigma}$, both denominators of B_F are at least $(\tau^{1-\sigma} + \tau^{1-\sigma} \psi + b^{1-\sigma})(1 + \tau^{1-\sigma} \psi + b^{1-\sigma})$, and

$$B_F \leq \frac{L_F \tau^{1-\sigma} \psi X}{\sigma (\tau^{1-\sigma} + \tau^{1-\sigma} \psi + b^{1-\sigma})(1 + \tau^{1-\sigma} \psi + b^{1-\sigma})},$$

so $\Delta W_H > 0$ whenever $\frac{\phi \tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma(\sigma-1)(1+\tau^{1-\sigma}+\psi)} \geq \frac{L_F \tau^{1-\sigma} \psi X}{\sigma(\tau^{1-\sigma}+\tau^{1-\sigma}\psi+b^{1-\sigma})(1+\tau^{1-\sigma}\psi+b^{1-\sigma})}$:

$$L_F \leq \frac{\phi(1-b^{1-\sigma})(\tau^{1-\sigma}+\tau^{1-\sigma}\psi+b^{1-\sigma})(1+\tau^{1-\sigma}\psi+b^{1-\sigma})}{(\sigma-1)\psi(1+\tau^{1-\sigma}+\psi)X}.$$

Because the following holds:

$$\begin{aligned} & (\tau^{1-\sigma} + \tau^{1-\sigma}\psi + b^{1-\sigma})(1 + \tau^{1-\sigma}\psi + b^{1-\sigma}) - \tau^{2(1-\sigma)}\psi(1 + \tau^{1-\sigma} + \psi) \\ &= \tau^{1-\sigma} + \tau^{1-\sigma}b^{1-\sigma} + b^{1-\sigma} + b^{2(1-\sigma)} + \tau^{1-\sigma}\psi(1 + 2b^{1-\sigma} - \tau^{2(1-\sigma)}) > 0, \end{aligned}$$

the right side of the condition above strictly exceeds $\bar{L}_F = \frac{\phi \tau^{2(1-\sigma)}(1-b^{1-\sigma})}{(\sigma-1)X}$ for every $\psi > 0$. Hence $L_F \leq \bar{L}_F$ implies $\Delta W_H > 0$ for every ψ . For the converse, multiply each term by ψ and let $\psi \rightarrow \infty$. Then, applying the L'Hôpital's rule, $\psi G \rightarrow \frac{\phi \tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma-1}$, $\psi B_H \rightarrow \frac{\phi \tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma}$, and $\psi B_F \rightarrow \frac{L_F X}{\sigma \tau^{1-\sigma}}$, so

$$\psi \Delta W_H \rightarrow \phi \frac{\tau^{1-\sigma}(1-b^{1-\sigma})}{\sigma(\sigma-1)} - \frac{L_F}{\sigma \tau^{1-\sigma}} X,$$

which is strictly negative if and only if $L_F > \bar{L}_F$. In that case there exists $\check{\psi}$ with $\Delta W_H < 0$ for every $\psi > \check{\psi}$.

(ii) As $\psi \rightarrow 0$, B_H and B_F vanish while $G \rightarrow \frac{\phi}{\sigma-1} \ln \frac{1+\tau^{1-\sigma}}{1+\tau^{1-\sigma}b^{1-\sigma}} > 0$, so $\Delta W_H > 0$ for every sufficiently small ψ . From the identity for X , $\bar{L}_F = \frac{\phi \tau^{2(1-\sigma)}(1-b^{1-\sigma})}{(\sigma-1)[(1-\tau^{1-\sigma})+\phi(1-b^{1-\sigma})]} < \frac{\tau^{2(1-\sigma)}}{\sigma-1}$ for every ϕ , so the bound $L_F > \frac{\tau^{2(1-\sigma)}}{\sigma-1}$ implies $L_F > \bar{L}_F$ and $\Delta W_H < 0$ for every sufficiently large ψ . It remains to show that a single threshold separates the two regions. Write $\Delta W_H = G(1-R)$ with $R \equiv (B_H + B_F)/G$. Since $\phi \in (0, 1)$, we have $G > 0$, so ΔW_H has the sign of $1-R$. Let $\Lambda \equiv \ln \frac{1+\tau^{1-\sigma}+\psi}{1+\psi+\tau^{1-\sigma}b^{1-\sigma}}$, so that $G = \frac{\phi}{\sigma-1}\Lambda$. Every term of $B_H + B_F$ is a positive constant times ψ over the product of its two market aggregates, so R is a weighted sum of the three functions

$$\begin{aligned} & \frac{\psi}{(1+\psi+\tau^{1-\sigma}b^{1-\sigma})(1+\tau^{1-\sigma}+\psi)\Lambda}, \quad \frac{\psi}{(\tau^{1-\sigma}+\tau^{1-\sigma}\psi+b^{1-\sigma})(2+\tau^{1-\sigma}\psi)\Lambda}, \\ & \frac{\psi}{(\tau^{1-\sigma}+\tau^{1-\sigma}\psi+b^{1-\sigma})(1+\tau^{1-\sigma}\psi+b^{1-\sigma})\Lambda}, \end{aligned}$$

with strictly positive weights. We claim that each of the three functions is strictly increasing in ψ . Given this claim, R is strictly increasing and $R \rightarrow 0$ as $\psi \rightarrow 0$, so ΔW_H changes sign at most once, from positive to negative. Together with the behavior at the two tails, this single sign change gives a unique crossing $\tilde{\psi}$, with $\Delta W_H > 0$ for $\psi < \tilde{\psi}$ and $\Delta W_H \leq 0$ for $\psi \geq \tilde{\psi}$, as stated.

To prove the claim, note that the logarithmic derivative of each function is $\frac{1}{\psi}$ minus the logarithmic derivatives of its two aggregates minus Λ'/Λ . Since $\Lambda = \ln(1+z) < z$ for $z = \frac{\tau^{1-\sigma}(1-b^{1-\sigma})}{1+\psi+\tau^{1-\sigma}b^{1-\sigma}}$ and $\Lambda' = -\frac{\tau^{1-\sigma}(1-b^{1-\sigma})}{(1+\psi+\tau^{1-\sigma}b^{1-\sigma})(1+\tau^{1-\sigma}+\psi)}$, we have $-\frac{\Lambda'}{\Lambda} > \frac{1}{1+\tau^{1-\sigma}+\psi}$. The logarithmic derivative of the first function therefore exceeds

$$\frac{1}{\psi} - \frac{1}{1+\psi+\tau^{1-\sigma}b^{1-\sigma}} = \frac{1+\tau^{1-\sigma}b^{1-\sigma}}{\psi(1+\psi+\tau^{1-\sigma}b^{1-\sigma})} > 0.$$

That of the second exceeds

$$\begin{aligned} & \left(\frac{1}{\psi} - \frac{\tau^{1-\sigma}}{\tau^{1-\sigma} + \tau^{1-\sigma}\psi + b^{1-\sigma}} \right) + \left(\frac{1}{1+\tau^{1-\sigma}+\psi} - \frac{\tau^{1-\sigma}}{2+\tau^{1-\sigma}\psi} \right) \\ &= \frac{\tau^{1-\sigma} + b^{1-\sigma}}{\psi(\tau^{1-\sigma} + \tau^{1-\sigma}\psi + b^{1-\sigma})} + \frac{2 - \tau^{1-\sigma} - \tau^{2(1-\sigma)}}{(1+\tau^{1-\sigma}+\psi)(2+\tau^{1-\sigma}\psi)} > 0. \end{aligned}$$

For the third the same steps give the lower bound

$$\frac{\tau^{1-\sigma} + b^{1-\sigma}}{\psi(\tau^{1-\sigma} + \tau^{1-\sigma}\psi + b^{1-\sigma})} + \frac{1 + b^{1-\sigma} - \tau^{1-\sigma} - \tau^{2(1-\sigma)}}{(1+\tau^{1-\sigma}+\psi)(1+\tau^{1-\sigma}\psi + b^{1-\sigma})},$$

whose second numerator can be negative. Once denominators are cleared, positivity of this bound is equivalent to

$$(\tau^{1-\sigma} + b^{1-\sigma})(1+\tau^{1-\sigma}+\psi)(1+\tau^{1-\sigma}\psi + b^{1-\sigma}) > (\tau^{1-\sigma} + \tau^{2(1-\sigma)} - 1 - b^{1-\sigma})\psi(\tau^{1-\sigma} + \tau^{1-\sigma}\psi + b^{1-\sigma}),$$

and we compare the two sides coefficient by coefficient in powers of ψ . At ψ^2 , the coefficients are $\tau^{1-\sigma}(\tau^{1-\sigma} + b^{1-\sigma})$ on the left and $\tau^{1-\sigma}(\tau^{1-\sigma} + \tau^{2(1-\sigma)} - 1 - b^{1-\sigma})$ on the right, and the left one is larger because $1 + 2b^{1-\sigma} > \tau^{2(1-\sigma)}$. At ψ , they are $(\tau^{1-\sigma} + b^{1-\sigma})(1 + b^{1-\sigma} + \tau^{1-\sigma} + \tau^{2(1-\sigma)})$

and $(\tau^{1-\sigma} + b^{1-\sigma})(\tau^{1-\sigma} + \tau^{2(1-\sigma)} - 1 - b^{1-\sigma})$, and the left one is larger because $2(1 + b^{1-\sigma}) > 0$. The constant term is positive on the left and absent on the right. This proves the claim.

Finally, consider the technology gap b . The product $\psi \Delta W_H$ is a rational-log expression in (ψ, b) whose expansion in $1/\psi$ can be differentiated in b term by term, so $\psi \partial \Delta W_H / \partial b$ converges to the b -derivative of the limit of $\psi \Delta W_H$ in part (i). Differentiating through $b^{1-\sigma}$,

$$\frac{\partial}{\partial b} \left[\phi \frac{\tau^{1-\sigma}(1 - b^{1-\sigma})}{\sigma(\sigma - 1)} - \frac{L_F}{\sigma \tau^{1-\sigma}} X \right] = \frac{\phi(\sigma - 1)b^{-\sigma}}{\sigma} \left[\frac{\tau^{1-\sigma}}{\sigma - 1} - \frac{L_F}{\tau^{1-\sigma}} \right],$$

which is strictly negative when $L_F > \frac{\tau^{2(1-\sigma)}}{\sigma - 1}$, so there exists $\hat{\psi}$ such that $\partial \Delta W_H / \partial b < 0$ for every $\psi > \hat{\psi}$. $\square$

Proof of Proposition 3.

Proof. By equation (C.2), the optimal innovation rates satisfy $\alpha x^N = D^N$ and $\alpha x^J = (1 - \phi)D^J(0) + \phi D^J(1)$, and $\alpha \geq \frac{1+L_F}{\sigma}$ keeps both rates interior. As $\psi \rightarrow 0$ and $b \rightarrow 1$ we therefore obtain $D^J(1) \rightarrow 0$ and

$$\begin{aligned} D^N &\rightarrow \frac{\gamma^{\sigma-1} - 1}{\sigma} \left[\frac{\tau^{1-\sigma}}{(\gamma^{\sigma-1} + \tau^{1-\sigma})(1 + \tau^{1-\sigma})} + \frac{\tau^{1-\sigma} L_F}{(\tau^{1-\sigma} \gamma^{\sigma-1} + 1)(1 + \tau^{1-\sigma})} \right], \\ D^J(0) &\rightarrow \frac{\gamma^{\sigma-1} - 1}{\sigma} \left[\frac{\tau^{1-\sigma}}{(\gamma^{\sigma-1} + \tau^{1-\sigma})(1 + \tau^{1-\sigma})} + \frac{L_F}{2(\gamma^{\sigma-1} + 1)} \right], \end{aligned}$$

whose difference is

$$M \equiv D^J(0) - D^N \rightarrow \frac{(\gamma^{\sigma-1} - 1)(1 - \tau^{1-\sigma})(1 - \tau^{1-\sigma} \gamma^{\sigma-1}) L_F}{2\sigma(\gamma^{\sigma-1} + 1)(\tau^{1-\sigma} \gamma^{\sigma-1} + 1)(1 + \tau^{1-\sigma})} > 0,$$

where the last inequality comes from $1 - \tau^{1-\sigma} \gamma^{\sigma-1} > 0$, which comes from $\gamma < \tau$.

(i) Every fringe profit is proportional to ψ and vanishes in the limit, so the welfare change from an innovation success at a given diffusion outcome is the consumer gain alone. Define

$$v^N \equiv \frac{1}{\sigma - 1} \ln \frac{\gamma^{\sigma-1} + \tau^{1-\sigma}}{1 + \tau^{1-\sigma}}, \quad v^D \equiv \frac{1}{\sigma - 1} \ln \frac{(1 + \tau^{1-\sigma}) \gamma^{\sigma-1}}{\gamma^{\sigma-1} + \tau^{1-\sigma}},$$

the consumer value of an exclusive innovation success and the incremental value of its

diffusion, both strictly positive since $\gamma > 1$. Without the JV, success moves the US aggregate from $1 + \tau^{1-\sigma}$ to $\gamma^{\sigma-1} + \tau^{1-\sigma}$, a welfare gain of v^N . Under the JV, the gain is v^N at $d = 0$ and $v^N + v^D$ at $d = 1$, since the aggregate then moves from $1 + \tau^{1-\sigma}$ to $(1 + \tau^{1-\sigma})\gamma^{\sigma-1}$. At the equilibrium rates the leader's private gain from success equals the marginal cost αx , so the net marginal welfare value of innovation is just the expected consumer gain, v^N without the JV and $v^N + \phi v^D$ under it. Both are strictly positive, and the latter is strictly larger.

(ii) With $D^J(1) \rightarrow 0$, $\alpha(x^J - x^N) = (1 - \phi)D^J(0) - D^N = D^J(0)[\bar{\phi}(\tau, L_F) - \phi]$ with $\bar{\phi}(\tau, L_F) \equiv 1 - \frac{D^N}{D^J(0)} = \frac{M}{D^J(0)}$. The JV therefore strictly raises US innovation when $\phi < \bar{\phi}$ and strictly lowers it when $\phi > \bar{\phi}$, and $M > 0$ puts $\bar{\phi}$ in $(0, 1)$.

We next check that $\bar{\phi}$ increases in both arguments. In L_F , the two limits are affine with the intercept $\frac{(\gamma^{\sigma-1}-1)\tau^{1-\sigma}}{\sigma(\gamma^{\sigma-1}+\tau^{1-\sigma})(1+\tau^{1-\sigma})} > 0$, and the identity above shows that the slope of D^N , $\frac{(\gamma^{\sigma-1}-1)\tau^{1-\sigma}}{\sigma(\tau^{1-\sigma}\gamma^{\sigma-1}+1)(1+\tau^{1-\sigma})}$, is smaller than the slope of $D^J(0)$, $\frac{\gamma^{\sigma-1}-1}{2\sigma(\gamma^{\sigma-1}+1)}$. The ratio $D^N/D^J(0)$ is then strictly decreasing in L_F , so $\bar{\phi}$ is strictly increasing. In τ , use $\bar{\phi} = M/D^J(0)$ and recall that $\tau^{1-\sigma}$ falls as τ rises. The limit of M is proportional to $\frac{(1-\tau^{1-\sigma})(1-\tau^{1-\sigma}\gamma^{\sigma-1})}{(\tau^{1-\sigma}\gamma^{\sigma-1}+1)(1+\tau^{1-\sigma})}$, whose numerator rises and whose denominator falls in τ , so M is strictly increasing in τ . In the limit of $D^J(0)$ the second term does not involve τ , and the first term satisfies

$$\frac{\partial}{\partial \tau} \frac{\tau^{1-\sigma}}{(\gamma^{\sigma-1} + \tau^{1-\sigma})(1 + \tau^{1-\sigma})} = \frac{(1 - \sigma) \tau^{-\sigma} (\gamma^{\sigma-1} - \tau^{2(1-\sigma)})}{[(\gamma^{\sigma-1} + \tau^{1-\sigma})(1 + \tau^{1-\sigma})]^2} < 0$$

because $\sigma > 1$ and $\gamma^{\sigma-1} > 1 > \tau^{2(1-\sigma)}$, so $D^J(0)$ is strictly decreasing in τ . Hence $\bar{\phi}$ is strictly increasing in τ .

Turning to formation, recall that fringe profits vanish in the limit. By part (i) of Proposition 1 the negotiating pair's joint operating-profit gain then vanishes as well, and the surplus reduces to the saving in innovation spending,

$$S \rightarrow -\frac{\alpha}{2} [(x^J)^2 - (x^N)^2].$$

Both rates are positive, so $S > 0$ exactly when $x^J < x^N$, that is, when $\phi > \bar{\phi}$. For every $\phi > \bar{\phi}$

the JV is thus established and strictly lowers US innovation.

The last step is the welfare claim. With zero leader bargaining power the leader is held to its outside option, so $\Delta W_H = \mathbb{E}_{x^J} [U^J(\omega)] - \mathbb{E}_{x^N} [U^N(\omega)]$ with $U^N(\omega) = -\ln P_H^N(\omega) + \Pi_{\tilde{h}}^N(\omega)$. At $\omega = 0$ both $(P_H^J(0, 1))^{1-\sigma}$ and $(P_H^N(0))^{1-\sigma}$ tend to $1 + \tau^{1-\sigma}$, so $U^J(0) - U^N(0) \rightarrow 0$. Meanwhile $U^J(1) - U^N(1) \rightarrow \phi v^D$ and $U^J(1) - U^J(0) \rightarrow v^N + \phi v^D$, using $v^N + v^D = \ln \gamma$. Adding and subtracting $\mathbb{E}_{x^N} [U^J(\omega)]$, substituting the rates, and using that $\alpha x^N = D^N$ and $\alpha x^J = (1 - \phi)D^J(0)$, we obtain that

$$\Delta W_H \rightarrow \frac{(1 - \phi) D^J(0)}{\alpha} (v^N + \phi v^D) - \frac{D^N}{\alpha} v^N.$$

Denote the right side by $Q(\phi)$. Note first that $v^N > v^D$, since this inequality reads $(\gamma^{\sigma-1} + \tau^{1-\sigma})^2 > \gamma^{\sigma-1} (1 + \tau^{1-\sigma})^2$ and factors as $(\gamma^{\sigma-1} - 1)(\gamma^{\sigma-1} - \tau^{2(1-\sigma)}) > 0$. Then $Q'(\phi) = \frac{D^J(0)}{\alpha} [(1 - 2\phi)v^D - v^N] < 0$, so Q is strictly decreasing. At $\phi = \bar{\phi}$ we have $(1 - \bar{\phi})D^J(0) = D^N$ and hence $Q(\bar{\phi}) = \frac{D^N \bar{\phi} v^D}{\alpha} > 0$, while $Q(1) = -\frac{D^N v^N}{\alpha} < 0$. So Q has a unique zero $\tilde{\phi}$, which lies strictly above $\bar{\phi}$, and in the limit the JV lowers US welfare exactly when $\phi > \tilde{\phi}$. $\square$

D. ADDITIONAL MATHEMATICAL EXPRESSIONS

Value Functions of Leaders. A Home leader's value function (regardless of $m^F > 0$ or not) without JV is expressed as follows:

$$\begin{aligned}
 r_{Ht} V_{ht}(\mathbf{m}; 0) - \dot{V}_{ht}(\mathbf{m}; 0) = & \max_{x_{ht}, d_{ht}} \left\{ \Pi_{ht}(\mathbf{m}) - \alpha_{Hr} \frac{(x_{ht})^\gamma}{\gamma} w_{Ht} - \alpha_{Hd} \frac{(d_{ht})^\gamma}{\gamma} w_{Ht} \right. \\
 & + x_{ht} \left(V_{ht}(\mathbf{m} + (1, 1, 0); 0) - V_{ht}(\mathbf{m}; 0) \right) + x_{ft} \left(V_{ht}(\mathbf{m} + (-1, 0, 1); 0) - V_{ht}(\mathbf{m}; 0) \right) \\
 & \left. + d_{ht} \left(V_{ht}(\mathbf{m}; 1) - V_{ht}(\mathbf{m}; 0) - C_t(\mathbf{m}) \right) + \sum_{\mathbf{m}'} \tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) \left(V_{ht}(\mathbf{m}'; 0) - V_{ht}(\mathbf{m}; 0) \right) \right\}, \quad (D.1)
 \end{aligned}$$

where $\tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m})$ denotes transition probabilities:

$$\tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) = \begin{cases} \delta^F & \text{if } \mathbf{m}' = \{0, |m^F| \times \mathbb{1}[m^F \leq 0] + m^{DH}, |m^F| \times \mathbb{1}[m^F > 0] + m^{DF}\} \\ \delta^D & \text{if } \mathbf{m}' = \{m^F, 0, m^{DF}\} \\ \delta^D & \text{if } \mathbf{m}' = \{m^F, m^{DH}, 0\} \\ 0 & \text{Otherwise,} \end{cases} \quad (D.2)$$

where $\mathbb{1}[\cdot]$ is an indicator function. By using an indicator function, we generalize equation (5.5) to apply in both cases: $m^F > 0$ and $m^F \leq 0$. A Home leader's value function with JV is

$$\begin{aligned}
 r_{Ht} V_{ht}(\mathbf{m}; 1) - \dot{V}_{ht}(\mathbf{m}; 1) = & \max_{x_{ht}} \left\{ \Pi_{ht}(\mathbf{m}) - \alpha_{Hr} \frac{(x_{ht})^\gamma}{\gamma} w_{Ht} + \kappa \Pi_{vt}(\mathbf{m}) \right. \\
 & + x_{ht} \left(V_{ht}(\mathbf{m} + (1, 1, 0); 1) - V_{ht}(\mathbf{m}; 1) \right) + x_{ft} \left(V_{ht}(\mathbf{m} + (-1, 0, 1); 1) - V_{ht}(\mathbf{m}; 1) \right) \\
 & + \phi \left(V_{ht}(0, |m^F| \times \mathbb{1}[m^F \leq 0] + m^{DH}, |m^F| \times \mathbb{1}[m^F > 0] + m^{DF}; 1) - V_{ht}(\mathbf{m}; 1) \right) \\
 & \left. + \chi \left(V_{ht}(\mathbf{m}; 0) - V_{ht}(\mathbf{m}; 1) \right) + \sum_{\mathbf{m}'} \tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) \left(V_{ht}(\mathbf{m}'; 1) - V_{ht}(\mathbf{m}; 1) \right) \right\}. \quad (D.3)
 \end{aligned}$$

A Foreign leader's value functions with and without JVs are

$$\begin{aligned}
r_{Ft}V_{ft}(\mathbf{m};0) - \dot{V}_{ft}(\mathbf{m};0) = \max_{x_{ft}} & \left\{ \Pi_{ft}(\mathbf{m}) - \alpha_{Fr} \frac{(x_{ft})^\gamma}{\gamma} w_{Ft} \right. \\
& + x_{ft} \left(V_{ft}(\mathbf{m} + (-1, 0, 1); 0) - V_{ft}(\mathbf{m}; 0) \right) + x_{ht} \left(V_{ft}(\mathbf{m} + (1, 1, 0); 0) - V_{ft}(\mathbf{m}; 0) \right) \\
& \left. + d_{ht} \left(V_{ft}(\mathbf{m}; 1) - V_{ft}(\mathbf{m}; 0) + C_t(\mathbf{m}) \right) + \sum_{\mathbf{m}'} \tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) \left(V_{ft}(\mathbf{m}'; 0) - V_{ft}(\mathbf{m}; 0) \right) \right\}.
\end{aligned} \tag{D.4}$$

$$\begin{aligned}
r_{Ft}V_{ft}(\mathbf{m};1) - \dot{V}_{ft}(\mathbf{m};1) = \max_{x_{ft}} & \left\{ \Pi_{ft}(\mathbf{m}) - \alpha_{Fr} \frac{(x_{ft})^\gamma}{\gamma} w_{Ft} + (1 - \kappa) \Pi_{vt}(\mathbf{m}) \right. \\
& + x_{ft} \left(V_{ft}(\mathbf{m} + (-1, 0, 1); 1) - V_{ft}(\mathbf{m}; 1) \right) + x_{ht} \left(V_{ft}(\mathbf{m} + (1, 1, 0); 1) - V_{ft}(\mathbf{m}; 1) \right) \\
& + \phi \left(V_{ft}(0, |m^F| \times \mathbb{1}[m^F \leq 0] + m^{DH}, |m^F| \times \mathbb{1}[m^F > 0] + m^{DF}; 1) - V_{ft}(\mathbf{m}; 1) \right) \\
& \left. + \chi \left(V_{ft}(\mathbf{m}; 0) - V_{ft}(\mathbf{m}; 1) \right) + \sum_{\mathbf{m}'} \tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) \left(V_{ft}(\mathbf{m}'; 1) - V_{ft}(\mathbf{m}; 1) \right) \right\}.
\end{aligned} \tag{D.5}$$

Value Functions of Fringe Firms. For both fringe firms in Home and Foreign, $i \in \{\tilde{h}, \tilde{f}\}$, the value functions without and with JVs are expressed as follows:

$$\begin{aligned}
r_{ct}V_{it}(\mathbf{m};0) - \dot{V}_{it}(\mathbf{m};0) = \Pi_{it}(\mathbf{m}) \\
& + x_{ht} \left(V_{it}(\mathbf{m} + (1, 1, 0); 0) - V_{it}(\mathbf{m}; 0) \right) + x_{ft} \left(V_{it}(\mathbf{m} + (-1, 0, 1); 0) - V_{it}(\mathbf{m}; 0) \right) \\
& + d_{ht} \left(V_{it}(\mathbf{m}; 1) - V_{it}(\mathbf{m}; 0) \right) + \sum_{\mathbf{m}'} \tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) \left(V_{it}(\mathbf{m}'; 0) - V_{it}(\mathbf{m}; 0) \right).
\end{aligned} \tag{D.6}$$

$$\begin{aligned}
r_{ct}V_{it}(\mathbf{m};1) - \dot{V}_{it}(\mathbf{m};1) = \Pi_{it}(\mathbf{m}) \\
& + x_{ft} \left(V_{it}(\mathbf{m} + (-1, 0, 1); 1) - V_{it}(\mathbf{m}; 1) \right) + x_{ht} \left(V_{it}(\mathbf{m} + (1, 1, 0); 1) - V_{it}(\mathbf{m}; 1) \right) \\
& + \phi \left(V_{it}(0, |m^F| \times \mathbb{1}[m^F \leq 0] + m^{DH}, |m^F| \times \mathbb{1}[m^F > 0] + m^{DF}; 1) - V_{it}(\mathbf{m}; 1) \right) \\
& + \chi \left(V_{it}(\mathbf{m}; 0) - V_{it}(\mathbf{m}; 1) \right) + \sum_{\mathbf{m}'} \tilde{\mathbb{T}}(\mathbf{m}'; \mathbf{m}) \left(V_{it}(\mathbf{m}'; 1) - V_{it}(\mathbf{m}; 1) \right).
\end{aligned} \tag{D.7}$$

Law of Motion for Technology Gaps and JV Status. The law of motion for $\mu_t(\mathbf{m}; \mathcal{J})$ is

$$\begin{aligned}
\dot{\mu}_t(\mathbf{m}; \mathcal{J}) = & \underbrace{x_{ht}(m^F - 1, m^{DH} - 1, m^{DF}; \mathcal{J})\mu_t(m^F - 1, m^{DH} - 1, m^{DF}; \mathcal{J})}_{\text{Innovation by Home leader}} \\
& + \underbrace{x_{ft}(m^F + 1, m^{DH}, m^{DF} - 1; \mathcal{J})\mu_t(m^F + 1, m^{DH}, m^{DF} - 1; \mathcal{J})}_{\text{Innovation by Foreign leader}} \\
& + \underbrace{d_{ht}(m^F, m^{DH}, m^{DF}; 0)\mathbb{1}[\mathcal{J} = 1]\mu_t(m^F, m^{DH}, m^{DF}; 0)}_{\text{JV investment}} + \underbrace{\chi\mathbb{1}[\mathcal{J} = 0]\mu_t(m^F, m^{DH}, m^{DF}; 1)}_{\text{JV exit}} \\
& + \underbrace{\delta^D\mathbb{1}[m^{DH} = 0]\sum_{n=1}^{n=\bar{m}}\mu_t(m^F, n, m^{DF}; \mathcal{J})}_{\text{Home domestic diffusion}} + \underbrace{\delta^D\mathbb{1}[m^{DF} = 0]\sum_{n=1}^{n=\bar{m}}\mu_t(m^F, m^{DH}, n; \mathcal{J})}_{\text{Foreign domestic diffusion}} \\
& + \underbrace{\delta^F\mathbb{1}[m^F = 0]\sum_{n=-\bar{m}}^{n=\bar{m}}\mu_t(n, m^{DH}, m^{DF}; \mathcal{J})}_{\text{International diffusion}} + \underbrace{\phi\mathbb{1}[m^F = 0, \mathcal{J} = 1]\sum_{n=-\bar{m}}^{n=\bar{m}}\mu_t(n, m^{DH}, m^{DF}; 1)}_{\text{JV direct diffusion}} \\
& - \underbrace{\left(x_{ht}(\mathbf{m}; \mathcal{J}) + x_{ft}(\mathbf{m}; \mathcal{J}) + 2\delta^D + d_{ht}(\mathbf{m})\mathbb{1}(\mathcal{J} = 0) + \delta^F + (\phi + \chi)\mathbb{1}(\mathcal{J} = 1)\right)\mu_t(\mathbf{m}; \mathcal{J})}_{\text{Subtracted mass}}.
\end{aligned} \tag{D.9}$$

The first four lines of the right hand side capture the mass that enters a state $(\mathbf{m}; \mathcal{J})$ from other states. The first line captures that in state $(m^F - 1, m^{DH} - 1, m^{DF})$, Home leader's successful innovation moves to (m^F, m^{DH}, m^{DF}) with rate x_{ht} . The second line captures evolution of states due to Foreign leader's innovation. In the third line, $\mathbb{1}[\mathcal{J} = 1]$ or $\mathbb{1}[\mathcal{J} = 0]$ are indicator functions of the JV status. For $\mathcal{J} = 1$, with rate d_{ht} , a JV is established moving from a state $(m^F, m^{DH}, m^{DF}; 0)$ to state $(m^F, m^{DH}, m^{DF}; 1)$. The second term in the third line captures the exogenous exit of existing JVs when $\mathcal{J} = 1$. The fourth line captures evolution of states due to direct diffusion through JVs, domestic and international diffusions. Finally, the last line captures the mass leaving the current state.

D.1 Nash-in-Nash Bargaining

We adopt the Nash-in-Nash solution, where each negotiating pair maximizes its Nash product, taking the actions of other pairs as given.

$$\begin{aligned}
C_t(\mathbf{m}) &= \operatorname{argmax}_C \left\{ (\Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C - C^E)^\xi \times (\Delta^{\text{JV}} V_{ft}(\mathbf{m}) + C)^{1-\xi} \right\} \\
\text{s.t.} \quad & \Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C - C^E \geq 0, \quad \Delta^{\text{JV}} V_{ft}(\mathbf{m}) + C \geq 0 \\
&= (1 - \xi)(\Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C^E) - \xi \Delta^{\text{JV}} V_{ft}(\mathbf{m})
\end{aligned} \tag{D.10}$$

$$\begin{aligned}
C_t^E(\mathbf{m}) &= \operatorname{argmax}_{C^E} \left\{ (\Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C - C^E)^{\xi^E} \times (\Delta^{\text{JV}} V_{\tilde{h}t}(\mathbf{m}) + C^E)^{1-\xi^E} \right\} \\
\text{s.t.} \quad & \Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C - C^E \geq 0, \quad \Delta^{\text{JV}} V_{\tilde{h}t}(\mathbf{m}) + C^E \geq 0 \\
&= (1 - \xi^E)(\Delta^{\text{JV}} V_{ht}(\mathbf{m}) - C) - \xi^E \Delta^{\text{JV}} V_{\tilde{h}t}(\mathbf{m})
\end{aligned} \tag{D.11}$$

Combining equations (D.10) and (D.11), we obtain

$$C = \frac{\xi^E(1 - \xi)}{\xi^E(1 - \xi) + \xi} \{ \Delta^{\text{JV}} V_{ht}(\mathbf{m}) + \Delta^{\text{JV}} V_{\tilde{h}t}(\mathbf{m}) \} - \frac{\xi}{\xi^E(1 - \xi) + \xi} \Delta^{\text{JV}} V_{ft}(\mathbf{m}) \tag{D.12}$$

$$C^E = \frac{\xi(1 - \xi^E)}{\xi(1 - \xi^E) + \xi^E} \{ \Delta^{\text{JV}} V_{ht}(\mathbf{m}) + \Delta^{\text{JV}} V_{ft}(\mathbf{m}) \} - \frac{\xi^E}{\xi(1 - \xi^E) + \xi^E} \Delta^{\text{JV}} V_{\tilde{h}t}(\mathbf{m}). \tag{D.13}$$

When we set $\xi^E = 1$, we obtain the expression used for our quantitative analysis on coordinated JV decisions.

D.2 Additional Figures and Tables

Table D1: Baseline vs. Banning Joint Venture Investments in 1999: Net Present Value of Real Profits and Labor Income. China

	Baseline	Shutting down JV	Changes (%)
CN leader profit (own + JV + JV fee)	0.042	0.038	−9.87
Own profit	0.042	0.038	−9.62
JV profit	0.004	0.000	n/a
JV fee payment	−0.004	0.000	n/a
CN fringe profit	0.063	0.059	−6.67
CN labor income	0.895	0.826	−7.71
CN total real income	1.000	0.923	−7.73

Notes. This table reports the net present value of real profits and labor income, deflated by China's price index and normalized by baseline total real income, under the counterfactual that JVs are banned in 1999 versus the baseline. Leader profits include own profits, JV profits, and JV fees.

Table D2: US Welfare Decomposition under Coordinated Joint Venture Decision: Baseline vs. JV Ban in 1999

	Baseline	JV Ban	Change (%)
US leader profit (own + JV + JV fee)	0.037	0.037	−0.65
Own profit	0.037	0.037	0.77
JV profit	0.001	0.000	n/a
JV fee revenue	0.000	0.000	n/a
US fringe profit	0.068	0.068	−0.38
US labor income	0.895	0.894	−0.05
US total real income	1.000	0.999	−0.09

Notes. This table reports net present values of real profits and labor income for the US, deflated by the US price index and normalized by total real income in the baseline with coordinated JV decision through multilateral bargaining in Section 7.2. The JV-ban counterfactual restricts new JVs from 1999 onward under the same coordinated-bargaining assumption. Leader profits include own profits, JV profits, and JV fees.

Table D3: The Effects of Banning Joint Venture in China under Different Scenarios

	Δ Fringe profit (%)	Δ Innovation rate (%)	Δ Real wage (%)	Δ Welfare (%)
<i>Panel A. JV ban in 1999</i>				
(i) Baseline	-6.67	54.53	-7.71	-8.90
(ii) Multilateral bargaining	0.33	2.02	-0.23	-0.14
(iii) Fixing innovation	-6.65	56.12	-7.69	-9.03
(iv-a) No leakage ($\phi = 0$)	12.05	61.65	-3.62	-5.73
(iv-b) + Larger profit share ($\phi = 0, \kappa = 1$)	6.86	43.92	-7.04	-8.26
<i>Panel B. JV ban in 2025</i>				
(v) Baseline	-2.42	19.75	-3.00	-2.33

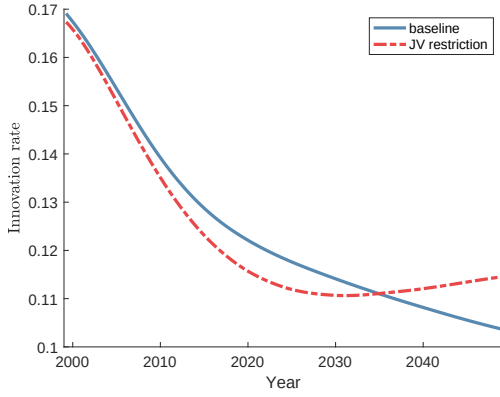
Notes. Each row reports the effect of a JV ban relative to the corresponding no-ban baseline, under different scenarios or parameterizations. Profits and the real wage are net present values deflated by the US price index, and welfare is in consumption-equivalent units. The innovation rate is the average intensity over the first 50 years. All columns are percent changes. Rows (i)–(iv) ban new JV investment from 1999 onward and evaluate outcomes from 1999, while row (v) bans from 2025 and evaluates from 2025. Row (i) is our main counterfactual. In row (ii), US leaders must compensate US fringe firms for their future profit losses in order to establish a JV, through the multilateral Nash-in-Nash bargaining, and the parenthetical excludes the fee paid to fringe firms so as to isolate the change in their operating profit. In row (iii), we adjust the US innovation rate under the counterfactual so that average innovation matches the baseline in each period, which shuts off the innovation margin. Row (iv-a) sets $\phi = 0$, eliminating direct diffusion through JVs and hence the leakage effect, and row (iv-b) additionally sets $\kappa = 1$, assigning US leaders the entire JV profit.

Table D4: Baseline vs. Banning Joint Venture in 1999. Robustness. Sensitivity Checks

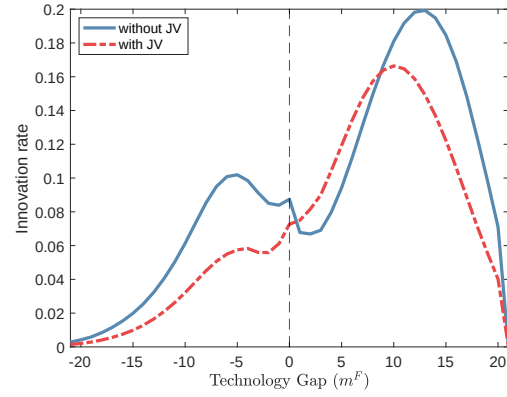
	Δ US Welfare (%)	Δ CN Welfare (%)	Δ US Innovation rate (%)	Δ CN Innovation rate (%)
Baseline	1.08	-8.90	3.62	54.53
<i>Panel A. Alternative assumptions</i>				
Love of variety	0.75	-9.20	3.62	54.53
Constant markup	1.24	-6.23	5.02	32.89
Allow JVs when US productivity is lower	0.95	-9.22	10.34	50.34
<i>Panel B. Direct diffusion (baseline: $\phi = 0.15$)</i>				
$\phi = 0.1$	0.75	-9.14	1.40	58.24
$\phi = 0.2$	1.21	-8.53	4.87	50.80
<i>Panel C. International diffusion (baseline: $\delta^F = 0.029$)</i>				
$\delta^F = 0.024$	1.25	-9.83	5.35	54.68
$\delta^F = 0.034$	0.88	-8.22	1.54	53.28
<i>Panel D. Domestic diffusion (baseline: $\delta^D = 0.029$)</i>				
$\delta^D = 0.024$	1.20	-8.77	5.06	57.47
$\delta^D = 0.034$	0.99	-9.00	2.61	52.29
<i>Panel E. Initial technology gap (baseline: $b = 20$)</i>				
$b = 23$	1.03	-9.68	-0.27	52.98
$b = 17$	1.07	-7.93	6.42	55.05
<i>Panel F. US JV profit share (baseline: $\kappa = 0.54$)</i>				
$\kappa = 0.75$	0.77	-9.50	2.47	53.61
$\kappa = 0.25$	1.48	-8.01	5.20	54.66

Notes. This table reports the effects of banning JV in 1999 with alternative parameterizations. Δ Welfare is expressed in consumption-equivalent units, and Δ innovation rate denotes the percentage change in average innovation rates over the first 50 years between the baseline and counterfactual scenarios.

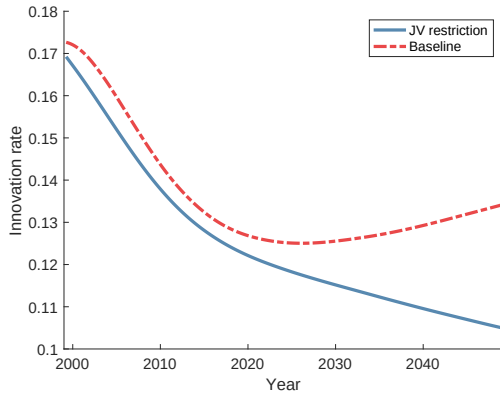
Figure D1: US Innovation Rate over Time and Technology Gap. The Cases of Higher Profit Share $\kappa = 1$ and No Direct Diffusion $\phi = 0$



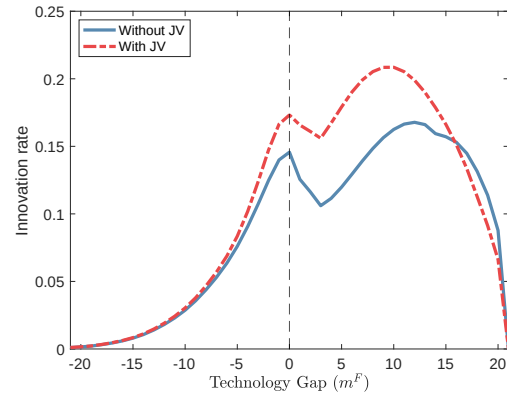
A. US average innovation rate



B. US Innovation rate over technology gap



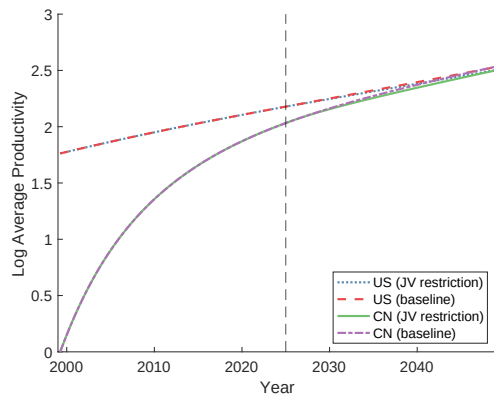
C. US average innovation rate



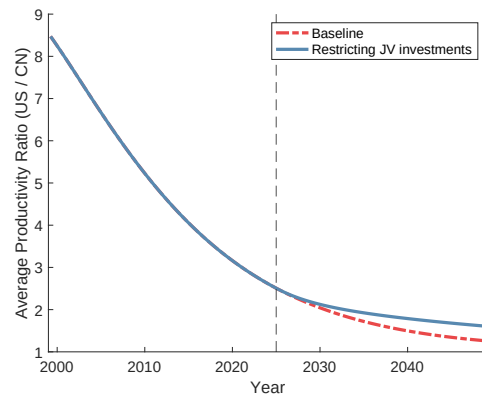
D. US Innovation rate over technology gap

Notes. In Panels A and B, we consider the case of larger profit share case by setting $\kappa = 1$. In Panels C and D, we shut down direct diffusion by setting $\phi = 0$. Panels A and C plot the average innovation rate in baseline and counterfactual scenarios. Panels B and D plot the innovation rate in the baseline scenario in 2025 over m^F , the technology gap between US and Chinese leader firms. $m^F > 0$ denotes the case when US firms exceed the Chinese leader in productivity. In this example, technology gaps between domestic firms are set to $m^{DF} = m^{DH} = \bar{m}$.

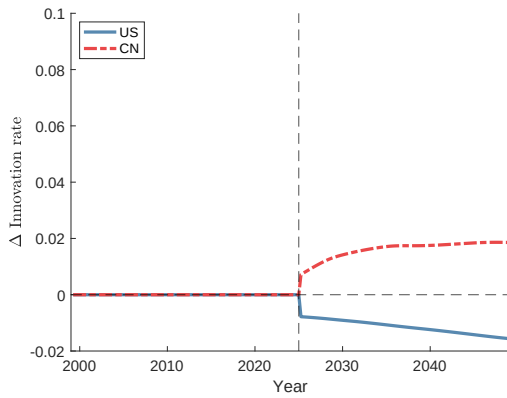
Figure D2: Baseline vs. Banning Joint Venture Investments in 2025: Dynamics of Productivity, Innovation, and Consumption



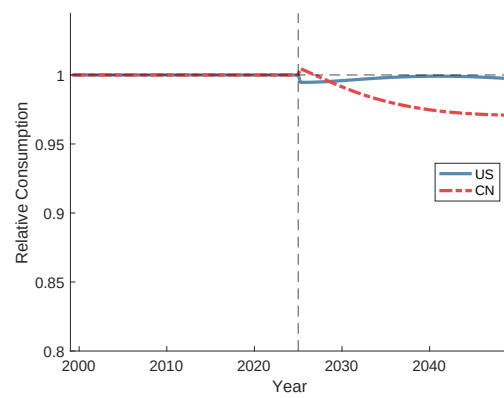
A. Log leader productivity



B. Productivity ratio (US/China)



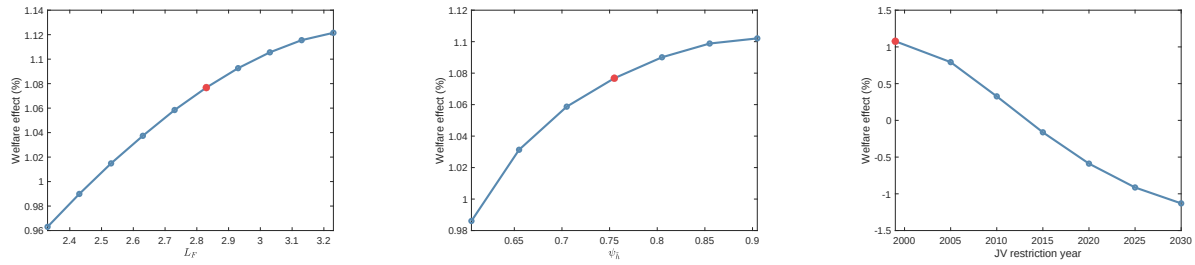
C. Innovation rate difference



D. Relative consumption

Notes. This figure compares dynamics under the counterfactual (JV ban beginning in 2025) to the baseline. The dashed vertical line is 2025. Panel A shows log average leader productivity for the US and China; Panel B shows the US–China productivity ratio; Panel C plots the difference in their average innovation rates (counterfactual minus baseline); and Panel D plots relative consumption (counterfactual over baseline).

Figure D3: Comparative Statics over Foreign Market Size, Fringe Demand Shifter, and Technology Gap (Timing). Welfare Effects of JV Ban (%)



A. Foreign market size L_F B. Fringe demand shifter ψ C. Technology gap b (timing)

Notes. This figure reports consumption-equivalent welfare changes for the US from banning JV investments over different values of the Foreign market size L_F , the fringe demand shifter ψ , and the timing of JV bans (the initial technology gap). The red dots mark the baseline calibrated values in Panels A and B, and the baseline ban timing in Panel C.

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